复旦数学资料汇编
代数拓扑

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第一章　代数拓扑

1.1　Hatcher

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For a quick review of Homology Theory, one can refer to the Appendix of *Characteristic Classes* written by Milnor and Stasheff.

1.0　Some Underlying Geometric Notions

定义 1.0.1. *A **retraction** of X onto a subspace A is a map $r : X \to A$ such that $r|A = \mathrm{id}$.*
*A **deformation retraction** of X onto A is a homotopy from id_X to r.*

定义 1.0.2. *For a subspace $A \subset X_1$ and a map $f : A \to X_0$, $X_0 \sqcup_f X_1$, X_0 **with** X_1 **attached along** A **via** f, is the quotient space $X_0 \sqcup X_1/\{a \in A \sim f(a) \in X_0\}$.*

定义 1.0.3 (Operations on Spaces)**.**
*The **cone** $CX = X \times I/X \times \{0\}$.*
*The **suspension** $SX = X \times I/X \times \{0\}/X \times \{1\}$.*
*The **mapping cylinder** M_f is $X \times I \supset X \times \{1\} \xrightarrow{f} Y$.*
*The **mapping cone** C_f is $CX \supset X \times \{1\} \xrightarrow{f} Y$.*
*The **join** $X * Y = X \times Y \times I/\{(x, y, 0) \sim (x, y', 0), (x, y, 1) \sim (x', y, 1)\}$, that is, collapsing $X \times Y \times \{0\}$ to X and $X \times Y \times \{1\}$ to Y.*
*The **wedge sum** $X \vee Y = X \sqcup Y/\{x_0 \sim y_0\}$ for chosen points $x_0 \in X, y_0 \in Y$.*
*The **smash product** $X \wedge Y = X \times Y/X * Y$.*

定义 1.0.4. *A **cell complex** or a **CW complex** X is a space obtained by $X^0 \subset X^1 \subset \cdots$, starting from a set of points X_0 and inductively forming the n-skeleton X^n by attaching n-cells $e^n_\alpha \supset S^{n-1} \to X^{n-1}$. Either set $X = X^n$ stopping at some finite n, or set $X = \bigcup_n X^n$ giving the weak topology: $A \subset X$ is open iff $A \cap X^n$ is open for each n.*

定义 1.0.5. *A **subcomplex** of X is a closed subspace A that is a union of cells of A. (X, A) is called a **CW pair**.*

例 1.0.1. *A **graph** is $X = X^1$, vertices with edges attached, two ends of which can be attached to the same vertex.*

例 1.0.2. *The sphere $S^n = D^n/\partial D^n$ has the structure $e^n \supset S^{n-1} \to e^0$.*

例 1.0.3. ***Real projective n-space*** *$\mathbb{R}P^n = S^n/\{v \sim -v\}$ is also the quotient space of a semisphere D^n with antipodal points identified (just $\mathbb{R}P^{n-1}$), or $e^n \supset S^{n-1} \to \mathbb{R}P^{n-1}$. So $\mathbb{R}P^n = e^0 \cup e^1 \cup \cdots \cup e^n$.*

例 1.0.4. ***Complex projective n-space*** *$\mathbb{C}P^n = e^0 \cup e^2 \cup \cdots \cup e^{2n}$.*

定义 1.0.6. *Orientable surface M_g*

Attach 2-cell to the wedge sum of $2g$ circles $a_1, b_1, \cdots a_g, b_g$ by the word $[a_1, b_1] \cdots [a_g, b_g] = a_1 b_1 a_1^{-1} b_1^{-1} \cdots a_g b_g a_g^{-1} b_g^{-1}$ we obtain an orientable surface M_g.

定义 1.0.7. *Nonorientable surface N_g*

Attach 2-cell to the wedge sum of g circles $a_1, \cdots a_g$ by the word $a_1^2 \cdots a_g^2$ we obtain a nonorientable surface N_g.

For example, N_1 is the projective plane $\mathbb{R}P^2$, and N_2 is the Klein bottle.

1.1 Fundamental Group

One can often show that two spaces are not homeomorphic by showing that their fundamental groups are not isomorphic, since it will be easy consequence of the definition of the fundamental group that homeomorphic spaces have isomorphic fundamental groups.

1.1.1 Basic Constructions

定义 1.1.1. *A homotopy of paths in X is a family $f_t : I \to X, 0 \le t \le 1$, such that*

1. *The endpoints $f_t(0) = x_0$ and $f_t(1) = x_1$ are independent of t.*
2. *The associated map $F : I \times I \to X$ defined by $F(s,t) = f_t(s)$ is continuous.*

When two paths f_0 and f_1 are connected in this way by a homotopy f_t, they are said to be ***homotopic****. The notation for this is $f_0 \sim f_1$.*

定义 1.1.2. *The relation of homotopy on paths with fixed endpoints in any space is an equivalence relation.*

The equivalence class of a path f under the equivalence relation of homotopy will be denoted $[f]$ and called the ***homotopy class*** *of f.*

定义 1.1.3. *The set of all homotopy classes $[f]$ of loops $f : I \to X$ at the basepoint x_0 is denoted $\pi_1(X, x_0)$.*

$\pi_1(X, x_0)$ is a group with respect to the product $[f][g] = [f \cdot g]$. This group is called the ***fundamental group*** *of X at the basepoint x_0.*

定义 1.1.4. *Define a* ***reparametrization*** *of a path f to be a composition $f\varphi$ where $\varphi : I \to I$ is any continuous map such that $\varphi(0) = 0$ and $\varphi(1) = 1$.*

Reparametrizing a path preserves its homotopy class since $f\varphi \sim f$ via the homotopy $f\varphi_t$, where $\varphi_t = (1-t)\varphi(s) + ts$.

定义 1.1.5. *Say h is a path from x_0 to x_1, then we define a* ***change-of-basepoint*** *map $\beta_h : \pi_1(X, x_1) \to \pi_1(X, x_0)$ by $\beta_h[f] = [h \cdot f \cdot \overline{h}]$.*

命题 1.1.1. *The map $\beta_h : \pi_1(X, x_1) \to \pi_1(X, x_0)$ is an isomorphism.*

Thus if X is path-connected, the group $\pi_1(X, x_0)$ is, up to isomorphism, independent of the choice of basepoint x_0. In this case the notation $\pi_1(X, x_0)$ is often abbreviated to $\pi_1(X)$.

定义 1.1.6. *A space is called **simply-connected** if it is path-connected and has trivial fundamental group.*

命题 1.1.2. *A space X is simply-connected iff there is a unique homotopy class of paths connecting any two points in X.*

定义 1.1.7. *Given a space X, a **covering space** of X consists of a space $\tilde{X}$ and a map $p : \tilde{X} \to X$ satisfying the following condition:*

For each point $x \in X$, there is an open neighborhood U of x in X such that $p^{-1}(U)$ is a union of disjoint open sets each of which is mapped homeomorphically onto U by p.

*Such a U will be called **evenly covered**.*

There are three easy lifting properties.

命题 1.1.3. *For each path $f : I \to X$ starting at a point $x_0 \in X$ and each $\tilde{x_0} \in p^{-1}(x_0)$ there is a unique lift $\tilde{f} : I \to \tilde{X}$ starting at $\tilde{x_0}$.*

命题 1.1.4. *For each homotopy $f_t : I \to X$ of paths starting at x_0 and each $\tilde{x_0} \in p^{-1}(x_0)$ there is a unique lifted homotopy $\tilde{f}_t : I \to \tilde{X}$ of paths starting at $\tilde{x_0}$.*

命题 1.1.5. *Given a map $F : Y \times I \to X$ and a map $\tilde{F} : Y \times \{0\} \to \tilde{X}$ lifting $F|_{Y\times\{0\}}$, then there is a unique map $\tilde{F} : Y \times I \to \tilde{X}$ lifting F and restricting to the given $\tilde{F}$ on $Y \times \{0\}$.*

定理 1.1.1. *$\pi(S^1)$ is an infinite cyclic group generated by the homotopy class of the loop $\omega(s) = (\cos 2\pi s, \sin 2\pi s)$ based at $(1, 0)$.*

In other words, $\pi(S^1) \cong \mathbb{Z}$.

定理 1.1.2. ***Algebraic fundamental theorem***

Every nonconstant polynomial with coefficients in $\mathbb{C}$ has a root in $\mathbb{C}$.

定理 1.1.3. ***Brouwer fixed point theorem** in dimension 2*

Every continunous map $h : D^2 \to D^2$ has a fixed point, that is, a point $x \in D^2$ with $h(x) = x$.

注 1.1.1. ***Brouwer fixed point theorem** in dimension n*

Every continunous map $h : D^n \to D^n$ has a fixed point, that is, a point $x \in D^n$ with $h(x) = x$.

定理 1.1.4. ***Borsuk-Ulam theorem** in dimension 2*

For every continuous map $f : S^2 \to \mathbb{R}^2$ there exists a pair of antipodal points x and $-x$ in S^2 with $f(x) = f(-x)$.

注 1.1.2. ***Borsuk-Ulam theorem** in dimension n*

For every continuous map $f : S^n \to \mathbb{R}^n$ there exists a pair of antipodal points x and $-x$ in S^n with $f(x) = f(-x)$.

推论 1.1.1. *There is no one-to-one continuous map from S^2 to $\mathbb{R}^2$, so S^2 is not homeomorphic to a subspace of $\mathbb{R}^2$.*

推论 1.1.2. *Whenever S^2 is expressed as the union of three closed sets A_1, A_2 and A_3, then at least one of these sets must contain a pair of antipodal points $\{x, -x\}$.*

Projecting the four faces of the tetrahedron radially onto the sphere, we obtain a cover of S^2 by four closed sets, none of which contains a pair of antipodal points.

注 1.1.3. *Assuming the higher-dimensional version of the Borsuk-Ulam theorem, the same arguments show that S^n cannot be covered by $n+1$ closed sets without antipodal pairs of points, though it can be covered by $n+2$ such sets, as the higher-dimensional analog of a tetrahedron shows.*

命题 1.1.6. *$\pi_1(X \times Y) \cong \pi_1(X) \times \pi_1(Y)$ if X and Y are path-connected.*

例 1.1.1. ***The Torus**: $\pi_1(S^1 \times S^1) \cong Z \times Z$.*

定义 1.1.8. *Suppose $\varphi : X \to Y$ is a map taking the basepoint $x_0 \in X$ to the basepoint $y_0 \in Y$. For brevity we write $\varphi : (X, x_0) \to (Y, y_0)$ in this situation.*

Then φ induces a homomorphism $\varphi_* : \pi_1(X, x_0) \to \pi_1(Y, y_0)$ defined by composing loops $f : I \to X$ based at x_0 with φ, that is, $\varphi_*[f] = [\varphi f]$.

命题 1.1.7. *Two basic properties of induced homomorphisms are:*

1. $(\varphi\psi)_* = (\varphi)_*(\psi)_*$ for a composition $(X, x_0) \xrightarrow{\psi} (Y, y_0) \xrightarrow{\varphi} (Z, z_0)$.

2. $1_* = 1$, which is a concise way of saying that the identity map $1 : X \to X$ induces the identity map $1 : \pi_1(X, x_0) \to \pi_1(X, x_0)$.

These two properties of induced homomorphisms are what makes the fundamental group a functor.

引理 1.1.1. *The decomposition of loops subordinated to an open covering.*

If a space X is the union of a collection of path-connected open sets A_α each containing the basepoint $x_0 \in X$ and if each intersection $A_\alpha \cap A_\beta$ is path-connected, then every loop in X at x_0 is homotopic to a product of loops each of which is contained in a single A_α.

定理 1.1.5. *$\pi_1(S^n) = 0$ if $n \geq 2$.*

推论 1.1.3. *$\mathbb{R}^2$ is not homeomorphic to $\mathbb{R}^n$ for $n \neq 2$.*

注 1.1.4. *The more general statement that $\mathbb{R}^m$ is not homeomorphic to $\mathbb{R}^n$ if $m \neq n$ can be proved in the same way using either the higher homotopy groups or homology groups.*

命题 1.1.8. *If a space X retracts onto a subspace A, then the homomorphism $i_* : \pi_1(A, x_0) \to \pi_1(X, x_0)$ induced by the inclusion $i : A \hookrightarrow X$ is injective. If A is a deformation retract of X, then i_* is an isomorphism.*

定义 1.1.9. *A **basepoint-preserving homotopy** $\varphi_t : (X, x_0) \to (Y, y_0)$ is the case that $\varphi_t(x_0) = y_0$ for all t.*

*Similarly, we can define **homotopy equivalence for spaces with basepoint**. One says $(X, x_0) \sim (Y, y_0)$ if there are maps $\varphi : (X, x_0) \to (Y, y_0)$ with homotopies $\varphi\psi \sim 1$ and $\psi\varphi \sim 1$ through maps fixing the basepoints.*

命题 1.1.9. *If $\varphi_t : (X, x_0) \to (Y, y_0)$ is a basepoint-preserving homotopy, then $\varphi_{0*} = \varphi_{1*}$.*

命题 1.1.10. *If $(X, x_0) \sim (Y, y_0)$, then $\pi_1(X, x_0) \cong \pi_1(Y, y_0)$.*

For homotopy equivalences one does not have to be quite so careful, as the conditions on basepoints can actually be dropped.

命题 1.1.11. *If $\varphi : X \to Y$ is a homotopy equivalence, then the induced homomorphism $\varphi_* : \pi_1(X, x_0) \to \pi_1(Y, \varphi(x_0))$ is an isomorphism for all $x_0 \in X$.*

引理 1.1.2. *If $\varphi_t : X \to Y$ is a homotopy and h is the path $\varphi_t(x_0)$ formed by the images of a basepoint $x_0 \in X$, then we have $\varphi_{0*} = \beta_h \varphi_{1*}$.*

$$\begin{array}{ccc} \pi_1(X, x_0) & \xrightarrow{\varphi_{1*}} & \pi_1(Y, \varphi_1(x_0)) \\ & \searrow^{\varphi_{0*}} & \downarrow{\scriptstyle \beta_h} \\ & & \pi_1(Y, \varphi_0(x_0)) \end{array}$$

1.1.2 Van Kampen's Theorem

The Van Kampen theorem gives a method for computing the fundamental groups of spaces that can be decomposed into simpler spaces whose fundamental groups are already known.

We shall see for example that for every group G there is a space X_G whose fundamental group is isomorphic to G.

定义 1.1.10. ***Free product**. Refer to Hatcher P41.*

定义 1.1.11. ***Free group** is the free product of any number of copies of $\mathbb{Z}$, finite or infinite. The generators are called a **basis** for the free group, and the number of basis element is the **rank** of the free group.*

定义 1.1.12. *A basic property of the free product $*_\alpha G_\alpha$ is that any collection of homomorphisms $\varphi_\alpha : G_\alpha \to H$ extends uniquely to a homomorphism $\varphi : *_\alpha G_\alpha \to H$.*

Namely, the value of φ on a word $g_1 \cdots g_n$ with $g_i \in G_{\alpha_i}$ must be $\varphi_{\alpha_1}(g_1) \cdots \varphi_{\alpha_n}(g_n)$, and using this formula to define φ gives a well-defined homomorphism.

Suppose a space X is decomposed as the union of a collection of path-connected open subsets A_α, each of which contains the basepoint $x_0 \in X$. The homomorphisms $j_\alpha : \pi_1(A_\alpha) \to \pi_1(X)$ induced by the inclusions $A_\alpha \hookrightarrow X$ extend to a homomorphism $\Phi : *_\alpha \pi_1(A_\alpha) \to \pi_1(X)$.

The Van Kampen theorem will say that Φ is very often surjective, but we can expect Φ to have a nontrivial kernel in general.

For $i_{\alpha\beta} : \pi_1(A_\alpha \cap A_\beta) \to \pi_1(A_\alpha)$ is the homomorphism induced by the inclusion $A_\alpha \cap A_\beta \hookrightarrow A_\alpha$ then $j_\alpha i_{\alpha\beta} = j_\beta i_{\beta\alpha}$, both these compositions being induced by the inclusion $A_\alpha \cap A_\beta \hookrightarrow X$, so the kernel of Φ contains all the elements of the form $i_{\alpha\beta}(\omega) i_{\beta\alpha}(\omega)^{-1}$ for $\omega \in \pi_1(A_\alpha \cap A_\beta)$.

定理 1.1.6. ***Van Kampen theorem***

If X is a union of path-connected open sets A_α each containing the basepoint $x_0 \in X$ and if each intersection $A_\alpha \cap A_\beta$ is path-connected, then the homomorphism $\Phi : *_\alpha \pi_1(A_\alpha) \to \pi_1(X)$ is surjective.

If in addition each intersection $A_\alpha \cap A_\beta \cap A_\gamma$ is path-connected, then the kernel of Φ is the normal subgroup N generated by all elements of the form $i_{\alpha\beta}(\omega)i_{\beta\alpha}(\omega)^{-1}$ for $\omega \in \pi_1(A_\alpha \cap A_\beta)$, and hence Φ induces an isomorphism $\pi_1(X) \cong *_\alpha \pi_1(A_\alpha)/N$.

例 1.1.2. ***Wedge sum**: $\pi_1(\vee_\alpha X_\alpha) \cong *_\alpha \pi_1(X_\alpha)$.*

注 1.1.5. *Van Kampen's theorem is often applied when there are just two sets A_α and A_β in the cover of X, so the condition on the triple intersections $A_\alpha \cap A_\beta \cap A_\gamma$ is superfluous and one obtains an isomorphism $\pi_1(X) \cong (\pi_1(A_\alpha) * \pi_1(A_\beta))/N$, under the assumption that $A_\alpha \cap A_\beta$ is path-connected.*

例 1.1.3. ***Linking of Circles***

The complement $\mathbb{R}^3 - A$ of a single circle A deformation retracts onto a wedge sum $S^1 \vee S^2$ embedded in $\mathbb{R}^3 - A$. Hence $\pi_1(\mathbb{R}^3 - A) \cong \pi_1(S^1 \vee S^2) \cong \mathbb{Z}$.

In similar fashion, the complement $\mathbb{R}^3 - (A \cup B)$ of two unlinked circles A and B deformation retracts onto $S^1 \vee S^1 \vee S^2 \vee S^2$, hence, $\pi_1(\mathbb{R}^3 - (A \cup B)) \cong \pi_1(S^1 \vee S^1 \vee S^2 \vee S^2) = \mathbb{Z} * \mathbb{Z}$.

On the other hand, if A and B are linked, then $\mathbb{R}^3 - (A \cup B)$ deformation retracts onto the wedge sum of S^2 and a torus $S^1 \times S^1$ separating A and B, hence $\pi_1(\mathbb{R}^3 - (A \cup B)) \cong \pi_1(S^1 \times S^1) \cong \mathbb{Z} \times \mathbb{Z}$.

For details and figures, please refer to Hatcher P46.

For the remainder of this section we shall be interested in cell complexes, and in particular in how the fundamental group is affected by attaching 2-cells.

Suppose we attach a collection of 2-cells e^2_α to a path-connected space X via maps $\varphi_\alpha : S^1 \to X$, producing a space Y. If s_0 is a basepoint of S^1 then φ_α determines a loop at $\varphi_\alpha(s_0)$ that we shall call φ_α.

For different α's the basepoint $\varphi_\alpha(s_0)$ of these loops φ_α may not all coincide. To remedy this, choose a basepoint $x_0 \in X$ and a path γ_α in X from x_0 to $\varphi_\alpha(s_0)$ for each α. Then $\gamma_\alpha \varphi_\alpha \overline{\gamma_\alpha}$ is a loop at x_0.

This loop may not be nullhomotopic in X, but it will certainly be nullhomotopic after the cell e^2_α is attached. Thus the normal subgroup $N \subset \pi_1(X, x_0)$ generated by all the loops $\gamma_\alpha \varphi_\alpha \overline{\gamma_\alpha}$ for varying α lies in the kernel of the map $\pi_1(X, x_0) \to \pi_1(Y, x_0)$ induced by the inclusion $X \hookrightarrow Y$.

命题 1.1.12. *If Y is obtained from X by attaching 2-cells as described above, then the inclusion $X \hookrightarrow Y$ induces a surjection $\pi_1(X, x_0) \to \pi_1(Y, x_0)$ whose kernel is N. Thus $\pi_1(Y) \cong \pi_1(X)/N$.*

命题 1.1.13. *If Y is obtained from X by attaching n-cells for a fixed $n > 2$, then the inclusion $X \hookrightarrow Y$ induces an isomorphism $\pi_1(X, x_0) \cong \pi_1(Y, x_0)$.*

For a path-connected cell complex X the inclusion of the 2-skeleton $X^2 \hookrightarrow X$ induces an isomorphism $\pi_1(X^2, x_0) \cong \pi_1(X, x_0)$.

注 1.1.6. *N is independent of the choice of the paths γ_α. If we replace γ_α by another path η_α having the same endpoints, then $\gamma_\alpha \varphi_\alpha \overline{\gamma_\alpha}$ changes to $\eta_\alpha \varphi_\alpha \overline{\eta_\alpha} = (\eta_\alpha \overline{\gamma_\alpha})\gamma_\alpha \varphi_\alpha \overline{\gamma_\alpha}(\gamma_\alpha \overline{\eta_\alpha})$, so $\gamma_\alpha \varphi_\alpha \overline{\gamma_\alpha}$ and $\eta_\alpha \varphi_\alpha \overline{\eta_\alpha}$ define conjugate elements of $\pi_1(X, x_0)$.*

定义 1.1.13. *$\langle g_\alpha \mid r_\beta \rangle$ denotes the group with generators g_α and relator r_β, in other words, the free group on the generator g_α modulo the normal subgroup generated by the words r_β in these generators.*

命题 1.1.14.

$$\pi_1(M_g) \cong \langle a_1, b_1, \cdots, a_g, b_g \mid [a_1, b_1] \cdots [a_g, b_g] \rangle.$$

推论 1.1.4. *The surface M_g is not homeomorphic, or even homotopy equivalent, to M_h if $g \neq h$.*

命题 1.1.15.

$$\pi_1(N_g) = \langle a_1, \cdots, a_g \mid a_1^2 \cdots a_g^2 \rangle.$$

This abelianizes to the direct sum of $\mathbb{Z}_2$ with $g-1$ copies of $\mathbb{Z}$ since in the abelianzation we can rechoose the generators to be $a_1, \cdots, a_{g-1}$ and $a_1 + \cdots + a_g$ with $2(a_1 + \cdots + a_g) = 0$.

推论 1.1.5. *N_g is not homotopy equivalent to N_h if $g \neq h$, nor is N_g homotopy equivalent to any orientable surface M_h.*

推论 1.1.6. *For every group G, there is a 2-dimensional cell complex X_G with $\pi_1(X_G) \cong G$.*

注 1.1.7. *However, the cell complex X_G may not be a surface. For more details, please refer to Hatcher P52.*

1.1.3 Covering Spaces

The connection between the fundamental group and covering spaces runs much deeper than this, and in many ways they can be regarded as two viewpoints toward the same thing. Algebraic aspects of the fundamental group can often be translated into geometric language of covering spaces.

定义 1.1.14. *A **covering space** of a space X is a space $\tilde{X}$ together with a map $p : \tilde{X} \to X$ satisfying the following condition: Each point $x \in X$ has an open neighborhood U in X such that $p^{-1}(U)$ is a union of disjoint open sets in $\tilde{X}$, each of which is mapped homeomorphically onto U by p.*

*Such a U is called **evenly covered** and the disjoint open sets in $\tilde{X}$ that project homeomorphically to U by p are called **sheets** of $\tilde{X}$ over U.*

We allow $p^{-1}(U)$ to be empty, so p need not be surjective. Remark that this definition varies from person to person.

The number of sheets over U is the cardinality of $p^{-1}(x)$ for $x \in U$. As x varies over X this number is locally constant, so it is constant if X is connected.

定义 1.1.15. *A **lift** of a map $f : Y \to X$ is a map $\tilde{f} : Y \to \tilde{X}$ such that $p\tilde{f} = f$.*

命题 1.1.16. ***Homotopy lifting property***

Given a covering space $p : \tilde{X} \to X$, a homotopy $f_t : Y \to X$, and a map $\tilde{f}_0 : Y \to \tilde{X}$ lifting f_0, then there exists a unique homotopy $\tilde{f}_t : Y \to \tilde{X}$ of $\tilde{f}_0$ that lifts f_t.

注 1.1.8. *Taking Y to be a point gives the **path lifting property**.*

命题 1.1.17. *The map $p_* : \pi_1(\tilde{X}, \tilde{x_0}) \to \pi_1(X, x_0)$ induced by a covering space $p : (\tilde{X}, \tilde{x_0}) \to (X, x_0)$ is injective.*

The image subgroup $p_*(\pi_1(\tilde{X}, \tilde{x_0}))$ in $\pi_1(X, x_0)$ consists of the homotopy classes of loops in X based at x_0 whose lift to $\tilde{X}$ starting at $\tilde{x_0}$ are loops.

命题 1.1.18. *The number of sheets of a covering space $p : (\tilde{X}, \tilde{x_0}) \to (X, x_0)$ with X and $\tilde{X}$ path-connected equals the index of $p_*(\pi_1(\tilde{X}, \tilde{x_0}))$ in $\pi_1(X, x_0)$.*

命题 1.1.19. ***Lifting criterion***

Suppose given a covering space $p : (\tilde{X}, \tilde{x_0}) \to (X, x_0)$ and a map $f : (Y, y_0) \to (X, x_0)$ with Y path-connected and locally path-connected.

Then a lift $\tilde{f} : (Y, y_0) \to (\tilde{X}, \tilde{x_0})$ of f exists iff $f_*(\pi_1(Y, y_0)) \subset p_*(\pi_1(\tilde{X}, \tilde{x_0}))$.

命题 1.1.20. ***Unique lifting property***

Given a covering sapce $p : \tilde{X} \to X$ and a map $f : Y \to X$, if two lifts $\tilde{f}_1, \tilde{f}_2 : Y \to \tilde{X}$ of f agree at one point of Y and Y is connected, then $\tilde{f}_1$ and $\tilde{f}_2$ agree on all of Y.

The Classification of Covering spaces

The main thrust of the classification will be a correspondence between connected covering spaces of X and subgroups of $\pi_1(X)$. This is often called the **Galois correspondence** because of its surprising similarity to another basic correspondence in the purely algebraic subject of Galois theory.

The Galois correspondence arises from the function that assigns to each covering space $p : (\tilde{X}, \tilde{x_0}) \to (X, x_0)$ the subgroup $p_*(\pi_1(\tilde{X}, \tilde{x_0}))$ of $\pi_1(X, x_0)$. Fisrt we consider whether this function is surjective. That is, we ask whether every subgroup of $\pi_1(X, x_0)$ is realized as $p_*(\pi_1(\tilde{X}, \tilde{x_0}))$ for some covering space $p : (\tilde{X}, \tilde{x_0}) \to (X, x_0)$. In particular we can ask whether the trivial subgroup is realized. Since p_* is always injective, this amounts to asking whether X has a simply-connected covering space.

定义 1.1.16. ***Semilocally simply-connected***

We say X is semilocally simply-connected if each point $x \in X$ has a neighborhood U such that the inclusion-induced map $\pi_1(U, x) \to \pi_1(X, x)$ is trivial.

A locally simply-connected space is certainly semilocally simply-connected. An example of a space that is not semilocally simply-connected is the **shrinking wedge of circles**.

Given a path-connected, locally path-connected, semilocally simply-connected space X with a basepoint $x_0 \in X$, we are therefore led to define

$$\tilde{X} = \{[\gamma] \mid \gamma \text{ is a path in } X \text{ starting at } x_0\}.$$

Let $\mathscr{U}$ be the collection of path-connected open sets $U \subset X$ such that $\pi_1(U) \to \pi_1(X)$ is trivial. Say that $\mathscr{U}$ is a basis for the topology on X if X is locally path-connected, semilocally simply-connected.

Given a set $U \in \mathscr{U}$ and a path γ in X from x_0 to a point in U, let

$$U_{[\gamma]} = \{[\gamma \cdot \eta] \mid \eta \text{ is a path in } U \text{ with } \eta(0) = \gamma(1)\}.$$

$U_{[\gamma]}$ depends only on the homotopy class $[\gamma]$ and $U_{[\gamma]} = U_{[\gamma']}$ if $[\gamma'] \in U_{[\gamma]}$. And the sets $U_{[\gamma]}$ form a basis for a topology basis for a topology on $\tilde{X}$.

The bijection $p : U_{[\gamma]} \to U$ is a homeomorphism. This implies $p : \tilde{X} \to X$ is continuous and then we can also deduce that this is a covering space.

It remains only to show that $\tilde{X}$ is simply-connected. Then we complete the construction of a simply-connected covering space $\tilde{X} \to X$.

The hypotheses for constructing a simply-connected covering space of X in fact suffice for constructing covering spaces realizing arbitrary subgroup of $\pi_1(X)$.

命题 1.1.21. *Suppose X is path-connected, locally path-connected, semilocally simply-connected. Then for every subgroup $H \subset \pi_1(X, x_0)$ there is a covering space $p : X_H \to X$ such that $p_*(\pi_1(X_H, \tilde{x_0})) = H$ for a suitably chosen basepoint $\tilde{x_0} \in X_H$.*

定义 1.1.17. *An isomorphism between covering spaces $p_1 : \tilde{X_1} \to X$ and $p_2 : \tilde{X_2} \to X$ is a homeomorphism $f : \tilde{X_1} \to \tilde{X_2}$ such that $p_1 = p_2 f$.*

命题 1.1.22. *If X is a path-connected and locally path-connected, then two path-connected covering spaces $p_1 : \tilde{X_1} \to X$ and $p_2 : \tilde{X_2} \to X$ are isomorphic via an isomorphism $f : \tilde{X_1} \to \tilde{X_2}$ taking a basepoint $\tilde{x_1} \in p_1^{-1}(x_0)$ to a basepoint $\tilde{x_2} \in p_2^{-1}(x_0)$ iff $p_{1*}(\pi_1(\tilde{X_1}, \tilde{x_1})) = p_{2*}(\pi_1(\tilde{X_2}, \tilde{x_2}))$.*

定理 1.1.7. ***Classification Theorem***

Let X be a path-connected, locally path-connected, semilocally simply-connected space. Then there is a bijection between the set of basepoint-preserving isomorphism classes of path-connected covering spaces $p : (\tilde{X}, \tilde{x_0}) \to (X, x_0)$ and the set of subgroups of $\pi_1(X, x_0)$, obtained by associating the subgroup $p_*(\pi_1(\tilde{X}, \tilde{x_0}))$ to the covering space $(\tilde{X}, \tilde{x_0})$.

If basepoints are ignored, this correspondence gives a bijection between isomorphism classes of path-connected covering sapce $p : \tilde{X} \to X$ and conjugacy classes of subgroups of $\pi_1(X, x_0)$.

定义 1.1.18. *A simply-connected covering space of a path-connected, locally path-connected space X is a covering sapce of every other path-connected covering space of X.*

*A simply-connected covering space of X is therefore called a **universal cover**. It is unique up to isomorphism, so one is justified in calling it **the** universal covering.*

定义 1.1.19. *For a covering space $p : \tilde{X} \to X$ the isomorphisms $\tilde{X} \to \tilde{X}$ are called **deck transformations** or **covering transformations**. These form a group $G(\tilde{X})$ under composition.*

例 1.1.4. *$p : \mathbb{R} \to S^1$ projecting a vertical helix onto a circle, $G(\tilde{X}) = \mathbb{Z}$ in this case.*

例 1.1.5. *The n-sheeted covering space $p : S^1 \to S^1, z \mapsto z^n$, the deck transformations are the rotations of S^1 through angles that the multiples of $2\pi/n$, so $G(\tilde{X}) = \mathbb{Z}_n$.*

By the unique lifting property, a deck transformation is completely determined by where it sends a single point, assuming $\tilde{X}$ is path-connected. In particular, only the identity deck transformation can fix a point of $\tilde{X}$.

定义 1.1.20. *A covering space $p : \tilde{X} \to X$ is called **normal** if for each $x \in X$ and each pair of lifts $\tilde{x}, \tilde{x}'$ of x there is a deck transformation taking $\tilde{x}$ to $\tilde{x}'$.*

For example, the covering space $\mathbb{R} \to S^1$ and the n-sheeted covering spaces $S^1 \to S^1$ are normal.

命题 1.1.23. *Let $p : (\tilde{X}, \tilde{x_0}) \to (X, x_0)$ be a path-connected covering space of the path-connected, locally path-connected space X, ane let H be the subgroup $p_*(\pi_1(\tilde{X}, \tilde{x_0})) \subset \pi_1(X, x_0)$. Then:*

(a) This covering space is normal iff H is a normal subgroup of $\pi_1(X, x_0)$.

(b) $G(\tilde{X})$ is isomorphic to the quotient $N(H)/H$ where $N(H)$ is the normalizer of H in $\pi_1(X, x_0)$.

In particular, $G(\tilde{X})$ is isomorphic to $\pi_1(X, x_0)/H$ if $\tilde{X}$ is a normal covering. Hence for the universal cover $\tilde{X} \to X$ we have $G(\tilde{X}) \cong \pi_1(X)$.

定义 1.1.21. *We call an action satisfying the following condition is a **covering space action** or **properly discontinuous action**.*

Each $y \in Y$ has a neighborhood U such that all the images $g(U)$ for varying $g \in G$ are disjoint. In other words, $g_1(U) \cap g_2(U) \neq \varnothing$ implies $g_1 = g_2$.

Equivalently, $U \cap g(U) \neq \varnothing$ only when g is the identity.

命题 1.1.24. *The action of the deck transformation group $G(\tilde{X})$ on $\tilde{X}$ is a covering space action.*

定义 1.1.22. *The **orbits** $Gy = \{g(y) \mid g \in G\}$ in Y, and Y/G is called the **orbit space** of the action.*

For example, for a normal covering space $\tilde{X} \to X$, the orbit space $\tilde{X}/G(\tilde{X})$ is just X.

命题 1.1.25. *If an action of a group G on a space Y satisfies covering space action condition, then*

(a) The quotient map $p : Y \to Y/G$, $p(y) = Gy$, is a normal covering space.

(b) G is the group of deck transformation of this covering space $Y \to Y/G$ if Y is path-connected.

(c) G is isomorphic to $\pi_1(Y/G)/p_*(\pi_1(Y))$ if Y is path-connected and locally path-connected.

推论 1.1.7. *For a covering space action of a group G on a simply-connected locally path-connected space Y, the orbit space Y/G has fundamental group isomorphic to G.*

例 1.1.6. *A covering sapce $M_{mn+1} \to M_{m+1}$. For details please refer to Hatcher P73.*

例 1.1.7. $\mathbb{R}P^n$.

The antipodal map of S^n, $x \mapsto -x$, generates an action of $\mathbb{Z}_2$ on S^n with orbit space $\mathbb{R}P^n$, real projective n-space.

Since S^n is simply-connected if $n \geq 2$, we deduce that $\pi_1(\mathbb{R}P^n) \cong \mathbb{Z}_2$ for $n \geq 2$.

1.2 Homology

1.2.1 Simplicial and Singular Homology

Simplicial homology

例 1.2.1. *$X = S^1$ with one vertex v and one edge e. Then $\Delta_0(S^1)$ and $\Delta_1(S^1)$ are both $\mathbb{Z}$ and the boundary map ∂_1 is zero since $\partial e = v - v$. Hence*

$$H_n^{\Delta}(S^1) \cong \begin{cases} \mathbb{Z}, & n = 0, 1 \\ 0, & n \geq 2 \end{cases}.$$

例 1.2.2. *$X = T$, the torus with the Δ-complex structure (a square identifying the opposite edges a, b and we draw one diagonal denoted by c.) Denote the 2-simplices U and L.*

$\partial_1 = 0$ so $H_1^\Delta(T) \cong \mathbb{Z}$. Since $\partial_2 U = a + b - c = \partial_2 L$ and $\{a, b, a + b - c\}$ is a basis for $\Delta_1(T)$, it follows that $H_1^\Delta(T) \cong \mathbb{Z} \oplus \mathbb{Z}$ with basis the homology classes $[a]$ and $[b]$.

Since there are no 3-complices, $H_2^\Delta(T)$ is equal to $\operatorname{Ker} \partial_2$, which is infinite cyclic generated by $U - L$. Thus

$$H_n^\Delta(T) \cong \begin{cases} \mathbb{Z} \oplus \mathbb{Z}, & n = 1 \\ \mathbb{Z}, & n = 0, 2 \\ 0, & n \geq 3 \end{cases}.$$

例 1.2.3. *$X = \mathbb{R}P^2$, the projective plane with the Δ-complex structure (a square identifying the opposite edges a, b with orientation revered and we draw one diagonal denoted by c.) Denote the 0-simplices w and v, the 2-simplices U and L.*

The image of ∂_1 is generated by $w - v$, so $H_0^\Delta(X) \cong \mathbb{Z}$ with either vertex as a generator.

Since $\partial_2 U = -a + b + c$ and $\partial_2 L = a - b + c$, we see that ∂_2 is injective, so $H_2^\Delta(X) = 0$.

Further, $\operatorname{Ker} \partial_1 \cong \mathbb{Z} \oplus \mathbb{Z}$ with basis $a - b$ and c and $\operatorname{Im} \partial_2$ is an index-two subgroup of $\operatorname{Ker} \partial_1$ since we can choose c and $a - b + c$ as a basis for $\operatorname{Ker} \partial_1$ and $a - b + c$ and $2c = (a - b + c) + (-a + b + c)$ as a basis of $\operatorname{Im} \partial_2$. Thus $H_1^\Delta(X) \cong \mathbb{Z}_2$.

$$H_n^\Delta(\mathbb{R}P^2) \cong \begin{cases} \mathbb{Z}_2, & n = 1 \\ \mathbb{Z}, & n = 0 \\ 0, & n \geq 2 \end{cases}.$$

例 1.2.4. *We can obtain a Δ-complex structure on S^n by taking two copies of Δ^n and identifying their boundaries via the identity map. Labeling these two n-simplices U and L, then it is obvious that $\operatorname{Ker} \partial_N$ is infinite cyclic generated by $U - L$. Thus $H_n^\Delta(S^n) \cong \mathbb{Z}$ for this Δ-complex structure on S^n.*

Singular homology

定义 1.2.1. *A singular n-simplex in a space X is by definition just a continuous map $\sigma : \Delta^n \to X$.*

注 1.2.1. *The word "singular" is used here to express the idea that σ need not be a nice embedding but can have "singularities" where its image does not look at all like a simplex.*

定义 1.2.2. *Let $C_n(X)$ be the free abelian group with basis the set of singular n-simplices in X. Elements of $C_n(X)$, called n-chains, are finite formal sums $\sum_i n_i \sigma_i$ for $n_i \in \mathbb{C}$ and $\sigma_i : \Delta^n \to X$.*

定义 1.2.3. ***Singular homology group*** $H_n(X) = \operatorname{Ker} \partial_n / \operatorname{Im} \partial_{n+1}$.

注 1.2.2. *It is evident from the definition that homeomorphic spaces have isomorphic singular homology groups H_n, in contrast with the situation for H_n^Δ.*

命题 1.2.1. *Corresponding to the decomposition of a space X into its path-components X_α there is an isomorphism of $H_n(X)$ with the direct sum $\bigoplus_\alpha H_n(X_\alpha)$.*

命题 1.2.2. *If X is nonempty and path-connected, then $H_0(X) \cong \mathbb{Z}$. Hence for any space X, $H_0(X)$ is a direct sum of $\mathbb{Z}$'s, one for each path-component of X.*

命题 1.2.3. *If X is a point, then $H_n(X) = 0$ for $n > 0$ and $H_0(X) \cong \mathbb{Z}$.*

定义 1.2.4. *Define the **reduced homology groups** $\tilde{H}_n(X)$ to be the homology groups of the augmented chain complex*

$$\cdots \to C_2(X) \xrightarrow{\partial_2} C_1(X) \xrightarrow{\partial_1} C_0(X) \xrightarrow{\varepsilon} \mathbb{Z} \to 0$$

where $\varepsilon(\sum_i n_i\sigma_i) = \sum_i n_i$.

注 1.2.3. *Formally, one can think of the extra $\mathbb{Z}$ in the augmented chain complex as generated by the unique map $[\varnothing] \to X$ where $[\varnothing]$ is the empty simplex, with no vertices. The augmentation map ε is then the usual boundary map since $\partial[v_0] = [\hat{v}_0] = [\varnothing]$.*

命题 1.2.4. *A chain map between chain complexes induces homomorphisms between the homology groups of the two complexes.*

定理 1.2.1. *If two maps $f, g : X \to Y$ are homotopic, then they induce the same homomorphism $f_* = g_* : H_n(X) \to H_n(Y)$.*

推论 1.2.1. *The maps $f_* : H_n(X) \to H_n(Y)$ induced by a homotopy equivalence $f : X \to Y$ are isomorphisms for all n.*

命题 1.2.5. *Chain-homotopic chain maps induce the same homomorphism on homology.*

The properties of induced homomorphisms we stated above hold equally well in the setting of reduced homology, with the same proofs.

1.2.2 Exact Sequences and Excision

If there was always a simple relationship between the homology groups of a space X, a subspace A, and the quotient space X/A, then this could be a very useful tool in understanding the homology groups of spaces such as CW complexes that can be built inductively from successively more complicated subspaces.

The novel feature of the actual relationship is that it involves the groups $H_n(X)$, $H_n(A)$ and $H_n(X/A)$ for all values of n simultaneously.

In practice this is not as bad as it might sound, and in addition it has the pleasant side effect of sometimes allowing high-dimensional homology groups to be computed in terms of lower-dimensional groups which may already be known, for example by induction.

定理 1.2.2. *If X is a space and A is a nonempty closed subspace that is a deformation retract of some neighborhood in X, then there is an exact sequence*

$$\cdots \to \tilde{H}_n(A) \xrightarrow{i_*} \tilde{H}_n(X) \xrightarrow{j_*} \tilde{H}_n(X/A) \xrightarrow{\partial} \tilde{H}_{n-1}(A) \xrightarrow{i_*} \tilde{H}_{n-1}(X) \to \cdots \to \tilde{H}_0(X/A) \to 0$$

where i is the inclusion $X \hookrightarrow X$ and j is the quotient map $X \to X/A$.

注 1.2.4. *The map ∂ will be constructed in the course of the proof. The idea is that an element $x \in \tilde{H}_n(X/A)$ can be represented by a chain α in X with $\partial\alpha$ a cycle in A whose homology class is $\partial x \in \tilde{H}_{n-1}(A)$.*

定义 1.2.5. *Pairs of spaces (X, A) satisfying the hypothesis of the theorem will be called **good pairs**.*

For example, if X is a CW complex and A is a nonempty subcomplex, then (X, A) is a good pair.

推论 1.2.2. *$\tilde{H}_n(S^n) \cong \mathbb{Z}$ and $\tilde{H}_i(S^n) = 0$ for $i \neq n$.*

证明. *For $n > 0$, take $(X, A) = (D^n, S^{n-1})$ so $X/A = S^n$. Then we could show that $\tilde{H}_i(S^n) \cong \tilde{H}_{i-1}(S^{n-1})$.*

推论 1.2.3. ***Brouwer theorem***

∂D^n is not a retract of D^n. Hence every map $f : D^n \to D^n$ has a fixed point.

定义 1.2.6. *Let $C_n(X, A)$ be the quotient group $C_n(X)/C_n(A)$. The boundary map ∂ is induced by the ∂ in $C_n(X)$.*

*$\partial^2 = 0$ holds then $C_*(X, A)$ becomes a complex, the homology group $H_n(X, A)$ corresponding to this complex is called **relative homology groups**.*

*Elements in $Z_n(X, A)$ are called **relative cycles**, n-chains $\alpha \in C_n(X)$ with $\partial\alpha \in C_{n-1}(A)$.*

*Elements in $B_n(X, A)$ are called **relative boundaries**, $\alpha = \partial\beta + \gamma$ for $\beta \in C_{n+1}(X)$ and $\gamma \in C_n(A)$.*

命题 1.2.6. *The commutative diagram with the columns are exact and the rows are chain complexes which we denote A, B and C. Such a diagram is called a **short exact sequence of chain complexes**.*

$$\begin{array}{ccccccccc}
 & & 0 & & 0 & & 0 & & \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
\cdots & \longrightarrow & A_{n+1} & \xrightarrow{\partial} & A_n & \xrightarrow{\partial} & A_{n-1} & \xrightarrow{\partial} & \cdots \\
 & & \downarrow{\scriptstyle i} & & \downarrow{\scriptstyle i} & & \downarrow{\scriptstyle i} & & \\
\cdots & \longrightarrow & B_{n+1} & \xrightarrow{\partial} & B_n & \xrightarrow{\partial} & B_{n-1} & \xrightarrow{\partial} & \cdots \\
 & & \downarrow{\scriptstyle j} & & \downarrow{\scriptstyle j} & & \downarrow{\scriptstyle j} & & \\
\cdots & \longrightarrow & C_{n+1} & \xrightarrow{\partial} & C_n & \xrightarrow{\partial} & C_{n-1} & \xrightarrow{\partial} & \cdots \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
 & & 0 & & 0 & & 0 & &
\end{array}$$

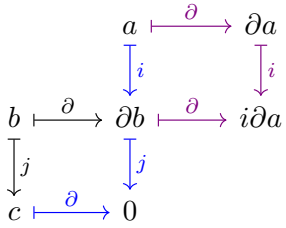
This short exact sequence of chain complexes stretches out into a long exact sequence of homology groups

$$\cdots H_n(A) \xrightarrow{i_*} H_n(B) \xrightarrow{j_*} H_n(C) \xrightarrow{\partial} H_{n-1}(A) \xrightarrow{i_*} H_{n-1}(B) \to \cdots$$

where $H_n(A)$ denotes the homology group $\operatorname{Ker}\partial/\operatorname{Im}\partial$ at A_n in the chain complex A, and $H_n(B)$ and $H_n(C)$ are defined similarly.

Now define the boundary map $\partial : H_n(C) \to H_{n-1}(A)$. Let $c \in C_n$ be a cycle, and $c = jb$ for some $b \in B_n$ as j is onto. Since $j\partial b = \partial c = 0$ implies $\partial b \in \operatorname{Ker} j = \operatorname{Im} i$, we have $\partial b = ia$ for some $a \in A_{n-1}$,

where $\partial a = 0$ since $i\partial a = \partial\partial b = 0$ and i is injective. Take $\partial[c] = [a]$.



The theorem represents the beginnings of the subject of homological algebra. The method of proof is sometimes called **diagram chasing**.

命题 1.2.7. *From the last proposition and the diagram*

$$\begin{array}{ccccccccc} 0 & \longrightarrow & C_n(A) & \xrightarrow{i} & C_n(X) & \xrightarrow{j} & C_n(X,A) & \longrightarrow & 0 \\ & & \downarrow\partial & & \downarrow\partial & & \downarrow\partial & & \\ 0 & \longrightarrow & C_{n-1}(A) & \xrightarrow{i} & C_{n-1}(X) & \xrightarrow{j} & C_{n-1}(X,A) & \longrightarrow & 0 \end{array}$$

we have the following exact sequence

$$\cdots \to H_n(A) \xrightarrow{i_*} H_n(X) \xrightarrow{j_*} H_n(X,A) \xrightarrow{\partial} H_{n-1}(A) \xrightarrow{i_*} H_{n-1}(X) \to \cdots .$$

命题 1.2.8. *There is a completely analogous long exact sequence of reduced homology groups for a pair (X, A) with $A \neq \varnothing$.*

In particular, $\tilde{H}_n(X, A)$ is the same as $H_n(X, A)$ for all n, where $A \neq \varnothing$.

例 1.2.5.

$$H_i(D^n, \partial D^n) \cong \begin{cases} \mathbb{Z}, & i = n \\ 0, & \textit{otherwise} \end{cases}.$$

例 1.2.6. *$H_n(X, x_0) \cong \tilde{H}_n(X)$ for all n.*

命题 1.2.9. *If two maps $f, g : (X, A) \to (Y, B)$ are homotopic through maps of pairs $(X, A) \to (Y, B)$, then $f_* = g_* : H_n(X, A) \to H_n(Y, B)$.*

命题 1.2.10. *An easy generalization of the long exact sequence of a pair (X, A) is the long exact sequence of a triple (X, A, B), where $B \subset A \subset X$:*

$$\cdots \to H_n(A, B) \to H_n(X, B) \to H_n(X, A) \to H_{n-1}(A, B) \to \cdots .$$

定理 1.2.3. ***Excision Theorem***

Given subspaces $Z \subset A \subset X$ such that $\overline{Z} \subset \circ A$, then the inclusion $(X - Z, A - Z) \hookrightarrow (X, A)$ induces isomorphisms $H_n(X - Z, A - Z) \to H_n(X, A)$ for all n.

Equivalently, for subspaces $A, B \subset X$ whose interiors cover X, the inclusion $(B, A \cap B) \hookrightarrow (X, A)$ induces isomorphism $H_n(B, A \cap B) \to H_n(X, A)$ for all n.

注 1.2.5. *In a metric space "smallness" can be defined in terms of diameters, but for general spaces it will be defined in terms of covers.*

定义 1.2.7. *Let $C_n^{\mathscr{U}}(X)$ be the subgroup of $C_n(X)$ consisting of chains $\sum_i n_i\sigma_i$ such that each σ_i has image contained in some set in the cover $\mathscr{U}$.*

命题 1.2.11. *The inclusion $\iota : C_n^{\mathscr{U}}(X) \hookrightarrow C_n(X)$ is a chain homotopy equivalence, that is, there is a chain map $\rho : C_n(X) \to C_n^{\mathscr{U}}(X)$ such that $\iota\rho$ and $\rho\iota$ are chain homotopic to the identity.*

Hence ι induces isomorphisms $H_n^{\mathscr{U}}(X) \cong H_n(X)$ for all n.

命题 1.2.12. *For good pairs (X, A), the quotient map $q : (X, A) \to (X/A, A/A)$ induces isomorphisms $q_* : H_n(X, A) \to H_n(X/A, A/A) \cong \tilde{H}_n(X/A)$ for all n.*

注 1.2.6. *This proposition shows that relative homology can be expressed as reduced absolute homology in the case of good pairs (X, A), but in fact there is a way of doing this for arbitrary pairs:*

$$\tilde{H}_n(X \cup CA) \cong H_n(X \cup CA, CA) \cong H_n(X \cup CA - \{p\}, CA - \{p\}) \cong H_n(X, A),$$

where $p \in CA$ is the tip of the cone.

例 1.2.7. *The identity map $i_n : \Delta^n \to \Delta^n$, viewed as a singular n-simplex, is a cycle generating $H_n(\Delta^n, \partial\Delta^n)$. We claim there are isomorphisms*

$$H_n(D^n, \partial D^n) \cong H_n(\Delta^n, \partial\Delta^n) \xrightarrow{\cong} H_{n-1}(\partial\Delta^n, \Lambda) \xleftarrow{\cong} H_{n-1}(\Delta^{n-1}, \partial\Delta^{n-1}),$$

where $\Lambda \subset \Delta^n$ is the union of all but one of the $(n-1)$-dimensional faces of Δ^n.

例 1.2.8. *To find a cycle generating $\tilde{H}_n(S^n)$ let us regard S^n as two n-simplices Δ_1^n and Δ_2^n with their boundaries identified in the obvious way, preserving the ordering of vertices.*

The difference $\Delta_1^n - \Delta_2^n$, viewed as a singular n-chain, is then a cycle, and we claim it represents a generator of $\tilde{H}_n(S^n)$. To see this, consider the isomorphisms

$$\tilde{H}_n(S^n) \xrightarrow{\cong} H_n(S^n, \Delta_2^n) \xleftarrow{\cong} H_n(\Delta_1^n, \partial\Delta_1^n).$$

推论 1.2.4. *If the CW complex X is the union of subcomplexes A and B, then the inclusion $(B, A\cap B) \hookrightarrow (X, A)$ induces isomorphisms $H_n(B, A \cap B) \to H_n(X, A)$ for all n.*

推论 1.2.5. *For a wedge sum $\bigvee_\alpha X_\alpha$, the inclusions $i_\alpha : X_\alpha \hookrightarrow \bigvee_\alpha X_\alpha$ induce an isomorphism $\bigoplus_\alpha i_{\alpha*} : \bigoplus_\alpha \tilde{H}_n(X_\alpha) \to \tilde{H}_n(\bigvee_\alpha X_\alpha)$, provided that the wedge sum is formed at basepoints $x_\alpha \in X_\alpha$ such that the pairs (X_α, x_α) are good.*

证明. *Since reduced homology is the same as homology relative to a basepoint, this follows from the proposition by taking $(X, A) = (\bigsqcup_\alpha X_\alpha, \bigsqcup_\alpha \{x_\alpha\})$.*

定理 1.2.4. *If nonempty open sets $U \subset \mathbb{R}^m$ and $V \subset \mathbb{R}^n$ are homeomorphic, then $m = n$.*

证明. *For $x \in U$ we have $H_k(U, U - \{x\}) \cong H_k(\mathbb{R}^m, \mathbb{R}^m - \{x\}) \cong \tilde{H}_{k-1}(\mathbb{R}^m - \{x\}) \cong \tilde{H}_{k-1}(S^{m-1})$ by excision, the long exact sequence for $(\mathbb{R}^m, \mathbb{R}^m - \{x\})$, and deformation retraction respectively.*

定义 1.2.8. *The **local homology groups** of a space X at a point $x \in X$ are defined to be the groups $H_n(X, X-\{x\})$.*

For any open neighborhood U of x, excision gives isomorphisms $H_n(X, X-\{x\}) \cong H_n(U, U-\{x\})$ assuming points are closed in X, and thus the groups $H_n(X, X-\{x\})$ depend only on the local topology of X near x.

A homeomorphism $f : X \to Y$ must induce isomorphisms $H_n(X, X-\{x\}) \cong H_n(Y, Y-\{f(x)\})$ for all x and n, so the local homology groups can be used to tell when spaces are not locally homeomorphic at certain points.

命题 1.2.13. ***Naturality***

For six chain complexes A, B, C, A', B', C' with

$$0 \to A_n \xrightarrow{i} B_n \xrightarrow{j} C_n \to 0$$

$$0 \to A'_n \xrightarrow{i'} B'_n \xrightarrow{j'} C'_n \to 0$$

for any n. Moreover, there are three chain maps $\alpha : A \to A'$, $\beta : B \to B'$, $\gamma : C \to C'$. Then the induced diagram

$$\begin{array}{ccccccccc}
\cdots & \longrightarrow & H_n(A) & \xrightarrow{i_*} & H_n(B) & \xrightarrow{j} & H_n(C) & \xrightarrow{\partial} & H_{n-1}(A) & \longrightarrow & \cdots \\
 & & \downarrow{\scriptstyle \alpha_*} & & \downarrow{\scriptstyle \beta_*} & & \downarrow{\scriptstyle \gamma_*} & & \downarrow{\scriptstyle \alpha_*} & & \\
\cdots & \longrightarrow & H_n(A') & \xrightarrow{i'_*} & H_n(B') & \xrightarrow{j'_*} & H_n(C') & \xrightarrow{\partial} & H_{n-1}(A') & \longrightarrow & \cdots
\end{array}$$

is commutative.

命题 1.2.14. *For $f : (X, A) \to (Y, B)$, the diagram*

$$\begin{array}{ccccccccc}
\cdots & \longrightarrow & H_n(A) & \xrightarrow{i_*} & H_n(X) & \xrightarrow{j} & H_n(X, A) & \xrightarrow{\partial} & H_{n-1}(A) & \longrightarrow & \cdots \\
 & & \downarrow{\scriptstyle f_*} & & \downarrow{\scriptstyle f_*} & & \downarrow{\scriptstyle f_*} & & \downarrow{\scriptstyle f_*} & & \\
\cdots & \longrightarrow & H_n(B) & \xrightarrow{i_*} & H_n(Y) & \xrightarrow{j_*} & H_n(Y, B) & \xrightarrow{\partial} & H_{n-1}(B) & \longrightarrow & \cdots
\end{array}$$

is commutative.

命题 1.2.15. *Recall Theorem 1.2.2, if X, Y are spaces and A, B are nonempty closed subspaces that are deformation retracts of some neighborhoods in X, Y respectively, then the following diagram commutes*

$$\begin{array}{ccccccccc}
\cdots & \longrightarrow & \tilde{H}_n(A) & \xrightarrow{i_*} & \tilde{H}_n(X) & \xrightarrow{q_*} & \tilde{H}_n(X/A) & \xrightarrow{\partial} & \tilde{H}_{n-1}(A) & \longrightarrow & \cdots \\
 & & \downarrow{\scriptstyle f_*} & & \downarrow{\scriptstyle f_*} & & \downarrow{\scriptstyle \overline{f}_*} & & \downarrow{\scriptstyle f_*} & & \\
\cdots & \longrightarrow & \tilde{H}_n(B) & \xrightarrow{i_*} & \tilde{H}_n(Y) & \xrightarrow{q'_*} & \tilde{H}_n(Y/B) & \xrightarrow{\partial} & \tilde{H}_{n-1}(B) & \longrightarrow & \cdots
\end{array}$$

1.2.3 The Equivalence of Simplicial and Singular Homology

定理 1.2.5. *The homomorphisms $H_n^{\Delta}(X, A) \to H_n(X, A)$ are isomorphisms for all n and all Δ-complex pairs (X, A).*

引理 1.2.1. ***The Five-Lemma***

In a commutative diagram of abelian groups as shown, if the two rows are exact and α, β, δ and ε are isomorphisms, then γ is an isomorphism also.

$$\begin{array}{ccccccccc} A & \xrightarrow{i} & B & \xrightarrow{j_*} & C & \xrightarrow{k} & D & \xrightarrow{l} & E \\ \downarrow{\scriptstyle\alpha} & & \downarrow{\scriptstyle\beta} & & \downarrow{\scriptstyle\gamma} & & \downarrow{\scriptstyle\delta} & & \downarrow{\scriptstyle\varepsilon} \\ A & \xrightarrow{i'} & B' & \xrightarrow{j'} & C' & \xrightarrow{k'} & D' & \xrightarrow{l'} & E' \end{array}$$

推论 1.2.6. *We can deduce from this theorem that $H_n(X)$ is **finitely generated** whenever X is a Δ-complex with finitely many n-simplices, since in this case the simplicial chain group $\Delta_n(X)$ is finitely generated, hence also its subgroup of cycles and therefore also the latter group's quotient $H_n^{\Delta}(X)$.*

定义 1.2.9. *If we write $H_n(X)$ as the direct sum of cyclic groups, then the number of $\mathbb{Z}$ summands is known traditionally as the nth **Betti number** of X, and integers specifying the orders of the finite cyclic summands are called **torsion coefficients**.*

1.2.4 Degree

定义 1.2.10. *For a map $f : S^n \to S^n$ with $n > 0$, the induced map $f_* : H_n(S^n) \to H_n(S^n)$ is a homomorphism from an infinite cyclic group to itself and so must be of the form $f_*\alpha = d\alpha$ for some integer d depending only on f. This integer is called the **degree** of f, with the notation* $\deg f$.

命题 1.2.16. *Here are some basic properties of degree:*

(a) $\deg \mathrm{id} = 1$.

(b) $\deg f = 0$ *if f is not surjective. (Hint: $f : S^n \to S^n - \{x\} \hookrightarrow S^n$.)*

(c) If $f \sim g$ then $\deg f = \deg g$ *since $f_* = g_*$. The converse statement, that $f \sim g$ if* $\deg f = \deg g$*, is a fundamental theorem of Hopf from around 1925.*

(d) $\deg fg = \deg f \cdot \deg g$ *since $(fg)_* = f_*g_*$. As a consequence,* $\deg f = \pm 1$ *if f is a homotopy equivalence.*

(e) $\deg f = -1$ *if f is a reflection of S^n.*

(f) $\deg - \mathrm{id} = (-1)^{n+1}$ *where* $-\mathrm{id}$ *is the antipodal map which is the composition of $n+1$ reflections.*

(g) If $f : S^n \to S^n$ has no fixed points, then $\deg f = (-1)^{n+1}$ *since it is not difficult to prove $f \sim -\mathrm{id}$.*

定理 1.2.6. *S^n has a continuous field of nonzero tangent vectors iff n is odd.*

证明. *Suppose that the tangent vectors $v(x)$, orthogonal to x, are all of length 1. The homotopy $f_t : S^n \to S^n, x \mapsto (\cos t)x + (\sin t)v(x)$ between* id *and* $-\mathrm{id}$ *implies $1 = (-1)^{n+1}$.*

命题 1.2.17. *$\mathbb{Z}_2$ is the only nontrivial group that can act freely on S^n if n is even.*

证明. *Since homeomorphisms have degree ± 1, an action of a group G on S^n determines a degree function $d : G \to \{\pm 1\}$. This is a homomorphism since* $\deg fg = \deg f \cdot \deg g$*. If the action is free, d sends each nontrivial element of G to $(-1)^{n+1}$ by property (g) above. Thus, when n is even, d has trivial kernel, so $G \subset \mathbb{Z}_2$.*

1.2.5 Cellular Homology

引理 1.2.2. *If X is a CW complex, then:*

(a) $H_k(X^n, X^{n-1})$ *is zero for* $k \neq n$ *and is free abelian for* $k = n$*, with a basis in one-to-one correspondence with the* n*-cells of* X*.*

(b) $H_k(X^n) = 0$ *for* $k > n$*. In particular, if* X *is finite-dimensional then* $H_k(X) = 0$ *for* $k > \dim X$*.*

(c) The map $H_k(X^n) \to H_k(X)$ *induced by the inclusion* $X^n \hookrightarrow X$ *is an isomorphism for* $k < n$ *and surjective for* $k = n$*.*

证明. *(a)* $H_k(X^n, X^{n-1}) = \tilde{H}_k(X^n/X^{n-1}) = \tilde{H}_k(\bigvee S^n) = \bigoplus \tilde{H}_k(S^n)$.

(b,c) Consider

$$H_{k+1}(X^n, X^{n-1}) \to H_k(X^{n-1}) \to H_k(X^n) \to H_k(X^n, X^{n-1})$$

If $k \neq n$*, the last term is* 0 *so* $H_k(X^{n-1}) \to H_k(X^n)$ *is surjective; if* $k \neq n, n-1$*, both the first term and last term are* 0 *so* $H_k(X^{n-1}) \cong H_k(X^n)$*.*
Therefore $H_k(X^n) \cong H_k(X^{n-1}) \cong \cdots \cong H_k(X^0) = 0$ *for* $k > n$*,* $H_k(X^n) \cong H_k(X^{n+1}) \cong \cdots \cong H_k(X)$ *for* $k < n$*, and* $H_k(X^n) \to H_k(X^{n+1}) \cong H_k(X)$ *is surjective for* $k = n$*.*

命题 1.2.18. *Let X be a CW compelx. Using 1.2.2, portions of the long exact sequence for the pairs* (X^{n+1}, X^n)*,* (X^n, X^{n-1}) *and* (X^{n-1}, X^{n-2}) *fit into a diagram*

$$\begin{array}{ccccccc} & & 0 & & & & \\ & & \searrow & & & & \\ & & & H_n(X^n) & & & \\ & & \overset{\partial_{n+1}}{\nearrow} & & & & \\ \cdots & \longrightarrow & H_{n+1}(X^{n+1}, X^n) & \xrightarrow{d_{n+1}} & & & \end{array}$$

where d_{n+1} *and* d_n *are defined as the compositions* $j_n\partial_{n+1}$ *and* $j_{n-1}\partial_n$*, which are just 'relativizations' of the boundary maps* ∂_{n+1} *and* ∂_n*.*

定义 1.2.11. *Note* $d_{n+1} \circ d_n = 0$*, the horizontal row in the diagram is a chain complex, called the* ***cellular chain complex*** *of* X*. Since* $H_n(X^n, X^{n-1})$ *is free with basis in one-to-one correspondence with the* n*-cells of* X*, one can think of elements of* $H_n(X^n, X^{n-1})$ *as linear combinations of* n*-cells of* X*.*

The homology groups of this cellular chain complex are called the ***cellular homology groups*** *of* X*, temporarily denoted by* $H_n^{CW}(X)$*.*

定理 1.2.7. $H_n^{CW}(X) \cong H_n(X)$.

命题 1.2.19. *Here are a few immediate applications:*

(i) $H_n(X) = 0$ *if* X *is a CW complex with no* n*-cells.*

(ii) More generally, if X *is a CW complex with* k n*-cells, then* $H_n(X)$ *is generated by at most* k *elements.*

(iii) If X is a CW complex having no two of its cells in adjacent dimensions, then $H_n(X)$ is free abelian with basis in one-to-one correspondence with the n-cells of X.

证明. *(ii)* $H_n(X) \cong \operatorname{Ker} d_n / \operatorname{Im} d_{n+1}$ *and* $\operatorname{Ker} d_n \subset H_n(X^n, X^{n-1}) = \mathbb{Z}^k$.

(iii) The maps d_n are automatically zero.

例 1.2.9. $H_i(\mathbb{C}P^n) \cong \begin{cases} \mathbb{Z}, & \text{for } i = 0, 2, 4, \cdots, 2n \\ 0, & \text{otherwise} \end{cases}$

例 1.2.10. *$S^n \times S^n$ with $n > 1$, using the product structure consisting of a 0-cell, two n-cells, and a $2n$-cell.*

命题 1.2.20. ***Cellular Boundary Formula***

$$d_n(e_\alpha^n) = \sum_\beta d_{\alpha\beta} e_\beta^{n-1}$$

where $d_{\alpha\beta}$ is the degree of the map $S_\alpha^{n-1} \to X^{n-1} \to S_\beta^{n-1}$ that is the composition of the attaching map of e_α^n with the quotient map collapsing $X^{n-1} - e_\beta^{n-1}$ to a point.

注 1.2.7. *In case X is connected and has only one 0-cell, then d_1 must be 0, otherwise $H_0(X)$ would not be $\mathbb{Z}$.*

例 1.2.11. *M_g: one 0-cell, $2g$ 1-cells, and one 2-cell attached by the product of commutators $[a_1, b_1] \cdots [a_g, b_g]$. The associated cellular chain complex is*

$$0 \to \mathbb{Z} \xrightarrow{d_2} \mathbb{Z}^{2g} \xrightarrow{d_1} \mathbb{Z} \to 0.$$

Note that both d_1 and d_2 are 0, then one can get

$$H_n(M_g) = \begin{cases} \mathbb{Z}, & n = 0, 2 \\ \mathbb{Z}^{2g}, & n = 1 \end{cases}.$$

例 1.2.12. *N_g: one 0-cell, g 1-cells, and one 2-cell attached by the word $a_1^2 \cdots a_g^2$. The associated cellular chain complex is*

$$0 \to \mathbb{Z} \xrightarrow{d_2} \mathbb{Z}^g \xrightarrow{d_1} \mathbb{Z} \to 0.$$

Note that $d_1 = 0$ and $d_2(1) = (2, 2, \cdots, 2)$, then one can get

$$H_n(N_g) = \begin{cases} \mathbb{Z}, & n = 0 \\ \mathbb{Z}^{g-1} \oplus \mathbb{Z}_2, & n = 1 \\ 0, & n = 2 \end{cases}.$$

注 1.2.8. *These two examples illustrate the general fact that the orientablility of a close connected manifold M of dimension n is detected by $H_n(M)$, which is $\mathbb{Z}$ if M is oriented and 0 otherwise. This will be shown in the next chapter.*

定义 1.2.12. *If X satisfies $\tilde{H}_i(X) = 0$ for all i, we call X an acyclic space.*

例 1.2.13. *The examples of $T^3 = S^1 \times S^1 \times S^1$ and $K \times S^1$ are also amazing. One can refer to Hatcher P142.*

定义 1.2.13. ***Moore Spaces***

Given an abelian group G and an integer $n \geq 1$, we can construct a CW complex X such that $H_n(X) \cong G$ and $\tilde{H}_i(X) = 0$ for $i \neq n$, commonly written $M(G, n)$ to indicate the dependence of G and n.

例 1.2.14. ***Real Projective Space*** $\mathbb{R}P^n$

$d_k = \deg \mathrm{id} + \deg(-\mathrm{id}_{S^{k-1}}) = 1 + (-1)^k$*, thus the cellular chain complex for* $\mathbb{R}P^n$ *is*

$$0 \to \mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \cdots \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \to 0, \quad \text{if } n \text{ is even}$$

$$0 \to \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{2} \cdots \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \to 0, \quad \text{if } n \text{ is odd}$$

From this it follows that

$$H_k(\mathbb{R}P^n) = \begin{cases} \mathbb{Z}, & \text{for } k = 0 \text{ and for } k = n \text{ odd} \\ \mathbb{Z}_2, & \text{for } k \text{ odd}, 0 < k < n \\ 0, & \text{otherwise} \end{cases}.$$

1.2.6 Euler Characteristic

定义 1.2.14. *For a finite CW complex X, the* ***Euler characteristic*** *$\chi(X)$ is defined to be the alternating sum $\sum_n (-1)^n c_n$ where c_n is the number of n-cells of X.*

注 1.2.9. *The following result shows that $\chi(X)$ can be defined purely in terms of homology, and hence depends only on the homotopy type of X. In particular, $\chi(X)$ is independent of the choice of CW structure on X.*

定理 1.2.8. $\chi(X) = \sum_n (-1)^n \operatorname{rank} H_n(X)$.

定义 1.2.15. *Here the* ***rank*** *of a finitely generated abelian group is the number of $\mathbb{Z}$ summands when the group is expressed as a direct sum of cyclic groups.*

If $0 \to A \to B \to C \to 0$ is a short exact sequence of finitely generated abelian groups, then $\operatorname{rank} B = \operatorname{rank} A + \operatorname{rank} C$.

例 1.2.15. $\chi(M_g) = 2 - 2g$ *and* $\chi(N_g) = 2 - g$.

1.2.7 Split Exact Sequences

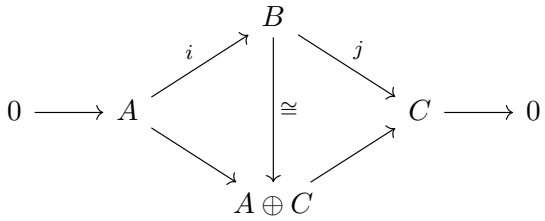
引理 1.2.3. ***Splitting Lemma***

For a short exact sequence $0 \to A \xrightarrow{i} B \xrightarrow{j} C \to 0$ of abelian groups the following statements are equivalent:

(a) There is a homomorphism $p : B \to A$ such that $p \circ i = \mathrm{id} : A \to A$.

(b) There is a homomorphism $s : C \to B$ such that $j \circ s = \mathrm{id} : C \to C$.

(c) There is a an isomorphism $B \cong A \oplus C$ making a commutative diagram, where the maps in the lower row are the obvious ones, $a \mapsto (a, 0)$ and $(a, c) \mapsto c$.



*If these conditions are satisfied, the exact sequence is said to be **split**.*

命题 1.2.21. *A retraction $r : X \to A$ gives a splitting $H_n(X) \cong H_n(A) \oplus H_n(X, A)$.*

例 1.2.16. *The last proposition can be used to show the nonexistence of such a retraction in some cases.*

(1) For the Brouwer fixed point theorem, a retraction $D^n \to S^{n-1}$ would give

$$H_{n-1}(D^n) \cong H_{n-1}(S^{n-1}) \oplus H_{n-1}(D^n, S^{n-1}).$$

(2) The mapping cylinder M_f of a degree m map $f : S^n \to S^n$ with $m > 1$. If M_f retracted onto the $S^n \subset M_f$ corresponding to the domain of f, we would have a split short exact sequence

$$\begin{array}{ccccccccc} 0 & \longrightarrow & H_n(S^n) & \longrightarrow & H_n(M_f) & \longrightarrow & H_n(M_f, S^n) & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle =} & & \downarrow{\scriptstyle =} & & \downarrow{\scriptstyle =} & & \\ 0 & \longrightarrow & \mathbb{Z} & \xrightarrow{m} & \mathbb{Z} & \longrightarrow & \mathbb{Z}_m & \longrightarrow & 0 \end{array}$$

But this sequence does not split since $\mathbb{Z}$ is not isomorphic to $\mathbb{Z} \oplus \mathbb{Z}_m$ if $m > 1$, so the retraction cannot exist.

The simplest case of degree 2 map $S^1 \to S^1$, $z \mapsto z^2$, this says that the Möbius band does not retract onto its boundary circle.

1.2.8 Mayer-Vietoris Sequences

命题 1.2.22. *For a pair of subspaces $A, B \subset X$ such that X is the union of the interiors of A and B, this exact sequence has the form*

$$\cdots \to H_n(A \cap B) \xrightarrow{\Phi} H_n(A) \oplus H_n(B) \xrightarrow{\Psi} H_n(X) \xrightarrow{\partial} H_{n-1}(A \cap B) \to \cdots \to H_0(X) \to 0$$

where $\Phi(x) = (x, -x), \Psi(x, y) = x + y$.

命题 1.2.23. *There is also a formally identical Mayer-Vietoris sequence for reduced homology groups, obtained by augmenting the previous short exact sequence of chain complexes in the obvious way:*

$$\begin{array}{ccccccccc} 0 & \longrightarrow & C_0(A \cap B) & \xrightarrow{\varphi} & C_0(A) \oplus C_0(B) & \xrightarrow{\psi} & C_0(A + B) & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \varepsilon} & & \downarrow{\scriptstyle \varepsilon \oplus \varepsilon} & & \downarrow{\scriptstyle \varepsilon} & & \\ 0 & \longrightarrow & \mathbb{Z} & \xrightarrow{\varepsilon} & \mathbb{Z} \oplus \mathbb{Z} & \xrightarrow{\psi} & \mathbb{Z} & \longrightarrow & 0 \end{array}$$

注 1.2.10. *Mayer-Vietoris sequence can be viewed as analogs of the van Kampen theorem since if $A \cap B$ is path-connected, the H_1 terms of the reduced Mayer-Vietoris sequence yield an isomorphism $H_1(X) \cong (H_1(A) \oplus H_1(B)) \,/\, \operatorname{Im} \Phi$. This is exactly the abelianized statement of the van Kampen theorem, and H_1 is the abelianization of π_1 for path-connected spaces.*

注 1.2.11. *There are also Mayer-Vietoris sequences for decompositions $X = A \cup B$ such that A and B are deformation retracts of neighborhoods U and V with $U \cap V$ deformation retracting onto $A \cap B$.*

例 1.2.17. *Take $X = S^n$ with A and B the northern and southern hemispheres, so that $A \cap B = S^{n-1}$. Then in the reduced Mayer-Vietoris sequence the terms $\tilde{H}_i(A) \oplus \tilde{H}_i(B)$ are zero, so we obtain isomorphisms $\tilde{H}_i(S^n) \cong \tilde{H}_{i-1}(S^{n-1})$.*

例 1.2.18. *We can decompose the Klein bottle K as the union of two Möbius bands A and B glued together by a homeomorphism between their boundary circles.*

Then A, B and $A \cap B$ are homotopy equivalent to circles, so the interesting part of the reduced Mayer-Vietoris sequence for the decomposition $K = A \cup B$ is the segment

$$0 \to H_2(K) \to H_1(A \cap B) \xrightarrow{\Phi} H_1(A) \oplus H_1(B) \to H_1(K) \to 0.$$

The map Φ is $\mathbb{Z} \to \mathbb{Z} \oplus \mathbb{Z}$, $1 \mapsto (2, -2)$, since the boundary circle of a Möbius band wraps twice around the core circle. Since Φ is injective we obtain $H_2(K) = 0$. Furthermore, we have $H_1(K) \cong \mathbb{Z} \oplus \mathbb{Z}_2$ since $\mathbb{Z} \oplus \mathbb{Z} = \langle (1, 0), (1, -1) \rangle$.

1.2.9 Homology with coefficients

定义 1.2.16. *The generalization consists of using chains of the form $\sum_i n_i \sigma_i$ where each σ_i is a singular n-simplex in X as before, but now the coefficients n_i are taken to be in a fixed abelian group G rather than $\mathbb{Z}$. Such n-chains form an abelian group $C_n(X; G)$ and there is the expected relative version $C_n(X, A; G) = C_n(X; G)/C_n(A; G)$.*

∂ defined similarly satisfies $\partial^2 = 0$ and hence $C_n(X; G)$ and $C_n(X, A; G)$ form chain complexes. The resulting homology groups $H_n(X; G)$ and $H_n(X, A; G)$ are called ***homology groups with coefficients in*** *G.*

Reduced groups $\tilde{H}_n(X; G)$ are defined via the augmented chain complex $\cdots \to C_0(X; G) \xrightarrow{\varepsilon} G \to 0$ with ε again defined by summing coefficients.

命题 1.2.24. *All the theory we developed for $\mathbb{Z}$ coefficients carries over directly to general coefficient group G.*

引理 1.2.4. *If $f : S^k \to S^k$ has degree m, then $f_* : H_k(S^k; G) \to H_k(S^k; G)$ is multiplication by m.*

Then one could define $d_{\alpha\beta}$ in the cellular homology.

证明.

$$\begin{array}{ccccccc}
\mathbb{Z} & \xrightarrow{\cong} & \tilde{H}_k(S^k; \mathbb{Z}) & \xrightarrow{f_*} & \tilde{H}_k(S^k; \mathbb{Z}) & \xrightarrow{\cong} & \mathbb{Z} \\
\downarrow{\scriptstyle \varphi} & & \downarrow{\scriptstyle \varphi_*} & & \downarrow{\scriptstyle \varphi_*} & & \downarrow{\scriptstyle \varphi} \\
G & \xrightarrow{\cong} & \tilde{H}_k(S^k; G) & \xrightarrow{f_*} & \tilde{H}_k(S^k; G) & \xrightarrow{\cong} & G
\end{array}$$

注 1.2.12. *The case $G = \mathbb{Z}_2$ is particularly simple since one is just considering sums of singular simplices with coefficients 0 or 1, so by discarding terms with coefficient 0 one can think of chains as just finite 'unions' of singular simplices.*

The boundary formulas also simplify since one on longer has to worry about signs. Since signs are an algebraic representation of orientation considerations, one can also ignore orientations. This means that homology with $\mathbb{Z}_2$ coefficients is often the most natural tool in the absence of orientability.

例 1.2.19. *It is instructive to see what happens to the homology of $\mathbb{R}P^n$ when the coefficient group G is chosen to be a field F. The cellular chain complex is*

$$\cdots \xrightarrow{0} F \xrightarrow{2} F \xrightarrow{0} F \xrightarrow{2} F \xrightarrow{0} F \to 0.$$

Hence if F has characteristic 2, for example $F = \mathbb{Z}_2$, then $H_k(\mathbb{R}P^n; F) \cong F$ for $0 \le k \le n$, a more uniform answer than with $\mathbb{Z}$ coefficients.

On the other hand, if F has characteristic different from 2, then the boundary maps $F \xrightarrow{2} F$ are isomorphisms, hence $H_k(\mathbb{R}P^n; F)$ is F for $k = 0$ and for $k = n$ odd, and is zero otherwise.

注 1.2.13. *In spite of the fact that homology with $\mathbb{Z}$ coefficients determines homology with other coefficient groups, there are many situations where homology with a suitably chosen coefficient group can provide more information than homology with $\mathbb{Z}$ coefficients.*

注 1.2.14. *In the next chapter we will see that there is a general algebraic formula expressing homology with arbitrary coefficients in terms of homology with $\mathbb{Z}$ coefficients.*

1.2.10 Homology and Fundamental Group

定理 1.2.9. *By regarding loops as singular 1-cycles, we obtain a homomorphism $h : \pi_1(X, x_0) \to H_1(X)$. If X is path-connected, then h is surjective and has kernel the commutator subgroup of $\pi_1(X)$, so h induces an isomorphism from the abelianization of $\pi_1(X)$ onto $H_1(X)$.*

1.2.11 Classical Applications

命题 1.2.25. *For an embedding $h : D^k \to S^n$, $\tilde{H}_i(S^n - h(D^k)) = 0$ for all i.*

命题 1.2.26. *For an embedding $h : S^k \to S^n$ with $k < n$, $\tilde{H}_i(S^n - h(S^k))$ is $\mathbb{Z}$ for $i = n - k - 1$ and 0 otherwise.*

证明. *Induction by k. Write S^k as the union of hemispheres D^k_+ and D^k_- intersecting in S^{k-1}. The Mayer-Vietoris sequence for $A = S^n - h(D^k_+)$ and $B = S^n - h(D^k_-)$, both of which have trivial reduced homology by the last proposition, then gives isomorphisms $\tilde{H}_i(S^n - h(S^k)) \cong \tilde{H}_{i+1}(S^n - h(S^{k-1}))$.*

推论 1.2.7. ***Jordan curve theorem***

A subspace of S^n homeomorphic to S^{n-1} separates it into two components, and these components have the same homology groups as a point.

推论 1.2.8. *An embedding $h : S^n \to S^n$ must be surjective.*

证明. *As in the last proof, consider the end of the Mayer-Vietoris sequence*

$$\tilde{H}_0(A) \oplus \tilde{H}_0(B) \to \tilde{H}_0(S^n - h(S^{n-1})) \to \tilde{H}_{-1}(S^n - h(S^n)).$$

For $\tilde{H}_0(S^n - h(S^{n-1})) \cong \mathbb{Z}$, we have $S^n - h(S^n)$ must be empty.

推论 1.2.9. *In particular, S^n cannot be embedded in $\mathbb{R}^n$ since this would yield a nonsurjective embedding in S^n.*

A consequence is that there is no embedding $\mathbb{R}^m \hookrightarrow \mathbb{R}^n$ for $m > n$ since this would restrict to an embedding of $S^n \subset \mathbb{R}^m$ into $\mathbb{R}^n$.

More generally there is no continuous injection $\mathbb{R}^m \to \mathbb{R}^n$ for $m > n$ since this too would give an embedding $S^n \hookrightarrow \mathbb{R}^n$.

定理 1.2.10. ***Invariance of Domain***

If U is an open set in $\mathbb{R}^n$ and $h : U \to \mathbb{R}^n$ is an embedding, or more generally just a continuous injection, then the image $h(U)$ is an open set in $\mathbb{R}^n$.

推论 1.2.10. *If M is a compact n-manifold and N is a connected n-manifold, then an embedding $h : M \to N$ must be surjective, hence a homeomorphism.*

命题 1.2.27. *An odd map $f : S^n \to S^n$, satisfying $f(-x) = -f(x)$ for all x, must have odd degree.*

命题 1.2.28. *Given a two-sheeted covering space $p : \tilde{X} \to X$,*

$$\cdots \to H_n(X;\mathbb{Z}_2) \xrightarrow{\tau_*} H_n(\tilde{X};\mathbb{Z}_2) \xrightarrow{p_*} H_n(X;\mathbb{Z}_2) \to H_{n-1}(X;\mathbb{Z}_2) \to \cdots$$

This is the long exact sequence of homology groups associated to a short exact sequence of chain complexes consisting of short exact sequences of chain groups

$$0 \to C_n(X;\mathbb{Z}_2) \xrightarrow{\tau} C_n(\tilde{X};\mathbb{Z}_2) \xrightarrow{p_\#} C_n(X;\mathbb{Z}_2) \to 0$$

The map $p_\#$ is surjective since singular simplices $\sigma : \Delta^n \to X$ always lift to $\tilde{X}$, as Δ^n is simply-connected. Each σ has in fact precisely two lifts $\tilde{\sigma}_1$ and $\tilde{\sigma}_2$. Because we are using $\mathbb{Z}_2$ coefficients, the kernel of $p_\#$ is generated by the sums $\tilde{\sigma}_1 + \tilde{\sigma}_2$. So if we define τ to send each σ to the sum of its two lifts, then the image of τ is the kernel of $p_\#$. Obviously τ is injective, so we have the short exact sequence indicated.

Since τ and $p_\#$ commute with boundary maps, we have a short exact sequence of chain complexes, yielding the long exact sequence of homology groups.

*The map τ_* is a special case of more general **transfer homomorphisms** considered in the next chapter, so we will refer to the long exact sequence involving the maps τ_* as the **transfer sequence**.*

推论 1.2.11. ***Borsuk-Ulam Theorem***

For every map $g : S^n \to \mathbb{R}^n$ there exists a point $x \in S^n$ with $g(x) = g(-x)$.

1.3 Cohomology

定义 1.3.1. *Consider a chain complex C of free abelian groups*

$$\cdots \to C_{n+1} \xrightarrow{\partial} C_n \xrightarrow{\partial} C_{n-1} \to \cdots$$

*The dual of C_n is the **cochain group** $C_n^* = \operatorname{Hom}(C_n, G)$, and the dual of ∂ is the **coboundary map** $\delta : C_{n-1}^* \to C_n^*$.*

$$\cdots \leftarrow C_{n+1}^* \xleftarrow{\delta} C_n^* \xleftarrow{\delta} C_{n-1}^* \leftarrow \cdots$$

*It follows from $\partial\partial = 0$ that $\delta\delta = 0$, and we can define the **cohomology group** $H^n(C;G) = \operatorname{Ker}\delta / \operatorname{Im}\delta$.*

定义 1.3.2. *A **free resolution** of an abelian group H is an exact sequence of free groups*

$$\cdots \to F_1 \xrightarrow{f_1} F_0 \xrightarrow{f_0} H \to 0$$

Appply $\mathrm{Hom}(-, G)$ *and we get a chain complex*

$$\cdots \leftarrow F_1^* \xleftarrow{f_1^*} F_0^* \xleftarrow{f_0^*} H^* \leftarrow 0$$

We may lose exactness, and use the temporary notation $H^n(F; G) = \mathrm{Ker}\, f_{n+1}^* / \mathrm{Im}\, f_n^*$.

引理 1.3.1. *For free resolutions F and F' of H, there are canonical isomorphisms $H^n(F; G) \cong H^n(F'; G)$ for all n.*

Every H has a free resolution of the form $0 \to F_1 \to F_0 \to H \to 0$. Suppose H is generated by $\{h_i\}$, and let F_0 be the free abelian group with basis $\{h_i\}$ and $f_0 : F_0 \to H, h_i \mapsto h_i$. Let $F_1 = \mathrm{Ker}\, f_0$, which is a subgroup of a free abelian group thus also free, and f_1 be the inclusion $F_1 \hookrightarrow F_0$. For this resolution we get $H^n(F; G) = 0$ for $n > 1$, which is hence also true for all free resolutions. Thus only $H^1(F; G)$ is nontrivial, written as $\mathrm{Ext}(H, G)$.

定理 1.3.1. ***Universal Coefficient Theorem for Cohomology*** *There is a split exact sequence*

$$0 \to \mathrm{Ext}(H_{n-1}(C), G) \to H^n(C; G) \xrightarrow{h} \mathrm{Hom}(H_n(C), G) \to 0$$

命题 1.3.1.

- $\mathrm{Ext}(H \oplus H', G) = \mathrm{Ext}(H, G) \oplus \mathrm{Ext}(H', G)$.
- $\mathrm{Ext}(H, G) = 0$ *if H is free.*
- $\mathrm{Ext}(\mathbb{Z}_n, G) = G/nG$.

推论 1.3.1. *If H_n and H_{n-1} of a chain complex C of free abelian groups are finitely generated, with torsion subgroups T_n and T_{n-1}, then $H^n(C; \mathbb{Z}) \cong (H_n/T_n) \oplus T_{n-1}$.*

引理 1.3.2. *If a sequence of abelian groups $A \xrightarrow{i} B \xrightarrow{j} C \to 0$ is exact, then so is $A \otimes 1 \xrightarrow{i\otimes 1} B \otimes 1 \xrightarrow{j\otimes 1} C \otimes 1 \to 0$.*

引理 1.3.3. *For free resolutions F and F' of H, there are canonical isomorphisms $H_n(F \otimes G) \cong H_n(F' \otimes G)$ for all n.*

The group $H_n(F \otimes G)$, which depends only on H and G, is denoted $\mathrm{Tor}_n(H, G)$. Since a free resolution $0 \to F_1 \to F_0 \to H \to 0$ always exists, only $\mathrm{Tor}_1(H, G)$ is nontrivial, and is written as $\mathrm{Tor}(H, G)$.

定理 1.3.2. ***Universal Coefficient Theorem for Homology*** *There is a natural exact sequence that splits though not naturally*

$$0 \to H_n(C) \otimes G \to H_n(C; G) \to \mathrm{Tor}(H_{n-1}(C), G) \to 0$$

命题 1.3.2.

- $\mathrm{Tor}(A, B) = \mathrm{Tor}(B, A)$.
- $\mathrm{Tor}(\bigoplus_i A_i, B) = \bigoplus_i \mathrm{Tor}(A_i, B)$.
- $\mathrm{Tor}(A, B) = 0$ *if A or B is free, or more generally torsionfree.*

- $\mathrm{Tor}(A,B)=\mathrm{Tor}(T(A),B)$ *where* $T(A)$ *is the torsion subgroup of* A.
- $\mathrm{Tor}(\mathbb{Z}_n,A)=\mathrm{Ker}(A\xrightarrow{n}A)$.

推论 1.3.2. *For finitely generated* A *and* B, $\mathrm{Tor}(A,B)\cong T(A)\otimes T(B)$.

推论 1.3.3. *(a)* $H_n(X;\mathbb{Q})\cong H_n(X)\otimes\mathbb{Q}$, *so when* $\mathrm{rank}\,H_n(X)<\infty$, *it equals* $\dim_{\mathbb{Q}}H_n(X;\mathbb{Q})$.
(b) If $H_n(X)$ *and* $H_{n-1}(X)$ *are finitely gererated, then for* p *prime,* $H_n(X;\mathbb{Z}_p)$ *consists a* $\mathbb{Z}_p$ *summand for: each* $\mathbb{Z}$ *summand of* $H_n(X)$, *each* $\mathbb{Z}_{p^k}$ *summand of* $H_n(X)$, *each* $\mathbb{Z}_{p^k}$ *summand of* $H_{n-1}(X)$.

注 1.3.1. *Here is a similar result given in Exercise 2.2.43:*
If X *is a CW complex with finitely many cells in each dimension, then* $H_n(X;G)$ *is the direct sum of: a* G *for each* $\mathbb{Z}$ *summand of* $H_n(X)$, *a* G/mG *for each* $\mathbb{Z}_m$ *summand of* $H_n(X)$, *a* $\mathrm{Ker}(G\xrightarrow{m}G)$ *for each* $\mathbb{Z}_m$ *summand of* $H_{n-1}(X)$.

定义 1.3.3. *A* ***local orientation*** *of* M *at* x *is a choice of generator* μ_x *of* $H_n(M|x)\cong\mathbb{Z}$, *and an* ***orientation*** *of* M *is a function* $x\mapsto\mu_x$ *assigning to each* $x\in M$ *a local orientation* $\mu_x\in H_n(M|x)$ *satisfying: each* $x\in M$ *has a neighborhood* B, *such that all* μ_y *at* $y\in B$ *are the images of one generator* $\mu_B\in H_n(M|B)\cong H_n(\mathbb{R}^n|B)$ *under the natural maps* $H_n(M|B)\to H_n(M|y)$. M *is* ***orientable*** *if an orientation exists.*

1.4　试卷: 代数拓扑 (H)

原网页

2025–2026 学年期末（研, 12.26）

注. 考试时间 2 小时 15 分钟.

一. (20 分)

1. (10 分) 设 m,n 都是正整数, 请构造一个道路连通的 3 维 CW 复形 X, 使得 $H_1(X)=\mathbb{Z}/m, H_2(X)=\underbrace{\mathbb{Z}\oplus\cdots\oplus\mathbb{Z}}_{n}, H_3(X)=\mathbb{Z}$.
2. 对上述构造的 X, 求解上同调群 $H^p(X;\mathbb{Z})$ 及 $H^p(X;\mathbb{Z}/6)$.

二. (26 分)

1. (8 分) 证明 $\mathbb{R}P^n$ 是可三角剖分的 CW 复形.
2. (10 分) 设 $h:|K|\to|K|$ 是连续映射, K 是有限复形. 设 $n>0$ 是偶数.
   (a) (5 分) 若 $|K|=\mathbb{R}P^n$, 证明 h 必有不动点.
   (b) (5 分) 若 $|K|=S^n$, 证明 $h\circ h$ 必有不动点.
3. (8 分) 证明 Lefschetz 数也可以用对应的上同调表达. 即, 记上同调群 $H^p(K)$ 的挠子群为 $T^p(K)$, 则

$$\begin{aligned}\Lambda(h)&=\sum_p(-1)^p\,\mathrm{Trace}(h_*:H_p(K)/T_p(K)\to H_p(K)/T_p(K))\\&=\sum_p(-1)^p\,\mathrm{Trace}(h^*:H^p(K)/T^p(K)\to H^p(K)/T^p(K)).\end{aligned}$$

三. (16 分) 设 $h \in C(|K|,|L|)$, K 是有限复形.

1. (8 分) 举例说明, 可以不存在任何单纯映射 $\phi : K \to L$ 逼近 h.
2. (8 分) 证明: 存在正整数 m, 使得存在 h 的单纯逼近 $\phi : \mathrm{sd}^m K \to L$.

四. (20 分)

1. (10 分) 设 K 是单纯复形.

   (a) (5 分) 证明 $|K|$ 是 CW 复形.

   (b) (5 分) 假设已知 $H_*(\mathcal{C}(K)) \cong H_*(\mathcal{S}(|K|))$, 证明 $H_*(\mathcal{C}(K)) \cong H_*(\mathcal{D}(|K|))$.
2. (10 分) 设 X 是拓扑空间, 有一族滤子 $X_0 \subset X_1 \subset \cdots \subset X_p \subset \cdots$, 满足 $X = \bigcup_p X_p$. 又设 $i \neq p$ 时 $H_i(X_p, X_{p-1}) = 0$.

   (c) (5 分) 对正整数 $q, r > p$, 证明 $H_p(X_q) \cong H_p(X_r, X_{p-2})$.（注：这里需要 X_0 是零调的，否则取非零调的 $X_i = X, \forall i$ 是一个反例）

   (d) (5 分) 若存在 $i \neq p$ 满足 $H_i(X_p, X_{p-1}) \neq 0$, 上面同构是否仍然成立? 此时对于 $H_p(X)$ 的计算, 说说你的想法.

五. (18 分)

1. (8 分) 对链复形 $\mathcal{C}$, 考虑上下同调群的 Kronecker 积

$$\begin{aligned} H^p(\mathcal{C}; G) \times H_p(\mathcal{C}) &\longrightarrow G, \\ (\alpha^p, \beta_p) &\longmapsto \langle \alpha^p, \beta_p \rangle, \end{aligned}$$

   其中 G 是 Abel 群. 给出 $\langle \alpha^p, \beta_p \rangle$ 的定义, 并证明上下同调群的 Kronecker 积是良定义的.
2. (8 分) 对自由链复形 $\mathcal{C}$, 上下同调群的 Kronecker 积提供给我们一个同态

$$\begin{aligned} \kappa : H^q(\mathcal{C}; G) &\longrightarrow \mathrm{Hom}(H_q(\mathcal{C}), G), \\ \alpha &\longmapsto \langle \alpha, \cdot \rangle. \end{aligned}$$

   证明存在同态 $\lambda : \mathrm{Hom}(H_q(\mathcal{C}), G) \longrightarrow H^q(\mathcal{C}; G)$, 使得 $\kappa \circ \lambda = \mathrm{id}_{\mathrm{Hom}(H_q(\mathcal{C}), G)}$.
3. (2 分) 举例说明 κ 不总是同构.

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1. (1) 给出一个道路连通的 4 维 CW 复形 X，使得 $H_1(X) = \mathbb{Z}/6\mathbb{Z}$，$H_2(X) \cong H_3(X) = \mathbb{Z}^4$，$H_4(X) = \mathbb{Z}$;

   (2) 对上题的 X，计算上同调群 $H^p(X; \mathbb{Z})$ 与 $H^p(X; \mathbb{Z}/3\mathbb{Z})$。
2. (1) 设 K 是有限单纯复形，且 $h \in C(|K|,|K|)$。请写出 h 的 Lefschetz 数 $\Lambda(h)$ 的定义;

   (2) 设 $f \in C(S^n, S^n)$，且 $\overline{f}$ 是被 f 经由 $p : S^n \longrightarrow \mathbb{R}P^n (= S^n/x \sim -x)$ 所诱导的 $\mathbb{R}P^n$ 到自身的连续映射，即下图可换：
$$\begin{array}{ccc} S^n & \xrightarrow{f} & S^n \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle p} \\ \mathbb{R}P^n & \xrightarrow{\overline{f}} & \mathbb{R}P^n. \end{array}$$
   求证:

   (i) 若 n 为偶数，则 $\overline{f}$ 必有不动点;

   (ii) 若 n 为奇数，且 $\deg f \neq (-1)^{n+1}$, 则 $\overline{f}$ 必有不动点; 而若 $\deg f = (-1)^{n+1}$, $\overline{f}$ 是否有不动点? 请给出理由。
3. (1) 设 K, L 是单纯复形，$h : |K| \to |L|$ 是连续映射，ϕ, ψ 均为 h 的单纯逼近。证明 $\phi_\# \simeq \psi_\#$;

   (2) 设 $f \simeq g : |K| \to |L|$ 是连续映射，且 K 为有限单纯复形。证明 $f_* = g_*$。

4. 设 X 是 CW 复形，记 X^p 为 X 的 p 维骨架。

(1) 对正整数 $l > q > r \geqslant 0$，证明我们有如下长正合列：

$$\cdots \longrightarrow H_p(X^q, X^r) \longrightarrow H_p(X^l, X^r) \longrightarrow H_p(X^l, X^q) \longrightarrow H_{p-1}(X^q, X^r) \longrightarrow \cdots$$

(2) 设 $l, q > p$ 为正整数，证明 $H_p(X^l, X^{p-2}) \cong H_p(X^q, X^{p-2})$。

5. (1) 设 X 为单点拓扑空间，证明 $\widetilde{H}_p(X) = 0$ 且对任意交换群 G, $\widetilde{H}^p(X; G) = 0$，

(2) 设 $f : X \to Y$ 为拓扑空间之间的连续映射，证明：

$$\langle f^*(\alpha), \beta \rangle = \langle \alpha, f_*(\beta) \rangle,$$

其中 $\alpha \in H^p(Y; G), \beta \in H_p(X)$。

6. 设 K 为有限单纯复形, 证明:

$$\chi(K) = \textstyle\sum_p (-1)^p f_p(K),$$

其中 $\chi(K)$ 为 K 的 Euler 数, $f_p(K)$ 为 K 的 p 阶整系数上同调群自由部分的秩.

2024–2025 学年期末

复旦大学数学科学学院

2024∼2025 学年第一学期本研合开课程期末考试试卷

■ A 卷　□ B 卷　□ C 卷

考试时间： 2024年12月19日

本科生课程名称： 代数拓扑（H）　**课程代码：** **MATH130112h.01**

研究生课程名称： 代数拓扑基础　**课程代码：** **MATH620001**

课程名称： 数学科学学院　**考试形式：** 闭卷

姓　名： _____　**学　号：** _____　**专　业：** _____

提示：请同学们秉持诚实守信宗旨，谨守考试纪律，摒弃考试作弊。学生如有违反学校考试纪律的行为，学校将按《复旦大学学生纪律处分条例》规定予以严肃处理。

题　目	1	2	3	4	5	6	总分
得　分							

1. (10 分, 5+5) K_1, K_2 是单纯复形 K 的子复形。
   (1) 证明：$K_1 \cap K_2, K_1 \cup K_2$ 仍然是单纯复形。
   (2) 证明：$\chi(K_1 \cup K_2) + \chi(K_1 \cap K_2) = \chi(K_1) + \chi(K_2)$，这里 $\chi(\cdot)$ 是 Euler 数。
2. (20 分, 10+10)
   (1) 证明：$\mathbb{R}P^n$ 每个维数 p 有一个 p 维胞腔，并且 $\mathbb{R}P^n$ 的 p 维骨架是 $\mathbb{R}P^p$。
   (2) $f : \mathbb{R}P^n \to \mathbb{R}P^n$ 是连续映射，f 有不动点吗？证明你的结论。
3. (12 分, 6+6) $f : S^n \to S^n$ 是连续映射。
   (1) 写出映射度 $\deg f$ 的定义。
   (2) $g : S^n \to S^n$ 是连续映射，且 $f(x) \neq g(x), \forall x \in S^n$，给出 $\deg f, \deg g$ 之间满足的方程，证明你的结论。
4. (12 分, 6+6) $T : \Delta_p \to X$ 是奇异单形。
   (1) 给出 ∂ 的定义。
   (2) 证明 $\partial^2 = 0$。
5. (10 分) 设 $q, r > p$, X^p 是 CW 复形 X 的 p 维骨架，证明：$H_p(X^q) \cong H_p(X^r)$。

下面开始本科生做第 1 题，研究生 2 选 1

6. (6-1, 12 分) 设 $0 \to \mathcal{C} \to \mathcal{D} \to \mathcal{E} \to 0$ 是自由链复形，$\varphi : \mathcal{C} \to \mathcal{D}$，证明: 若 φ_* 是同构，则 φ^* 亦是同构.
7. (6-2, 12 分) 设 $h \in C(|K|, |L|)$，其中 K 是一个有限单纯复形。求证：存在正整数 m 以及一个单纯映射 $\phi : \mathrm{sd}^m K \longrightarrow L$，使得对 $\mathrm{sd}^m K$ 的任意顶点 v 满足

$$h(\mathrm{Star}(v, \mathrm{sd}^m K)) \subset \mathrm{Star}(\phi(v), L).$$

   此时，称 ϕ 为 h 的从 $\mathrm{sd}^m K$ 到 L 的一个单纯逼近。
8. (7-1, 12 分, 6+6)
   (1) 给出环面 T^2, $X = S^1 \vee S^1 \vee S^2$ 的上同调环结构。
   (2) 设 $f : X \longrightarrow T^2$ 是 X 到 2 维环面的一个连续映射。证明：$f^* : H^2(T^2) \longrightarrow H^2(X)$ 是平凡同态。
9. (7-2, 12 分, 4+4+4)
   (1) 构造只有 $0, 1, 2$ 维胞腔各 1 个的胞腔复形 X，使得 $H_1(X) = \mathbb{Z}/n\mathbb{Z}$。
   (2) X 是 2 维流形吗，请证明。
   (3) 证明 $f : X \to T^2$ 诱导的 $f_* : H_p(X) \to H_p(T^2)$ 在 $p = 1, 2$ 是平凡的。
10. (8-1, 12 分) $H^p(X)$ 是 X 的 $\mathbb{Z}$ 系数上同调群，证明：$H_p(X)$ 自由部分的秩 $= H^p(X)$ 自由部分的秩。
11. (8-2, 12 分) 假设有限单纯复形上的某自同态的不动点都是孤立的，证明：该自同态的不动点数量是它的 Lefschetz 数。

2023–2024 学年期末

warning| 本次考试内容覆盖到前 16 周。以下只回忆了本科生的试题，只上到前 12 周的研究生考试内容与此有部分试题不同。

复旦大学数学科学学院

2023∼2024 学年第一学期本研合开课程期末考试试卷

■ A 卷　□ B 卷　□ C 卷

考试时间： 2023年12月18日

本科生课程名称： 代数拓扑（H）　**课程代码：** **MATH130112h.01**

研究生课程名称： 代数拓扑基础　**课程代码：** **MATH620001**

课程名称： 数学科学学院　**考试形式：** 闭卷

姓　名： _____　**学　号：** _____　**专　业：** _____

提示：请同学们秉持诚实守信宗旨，谨守考试纪律，摒弃考试作弊。学生如有违反学校考试纪律的行为，学校将按《复旦大学学生纪律处分条例》规定予以严肃处理。

题　目	1	2	3	4	5	6	总分
得　分							

1. 20 分，4+8+8
   (1) 请给出一种 $T^2\#\mathbb{R}P^2$ 的胞腔剖分中，所用到每个维数的胞腔数目都是最小可能值的剖分.
   (2) 请给出 (1) 中胞腔复形的同调群.
   (3) 构造只有 $0,1,2$ 维胞腔各 1 个的胞腔复形 X，使得 $H_1(X)=\mathbb{Z}/m\mathbb{Z}$.
2. 18 分，10+8
   (1) 如果两个保持增广的链映射 ϕ,ψ 都被同一个零调承载子 Φ 所承载，证明 ϕ 和 ψ 同伦.
   (2) 如果两个保持增广的上链映射 ϕ,ψ 都被同一个零调承载子 Φ 所承载，证明 ϕ 和 ψ 同伦.
3. 18 分，4+14
   (1) 给出 S^n 到 S^n 的映射的映射度的定义.
   (2) 构造映射度是任意给定整数的映射.
4. 16 分，8+8
   (1) 证明 CW 复形中紧集至多与有限个开胞腔相交.

(2) 如果 U 是 X 中的开集，A 是紧集，且为 U 的形变收缩核. 证明 $H_*(X,A)\cong \tilde{H}_*(X/A)$.

5. 8 分

一个单纯 n 单形和它的面构成一个复形 X. X 到自身的单纯映射如果不动点是孤立的，证明它只有一个不动点.

6. 20 分，4+8+8

(1) 给出奇异上链中杯积的定义.

(2) 证明上述的杯积诱导上同调群之间的杯积.

(3) 证明任何从 $S^1\vee S^1\vee S^1\vee S^2$ 到 T^2 的映射诱导的 2 维上同调群的映射是零映射.

2022–2023 学年期末

复旦大学数学科学学院

2022∼2023 学年第一学期本研合开课程期末考试试卷

■ A 卷　□ B 卷　□ C 卷

考试时间： 2022年12月23日

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题　目	1	2	3	4	5	6	总分
得　分							

一、 (共计 30 分)

1. (6 分) 设 X 是一个 CW 复形。论述它的胞腔链复形 $\mathcal{D}(X)=\{(D_p(X),\partial)\}$ 是如何定义的?

2. (24 分) 设 $Y=\mathbb{R}P^2\sharp\mathbb{R}P^2\sharp\mathbb{R}P^2$ 是三个实射影平面的连通和。

(i) (8 分) 请画图给出 Y 的一个可三角刨分的 CW 复形结构，并用这个结构给出其对应的胞腔链复形 $\mathcal{D}(Y)$（包括链群及边缘算子的确定）；

(ii) (8 分) 计算 Y 的整系数同调群 $H_p(Y)$;

(iii) (8 分) 计算 Y 的系数群为整数模 6 的上同调群 $H^p(Y;\mathbb{Z}/6\mathbb{Z})$。

二、 (18 分)

1. (10 分) 设 $h \in C(|K|,|L|)$，其中 K 是一个有限单纯复形。求证：存在正整数 m 以及一个单纯映射 $\phi : \mathrm{sd}^m K \longrightarrow L$，使得对 $\mathrm{sd}^m K$ 的任意顶点 v 满足

$$h(\mathrm{Star}(v, \mathrm{sd}^m K)) \subset \mathrm{Star}(\phi(v), L).$$

此时，称 ϕ 为 h 的从 $\mathrm{sd}^m K$ 到 L 的一个单纯逼近。

2. (8 分) 设 $K = \{1, 6, (1, 6)\}$ 和 $L = \{1, 3, 6, (1, 3), (3, 6)\}$ 是 $\mathbb{R}$ 中的两个一维单纯复形，满足 $|K| = |L|$。求证：恒同映射 $id_{|K|} : |K| \longrightarrow |L|$ 不存在从 K 到 L 的单纯逼近。进一步, 求最小的正整数 m, 使得恒同映射 $id_{|K|}$ 存在单纯逼近 $\phi : \mathrm{sd}^m K \longrightarrow L$。

三、 (12 分)

1. (6 分) 设 M^n 是 n 维拓扑流形，计算其局部同调群

$$H_p(M^n, M^n - x) = ?$$

其中 $x \in M^n$。

2. (6 分) 求证：$\mathbb{R}^m$ 同胚于 $\mathbb{R}^n$ 当且仅当 $m = n$。

四、 (共计 20 分) 设 X 是一个拓扑空间, 且 $S^p(X;R) = \mathrm{Hom}(S_p(X), R)$ 是其 p 维奇异上链群。

1. (4 分) 问：杯积

$$\cup : S^p(X;R) \times S^q(X;R) \longrightarrow S^{p+q}(X;R)$$

是如何定义的？

2.(8 分) 求证：$\delta(c^p \cup c^q) = (\delta c^p) \cup c^q + (-1)^p c^p \cup (\delta c^q)$，其中 $c^p \in S^p(X;R)$ 且 $c^q \in S^q(X;R)$。

3. (8 分) 设 $f : S^2 \longrightarrow T^2$ 是 2 维球面到 2 维环面的一个连续映射。证明：$f^* : H^2(T^2) \longrightarrow H^2(S^2)$ 是平凡同态。

五、 (共计 20 分)

1. (4 分) 设 K 是有限单纯复形, 且 $h \in C(|K|,|K|)$。请写出 h 的 Lefschetz 数 $\Lambda(h)$ 的定义。

2. (16 分) 设 $f \in C(S^n, S^n)$，且 $\overline{f}$ 是被 f 经由 $p : S^n \longrightarrow \mathbb{R}P^n (= S^n/x \sim -x)$ 所诱导的 $\mathbb{R}P^n$ 到自身的连续映射，即下图可换：

$$\begin{array}{ccc} S^n & \xrightarrow{f} & S^n \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle p} \\ \mathbb{R}P^n & \xrightarrow{\overline{f}} & \mathbb{R}P^n. \end{array}$$

求证:

(i) (8 分) 若 n 为奇数，则 $\Lambda(f) = \Lambda(\overline{f})$。

(ii) (8 分) 若 n 为偶数，则 $\overline{f}$ 必有不动点。

六、 (选做题，可不做)

对任何一个大于 1 的正整数 m，构造一个仅由一个 0 维开胞腔，一个 1 维开胞腔以及一个 2 维开胞腔做成的 2 维 CW 复形 $X = e^0 \cup e^1 \cup e^2$，使得 $H^2(X) \cong \mathbb{Z}/m\mathbb{Z}$。

1.5　试卷: 拓扑学 (II)

原网页

2025 秋试卷

1. (25 分) (1) 对所有可定向与不可定向闭曲面 X 进行剖分，并计算各阶同调群 $H_i(X)$。
   (2) 计算 $\mathbb{Z}_4$ 系数的同调群 $H_i(X;\mathbb{Z}_4)$。
   (2) 计算 $\mathbb{Q}$ 系数的同调群 $H_i(X;\mathbb{Q})$。

2. (15 分) (1) 计算 n 维单位球面 $\mathbb{S}^n$ 的同调群 $H_i(\mathbb{S}^n)$。
   (2) 证明 Brouwer 不动点定理。
   (3) 设 $A=(a_{ij})$ 是非异的 n 阶方阵，所有的 $a_{ij}\geq 0$，证明：A 有正的特征值。
3. (15 分) 设 $f:\mathbb{S}^{2n}\to\mathbb{S}^{2n}$ 是连续映射，证明：f 要么有不动点，要么将某点映为对径点。
4. (15 分) 设 X 是道路连通的拓扑空间，它的双角锥 $S(X)$ 是通过将 $X\times[-1,1]$ 中 $X\times\{1\}$ 与 $X\times\{-1\}$ 分别捏为一点得到的商空间。证明：对 $p\geq 2$，$H_p(S(X))\cong H_{p-1}(X)$。（提示：使用 Mayor-Vietoris 序列）
5. (16 分) (1) 叙述链映射 $f:C=\{C_q,\partial_q\}\to D=\{D_q,\partial_q\}$ 的定义。
   (2) 证明：对于链映射 $f:C=\{C_q,\partial_q\}\to D=\{D_q,\partial_q\}$，它诱导了同调群之间的同态。
   (3) 对两个链映射 $f,g:C=\{C_q,\partial_q\}\to D=\{D_q,\partial_q\}$，叙述它们是链同伦的定义。
   (4) 证明：若两个链映射 $f,g:C=\{C_q,\partial_q\}\to D=\{D_q,\partial_q\}$ 是链同伦的，则它们诱导了相同的同调群之间的同态。
6. (14 分) 设 $\mathbb{S}^1_a$ 和 $\mathbb{S}^1_b$ 是单位圆周 $\mathbb{S}^1$ 的两个拷贝，$Y=\mathbb{S}^1_a\vee\mathbb{S}^1_b$ 是通过将它们在一点 A 处粘连得到的空间，分别用 a 和 b 代表基点为 A 的 $\mathbb{S}^1_a$ 和 $\mathbb{S}^1_b$ 的闭道路，则 Y 的基本群可由 a 和 b 生成。$f:\partial\mathbb{D}^2\cong\mathbb{S}^1\to Y$ 是连续映射，$f(\mathbb{S}^1)$ 在 Y 中的道路类是 $a*a*b*a*a*\bar{b}$，其中 $\bar{b}$ 是 b 的逆道路。定义 $W=Y\sqcup\mathbb{D}^2/\sim$，其中 $x\sim y$ 当且仅当 $x=f(y)$ 且 $y\in\partial\mathbb{D}^2$。求 W 的同调群。

2024 秋试卷

1. 证明 Euler–Poincaré 公式。
2. 证明：f 有单纯逼近等价于 f 具有星形性质。
3. 对可定向与不可定向闭曲面 X 进行剖分。再计算 $H^*(X;\mathbb{Z}_{13})$。
4. n 维单纯复形 K，L 是 K 的锥形。对任何 Abel 群 G，求 $H^*(L;G)$。
5. 设 f 是 S^1 到 S^3 的光滑嵌入，$f(S^1)$ 存在邻域 $N\simeq S^1\times D^2$。令 $W=S^3-\operatorname{Int}N$，有 $\partial N=\partial W\simeq S^1\times S^1$。已知 S^3 存在单纯剖分，相对应有 W,N 上的单纯剖分。求 W 的同调群。
6. (i) 给出上积、卡积的定义。
   (ii) 证明上积诱导了上同调群的乘法。
   (iii) 证明卡积诱导 $H^q(X)\times H_{p+q}(X)\to H_p(X)$。

2023 秋试卷

1. 证明：任意抽象单纯复形都存在几何实现。
2. 设 K 是有限 n 维复形，底空间 $|K|$ 为 n 维流形。证明：
$$H_n(K)=\begin{cases}\mathbb{Z}, & K\text{ 可定向}\\ 0, & K\text{ 不可定向}\end{cases}.$$
3. 设对径映射 $r\colon S^n\to S^n$，$r(x_1,\cdots,x_{n+1})=(-x_1,\cdots,-x_{n+1})$。求 $\deg r$。
4. 对可定向与不可定向闭曲面 X 进行剖分。再计算 $H^*(X;\mathbb{Z}_{11})$。
5. 计算 $H_p(\mathbb{R}P^n),H_p(\mathbb{C}P^n)$。
6. 设整数 $0<k<n$，证明：任意连续的 $f\colon S^k\to S^n$ 零伦。

2022 秋试卷

1. 求同调群 $H_*(S^1\times S^2)$。
2. 证明：任意 n 维抽象单纯复形都存在 $\mathbb{R}^{2n+1}$ 中的几何实现。
3. 已知 f 有单纯逼近等价于 f 具有星形性质，试证单纯逼近定理。
4. 构造空间 X，使得 $H_0(X)=H_2(X)=H_3(X)=H_6(X)=\mathbb{Z}$，且 $H_1(X)=\mathbb{Z}\oplus\mathbb{Z}_2\oplus\mathbb{Z}_2$，其余阶的同调群平凡。
5. 设 D 是复平面上的单位圆盘，$p\geq 2$ 是一个固定的整数。取等价关系 $\sim$ 为：对于边缘上的两点 x,y，若 $e^{\frac{2\pi i}{p}}x=y$，则 $x\sim y$。求商空间的同调群。
6. (i) 给出上积、卡积的定义，说明其如何诱导链复形之间的映射。
   (ii) 说明上积诱导了上同调群的乘法。
   (iii) 陈述 Poincaré 对偶定理，并证明: 对奇数维可定向闭流形 M，Euler 数 $\chi(M)=0$。

2021 秋试卷一

1. 求亏格为 n 的可定向闭曲面的 Euler 数，亏格为 n 的不可定向闭曲面的 Euler 数。
2. 求同调群 $H_*(\mathbb{R}P^n;\mathbb{Z}_2)$。
3. 对紧致无边 n 维流形 M 进行有限剖分。证明：
$$H_n(M)=\begin{cases}\mathbb{Z}, & M\text{ 可定向}\\ 0, & M\text{ 不可定向}\end{cases}.$$
4. 设 $A=(a_{ij})$ 是非异的 n 阶方阵，所有的 $a_{ij}\geq 0$，证明：A 有正的特征值。
5. 同 22 年题 6。
6. 设 f 是 S^1 到 S^3 的光滑嵌入，像 $f(S^1)$ 存在邻域 $N\simeq S^1\times D^2$。令 $W=S^3-\operatorname{Int}N$，有 $\partial N=\partial W\simeq S^1\times S^1$。求 W 的同调群。

2021 秋试卷二

1. 设 $X=S^2\vee S^2\vee S^2$，求同调群 $H_*(X)$。
2. 设 X 是 4 维单形的闭包复形的 2 维骨架，求同调群 $H_*(X;\mathbb{Z}_3)$。

3. 证明：f 有单纯逼近等价于 f 具有星形性质。
4. 证明：n 维伪流形可定向等价于其 n 维同调群是 $\mathbb{Z}$。
5. 用局部同调群证明：$X = S^1 \vee S^2 \vee S^1$ 与环面 T^2 不同胚。
6. 同 22 年题 6。

1.6　讲义: 代数拓扑 (H)

原网页

- 单纯同调论
- 同调群的同伦不变性
- 奇异同调论
- CW 复形的同调
- 上同调理论
- 附录

这是《代数拓扑基础》（本科生课《代数拓扑 (H)》）课程的讲义. 这一课程从单纯同调论开始，讨论复形的同调和上同调相关的内容. 谢谢从前有一只呜喵的授权.

在课程讲授的安排中, 有一些内容需要解释:

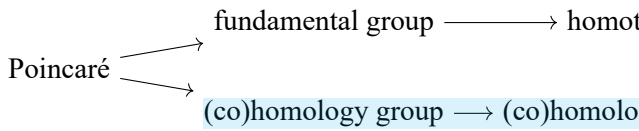
(1) 单纯同调论. 本课程相较于一些经典的代数拓扑教材, 花费了相当多的时间在单纯复形的同调以及单纯逼近定理和代数重分定理等细节上. 这一方面是因为使用的教材由 Munkres 所著, 在其 *Elements of Algebraic Topology* 一书的前两章中花费了大量篇幅事无巨细地讨论了单纯同调中的细节, 另一方面是因为希望在单纯同调的讨论中建立起的思想和技巧可以类似地推广到奇异复形和胞腔复形上, 以降低理解难度.

(2) 正合序列和一些相关结论. 囿于开课时间, 抽象代数的课程尚未进行到正合序列的内容, 为了内容的完整性, 在第 3 章结束对相对同调的讨论中, 需要补充正合序列的相关结论以供后续使用.

(3) 上同调理论. 限于课程时间, 我们没有进一步讨论上同调环的内容. 这篇讲义的本意是忠实地记录课程中授课教师的板书, 因此在结束万有系数定理的证明之后没有再做进一步讨论.

注：在 2022,2023 学年的实际授课（吕志）中，上同调部分教到了上同调环，并且期末考核也考到了.

课程的主线按照下图进行, 我们主要讨论单纯同调及其导出的同调群, 奇异同调理论和



CW 复形的同调. 在课程的最后, 我们讨论一些上同调的理论.

1.7　CW 复形的同调

原网页

1.8　CW 复形及其结构

与 $\mathbb{D}^n$ 同胚的几何体称作一个 n–**胞腔** (n–cell), 与 $\mathbb{B}^n$ 同胚的几何体称作一个 n–**开胞腔**. 习惯上记 n–开胞腔为 e_n, n–胞腔为 $\bar{e}_n$. 特别地, 约定 0–胞腔是单点 pt, 并用 e_0 记它.

例 1.8.1. 球面 $\mathbb{S}^n$ 可以看作是一个 0–胞腔 e_0 和一个 n–开胞腔 e_n 构成的, 这因 $\mathbb{S}^n$ 去掉北极点 N 后

$$\mathbb{S}^n \setminus N \cong \mathbb{R}^n \cong \mathbb{B}^n \cong e_n.$$

那么多个球面的一点并从胞腔的角度来看是一件非常平凡的事情. k 个不同维数的球面一点并 $\mathbb{S}^{n_1} \vee \mathbb{S}^{n_2} \vee \cdots \vee \mathbb{S}^{n_k}$ 无非就是 $e_0, e_{n_1}, e_{n_2}, \ldots, e_{n_k}$ 构成的.

例 1.8.2 (环面和 Klein 瓶)**.** 环面 $\mathbb{T}^2$ 和 *Klein* 瓶 S 都可以看作一个 0–胞腔 e_0, 两个 1–开胞腔 e_1^1 和 e_1^2, 以及一个 2–开胞腔 e_2 构成的复形*:*

定义 1.8.1 (CW 复形)**.** 称 X 是一个 ***CW* 复形**, 若 X 是一个拓扑空间, 一族不交的开胞腔 $\{e_\alpha\}_{\alpha\in\mathcal{A}}$ 使得 $X = \bigcup\limits_{\alpha\in\mathcal{A}} e_\alpha$, 满足

1. X 是 *Hausdorff* 空间*;*
2. 对任何 m–开胞腔 e_α^m, 存在其**示性映射** $f_\alpha : \mathbb{D}^m \to X$ 使得 $f_\alpha : \mathbb{B}^m \to e_\alpha^m$ 是一个同胚, 且 $f_\alpha(\mathrm{Bd}\,\mathbb{B}^m)$ 落在有限多个维数低于 m 的开胞腔的并中*; (*注意这里不需要边缘与开胞腔的并相等, 只需要落在其中即可*.)*
3. 对 $A \subset X$, 若对任意 e_α 都有 $A \cap \bar{e}_\alpha$ 是 $\bar{e}_\alpha$ 中的闭子集, 则 A 是 X 中的闭子集*.(*称作是 X 上的弱拓扑*).*

特别地, 若 $|\mathcal{A}| < \infty$, 称 X 是一个**有限 *CW* 复形**, 其**维数**可由 $\dim X = \max\limits_{\alpha} \dim e_\alpha$ 定义. 此时 X 是 T_4 空间.

CW 复形中, C 意为 closure–finite, 闭有限性, 而 W 意为 weak topology, 即 3 中定义的拓扑.

命题 1.8.1. 设 X 是一个 *CW* 复形, e_α 是其一个胞腔, f_α 是 e_α 的示性映射. 有 $f_\alpha(\mathbb{D}^m) = \bar{e}_\alpha$ 且 $f_\alpha(\mathrm{Bd}\,\mathbb{B}^m) = \partial\bar{e}_\alpha$.

证明. 由 f_α 的连续性可知

$$f_\alpha(\mathbb{D}^m) = f_\alpha(\bar{\mathbb{B}}^m) \subset \overline{f_\alpha(\mathbb{B}^m)} = \bar{e}_\alpha.$$

反之, 因 $\mathbb{D}^m$ 是紧的, 于是 $f_\alpha(\mathbb{D}^m)$ 是 X 中的紧集. 因 X 是 T_3 的, 有 $f_\alpha(\mathbb{D}^m)$ 是闭集, 因此 $\bar{e}_\alpha \subset f_\alpha(\mathbb{D}^m)$. 这就说明了 $f_\alpha(\mathbb{D}^m) = \bar{e}_\alpha$.

而

$$\emptyset = f_\alpha(\mathbb{B}^m) \cap f_\alpha(\mathrm{Bd}\,\mathbb{B}^m) = e_\alpha \cap f_\alpha(\mathrm{Bd}\,\mathbb{B}^m),$$

于是

$$f_\alpha(\mathrm{Bd}\,\mathbb{B}^m) = \bar{e}_\alpha \setminus e_\alpha = \partial e_\alpha.$$

命题得证. □

命题 1.8.2. *设 X 是 CW 复形, 那么 $f : X \to Y$ 连续当且仅当 f 限制到每一个开胞腔 $f|_{e_\alpha}$ 都连续.*

证明. 必要性是显然的, 下面证明充分性. 任取 $B \subset Y$ 是一个闭集, 对任意的 $e_\alpha \subset X$, 因

$$f^{-1}(B) \cap e_\alpha = (f|_{e_\alpha})^{-1}(B)$$

是 $\bar{e}_\alpha$ 中的闭子集, 于是 $f^{-1}(B)$ 是闭集. □

为了处理 CW 复形的同伦, 我们把上面的命题简单推广一下, 对 X 做一次加厚:

推论 1.8.1. *设 X 是 CW 复形, 那么 $f : X \times I \to Y$ 连续当且仅当对 X 的每一个开胞腔 e_α, $f|_{e_\alpha \times I}$ 是连续的.*

定义 1.8.2 (CW 子复形)**.** *设 X 是一个 CW 复形, $Y \subset X$ 是一族 X 中开胞腔的并, 且满足若 $e_\alpha \subset Y$, 则 $\bar{e}_\alpha \subset Y$. 则 Y 也是一个 CW 复形, 称作是 X 的* ***CW 子复形****. 它是 X 的闭集.*

证明. 首先 Y 作为 X 的子空间是 Hausdorff 的, 且对于任意 $e_\alpha \subset Y \subset X$, 因 X 是 CW 复形, 存在其示性映射 $f_\alpha : \mathbb{D}^m \to X$ 使得 $f_\alpha(\mathbb{D}^m) = \bar{e}_\alpha \subset Y$, 于是

$$f_\alpha(\mathrm{Bd}\,\mathbb{B}^m) \cap \{e_\beta\}_{\beta \in I_\alpha} \subset Y,$$

这里 I_α 是那些与 $\bar{e}_\alpha$ 有非空交的 β 全体. 那么 $f_\alpha(\mathrm{Bd}\,\mathbb{B}^m)$ 落在 Y 中有限多个开胞腔的并中. 这说明了闭有限性.

设 $B \subset Y$ 使得对任意 $e_\alpha \subset Y$ 都有 $B \cap \bar{e}_\alpha$ 在 $\bar{e}_\alpha$ 中闭, 我们需要证明它在 Y 中也是闭集. 设 $e_\beta \subset X$ 但 $e_\beta \not\subset Y$, 则 $e_\beta \cap Y = \emptyset$, 进而 $\bar{e}_\beta \cap Y = \partial e_\beta$. 由定义可知 $\bar{e}_\beta \cap Y$ 落在有限多个 Y 的开胞腔之并中, 不妨设

$$\bar{e}_\beta \cap Y \subset e_{i_1} \cup \cdots \cup e_{i_k},$$

那么

$$B \cap \bar{e}_\beta = ((B \cap \bar{e}_{i_1}) \cup \cdots \cup (B \cap \bar{e}_{i_k})) \cap \bar{e}_\beta.$$

于是 $B \cap \bar{e}_\beta$ 在 $\bar{e}_\beta$ 中是闭集 (因后面的每一个 $B \cap \bar{e}_{i_j}$ 都是闭的), 于是 B 是 X 中的闭子集. 从而 $B \cap Y = B$ 在 Y 中是闭子集. 由定义可知 Y 是 □

推论 1.8.2 (骨架)**.** *记使得 $\dim e_\alpha = p$ 的 α 全体为 $\mathcal{A}_p$, 再记 $\mathcal{A}_{\leqslant p} = \bigcup_{k \leqslant p} \mathcal{A}_k$. 那么由 1.8.2 可知*

$$X^p := \bigcup_{\alpha \in \mathcal{A}_{\leqslant p}} e_\alpha$$

是 X 的一个 CW 子复形, 满足 $\dim X^p = p$, 称作是 X 的 p **维骨架**.

借助骨架的概念, CW 复形可以如此归纳地定义: 在此之前先定义拓扑空间的拓扑和:

定义 1.8.3 (拓扑和)**.** 给定两个拓扑空间 (X_1,τ_1), (X_2,τ_2), 在它们的无交并 $X_1\sqcup X_2$ 上定义拓扑

$$\tau=\{U\subset X_1\sqcup X_2:U\cap X_i\in\tau_i\},$$

称如此构成的拓扑空间 $(X_1\sqcup X_2,\tau)$ 是 X_1 和 X_2 的**拓扑和**.

定义 1.8.4 (正则 CW 复形)**.** 若在 *CW* 复形的定义中, 对每一个开胞腔 e_α, ∂e_α 都恰是有限多个维数更低的开胞腔之并, 则称 X 是一个**正则 *CW* 复形**.

例 1.8.3. 设 K 是一个单纯复形, 那么 $|K|=\bigcup\limits_{\sigma\in K}\operatorname{Int}\sigma$ 是一个 *CW* 复形, 其开胞腔即为 $\{\operatorname{Int}\sigma\}_{\sigma\in K}$. 特别地, 它还是一个正则 *CW* 复形. 设另有 $|L|=\bigcup\limits_{\tau\in L}\operatorname{Int}\tau$ 也是 *CW* 复形, 则

$$|K|\times|L|=\bigcup_{\sigma\in K}\bigcup_{\tau\in L}(\operatorname{Int}\sigma\times\operatorname{Int}\tau)=\bigcup_{\sigma\in K}\bigcup_{\tau\in L}\operatorname{Int}(\sigma\times\tau)$$

仍是一个 *CW* 复形, 且它是一个正则 *CW* 复形.

单纯复形对应的空间总是可以三角剖分的, 我们自然地可以去考虑 CW 复形的三角剖分. 仍然希望一些比较好的 CW 复形可以和单纯复形扯上那么一点儿关系.

定义 1.8.5 (三角剖分)**.** 称 K 是一个**可三角剖分**的 *CW* 复形, 若存在同胚 $h:X\to|K|$ 满足对任意 $p\in\mathbb{Z}$, 存在 K 的子复形 L, $\dim L\leqslant p$ 使得 $h|_{X^p}:X^p\to|L|$ 是同胚.

需要注意的是, 单纯复形与 CW 复形本身是有一些区别的. 对 n 维单纯复形 K, 总有

$$K^{(0)}\subsetneqq K^{(1)}\subsetneqq\cdots\subsetneqq K^{(n)}=K,$$

其中的每一项与前一项的差都是非零的, 每一个包含关系都是严格的. 但对 n 维 CW 复形 X, 上述关系变成

$$X^0\subset X^1\subset\cdots\subset X^n=X,$$

只有 X^0 一定非零, X^n 与前一项的差一定非零, 中间的包含关系都有可能相等. 例如对 CW 复形 $\mathbb{S}^n$ 就有 $X^0=X^1=\cdots=X^{n-1}$.

例 1.8.4. 对球面 $X=\mathbb{S}^n$, 它有三角剖分 $h:X\to|K|$ 使得对任何的 $p\leqslant n-1$, 都有 $h|_{X^p}:X^p\to|K|^{(0)}$, 这因 $X^0=X^1=\cdots=X^{n-1}$, 而 $h:X\to|K|$.

例 1.8.5 (不可三角剖分的 CW 复形)**.** 下面构造一个不可三角剖分的 *CW* 复形, 考虑一个正方形粘上一个三角形构成的单纯复形 K, 其几何实现记作 A. 在正方形上存在一条曲线 $(x,\sin 1/x)$, 记作 C, 那么 $\bar{C}\subset A$. 将正方形与三角形相交的线记作 D, 并如此定义 *CW* 复形: 黏合映射 $g:\operatorname{Bd}\mathbb{B}^3\to\bar{C}\subset A$, 此时 $X=A\sqcup_g\mathbb{B}^3$ 具有 *CW* 复形结构, 并且映射

$$\pi:A\sqcup\mathbb{B}^3\to A\sqcup_g\mathbb{B}^3$$

满足 $\pi|_{\operatorname{Bd}\mathbb{B}^3}=g$, 于是 $\pi(\operatorname{Int}\mathbb{B}^3)=e_3$ 就是三角形内部构成的胞腔.

由此确定的 X 是不可三角剖分的, 为了证明这一断言, 我们使用反证法. 否则, 存在单纯复形 L 使得

$$h : |L| \to X = (A \setminus C) \cup C \cup e_3$$

是一个同胚.

首先考虑 $\bar{e}_3$, 取 $x \in e_3$ 有

$$H_3(X, X \setminus x) \cong H_3(\mathbb{B}^3, \mathbb{B}^3 \setminus 0) \cong H_3(\mathbb{B}^3, \mathbb{S}^2) \cong \mathbb{Z}.$$

取 $x \in A \setminus C$ 有

$$H_3(X, X \setminus X) \cong H_3(\mathbb{B}^2, \mathbb{B}^2 \setminus 0) = 0$$

中间的一项也可能是 $H_3(\mathbb{B}^2_+, \mathbb{B}^2_+ \setminus 0)$, 这源于切除公理中取 $U = X \setminus \mathbb{B}^2$ 或是 $U = X \setminus \mathbb{B}^2_+$, 取决于 x 位于内部还是边界. 而对 $\sigma \in L$, $h(\operatorname{Int}\sigma)$ 不可能同时与 e_3 和 $A \setminus C$ 相交. 这因对任何 $x, y \in \operatorname{Int}\sigma \subset |K|$, 都有

$$H_p(|K|, |K| \setminus x) \cong H_p(|K|, |K| \setminus y).$$

于是 $h(\operatorname{Int}\sigma)$ 或者是属于 $\bar{e}_3$ 的, 或者是属于 $A = \overline{A \setminus C}$ 的. 由此, $\bar{e}_3$ 和 A 都分别可以三角剖分.

同样地, 断言 D 也可以被 h 三角剖分. 因 $D \subset A$, 取 $x \in \operatorname{Int} D$. 注意到 $A = x * \partial A$ 可缩, 由长正合序列可得

$$H_2(A, A \setminus x) \cong H_1(A \setminus x) \cong \mathbb{Z} \oplus \mathbb{Z},$$

这因此时 $A \setminus x$ 实际上同伦于 8 字形空间. 取 $x \in A \setminus D$, 就有

$$H_2(A, A \setminus x) \cong \begin{cases} H_2(\mathbb{B}^2_+, \mathbb{B}^2_+ \setminus 0), & x \in \partial A \setminus D \\ H_2(\mathbb{B}^2, \mathbb{B}^2 \setminus 0), & x \notin \partial A \end{cases}$$

于是对任意 $\sigma \in L$, 也有 $h(\operatorname{Int}\sigma) \cap D = \emptyset$, 那么 $h(\sigma) \subset D$, 于是 D 也可以被 h 三角剖分.

进而 $\bar{C} \cap D$ 也可被 h 三角剖分. 但这不可能成立, 因为 $\bar{C}$ 与 D 相交于无穷多个点.

直观上讲, 从一个离散点集 X^0 开始, 在 X^0 上定义离散拓扑. 按照如下的方式从 X^{p-1} 构造 X^p: 将 $\mathbb{D}^p$ 通过黏合映射 $\varphi : \operatorname{Bd}\mathbb{D}^p \to X^{p-1}$ 粘到 X^{p-1} 上, 那么在拓扑和 $X^{p-1} \sqcup \mathbb{D}^p$ 上定义等价关系:

$$\forall x \in \operatorname{Bd}\mathbb{D}^p\,(x \sim \varphi(x)),$$

从而 $X^p = X^{p-1} \sqcup \mathbb{D}^p / \sim$.

1.9　CW 复形的结构

定理 1.9.1. 设 X 是 p 维的 *CW* 复形, 则

(1) $X = X^{p-1} \sqcup_g (\sum \mathbb{D}^p_\alpha)$, 这里 $\sum$ 表示拓扑和, $g : \sum \operatorname{Bd}\mathbb{D}^p_\alpha \to X^p$.

(2) 设 Y 是一个 *CW* 复形, $\dim Y \leqslant p-1$, $g : \sum \operatorname{Bd}\mathbb{D}^p_\alpha \to Y$ 使得 g 将每一个 $\operatorname{Bd}\mathbb{D}^p_\alpha$ 映到有限多个 Y

1.10　上同调理论

原网页

1.11　Hom 函子和单纯上同调

我们考虑 Abel 群之间的群同态:

定义 1.11.1 (Hom 函子)**.** 设 A, G 都是 *Abel* 群, 记 A 到 G 的群同态全体是 $\operatorname{Hom}(A, G)$. G 上的加法结构可以诱导 $\operatorname{Hom}(A, G)$ 上的加法结构如下*:*

$$(f+g)(a) := f(a) + g(a).$$

于是 $(\operatorname{Hom}(A, G), +)$ 构成一个 *Abel* 群. 此时, 群同态 $h : A \to B$ 可以如下诱导一个 $\operatorname{Hom}(B, G)$ 到 $\operatorname{Hom}(A, G)$ 的同态*:*

$$\varphi h : A \xrightarrow{h} B \xrightarrow{\varphi} G \in \operatorname{Hom}(A, G),$$

那么 $\tilde{h} := \operatorname{Hom}(h, G) = [\varphi \mapsto \varphi h]$ 是一个同态.

命题 1.11.1. $\operatorname{Hom} : A \mapsto \operatorname{Hom}(A, G)$, $h \mapsto \tilde{h}$ 给出了一个从 Ab 到 Ab 的反变函子.

证明. 对 $\operatorname{id} : A \to A$, $\widetilde{\operatorname{id}} = \operatorname{Hom}(\operatorname{id}, G)$ 自然是 $\operatorname{Hom}(A, G)$ 上的恒等映射. 而对 $A \xrightarrow{h} B \xrightarrow{k} C$, 只需计算验证

$$\widetilde{kh}(\varphi) = \varphi kh = \tilde{k}\varphi h = \tilde{h}\tilde{k}\varphi$$

可知 $\widetilde{kh} = \tilde{h}\tilde{k}$, 也即 Hom 是一个反变函子. □

命题 1.11.2. 设 h 是一个 *Abel* 群同态, 那么

1. h 是同构 $\Longrightarrow \tilde{h}$ 是同构;
2. $h=0 \Longrightarrow \tilde{h}=0$;
3. h 是满同态 $\Longrightarrow \tilde{h}$ 是单同态;
4. 若序列 $0 \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0$ 正合且分裂, 则序列 $0 \longleftarrow \mathrm{Hom}(A,G) \xleftarrow{\tilde{f}} \mathrm{Hom}(B,G$ 也正合且分裂. 也即 f 是单同态推 $\tilde{f}$ 是满同态需要添加序列分裂的条件.

证明. (1), (2) 的证明是容易的.

(3) 任取 $\varphi \in \mathrm{Hom}(B,G)$ 使得 $\tilde{h}(\varphi)=\varphi h=0$, 那么对任意 $a \in A$ 成立

$$\tilde{h}(\varphi(a))=\varphi h(a)=0,$$

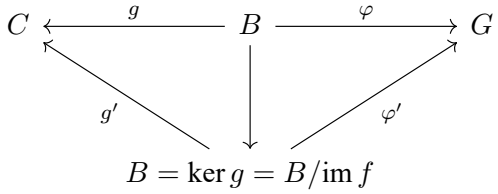
因 h 是满同态, 这即对任意 $b \in B$ 有 $\varphi(b)=0$, 故 $\varphi=0$, 这说明 $\tilde{h}$ 是单的.

(4) 只需证明在 $\mathrm{Hom}(B,G)$ 处正合, 也即 $\operatorname{im}\tilde{g}=\ker\tilde{f}$.

因第一个序列正合, 故 $fg=0$, 由 (2) 可知 $\widetilde{gf}=\tilde{f}\tilde{g}=0$. 于是 $\operatorname{im}\tilde{g} \leqslant \ker\tilde{f}$. 再取 $\varphi \in \ker\tilde{f} \leqslant \mathrm{Hom}(B,G)$, 由 $\tilde{f}(\varphi)=\varphi f=0$ 可知 $\operatorname{im} f \leqslant \ker\varphi$, 那么

$$\varphi' : B/\operatorname{im} f \to G$$

是由 φ 诱导的群同态, 满足



因为 $B \to C$ 是满同态, 于是 g' 是同构, 那么由交换图的可交换性得到

$$\varphi=\varphi'(g')^{-1}g=\tilde{g}(\varphi'(g')^{-1}) \in \operatorname{im}\tilde{g},$$

于是 $\ker\tilde{f} \leqslant \operatorname{im}\tilde{g}$.

再考虑另一部分的证明. 因第一个序列分裂, 于是存在 $\pi : B \to A$ 使得 $\pi f=\mathrm{id}_A$, 那么 $\widetilde{\pi f}=\widetilde{\mathrm{id}}_A$, 也即 $\tilde{f}\tilde{\pi}=\widetilde{\mathrm{id}}_A$, 故 $\tilde{f}$ 满, $\tilde{\pi}$ 单, 并且第二个序列也是分裂的. □

在 (4) 中如果去掉序列分裂的条件, 那么结论往往不成立. 考虑 $h : \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z}$, 这是一个单同态, 那么 $\tilde{h} : \mathrm{Hom}(\mathbb{Z},\mathbb{Z}) \xrightarrow{\tilde{2}} \mathrm{Hom}(\mathbb{Z},\mathbb{Z})$, 对 $\varphi : \mathbb{Z} \to \mathbb{Z}$ 是一个同态, 就有

$$\tilde{2}\varphi=\varphi(\times 2)=2\varphi,$$

而 $\mathrm{Hom}(\mathbb{Z},\mathbb{Z}) \cong \mathbb{Z}$, 容易看出 $\tilde{h}$ 并不是满同态.

命题 1.11.3. 设 A, G, A_i, G_i 是 *Abel* 群, 则

1. $\mathrm{Hom}(\bigoplus\limits_{i\in\alpha} A_i, G) \cong \prod\limits_{i\in\alpha} \mathrm{Hom}(A_i, G)$;
2. $\mathrm{Hom}(A, \prod\limits_{i\in\alpha} G_i) \cong \prod\limits_{i\in\alpha} \mathrm{Hom}(A, G_i)$.

证明. (1) 对 $f : \bigoplus\limits_{i\in\alpha} A_i \to G$, 取 $f_i=f|_{A_i}$, 那么 $\Phi : f \mapsto \prod\limits_{i\in\alpha} f_i$ 就是一个 $\mathrm{Hom}(\bigoplus\limits_{i\in\alpha} A_i, G) \to \prod\limits_{i\in\alpha} \mathrm{Hom}(A_i, G)$ 的同态.

反之, 对 $f=\prod\limits_{i\in\alpha} f_i$, $f_i : A_i \to G$, 定义 $\Psi(f) : \bigoplus\limits_{i\in\alpha} A_i \to G$ 满足 $\Psi(f)(a)=\sum\limits_{i\in\alpha} f_i(a_i)$ 即可. 这里限制到代数直和是因为有限和的限制让我们无需考虑级数.

(2) 类似. □

命题 1.11.4. *设 G 是 Abel 群, 则*

$$\mathrm{Hom}(\mathbb{Z},G)\cong G,\qquad \mathrm{Hom}(\mathbb{Z}/m\mathbb{Z},G)\cong \ker[G\xrightarrow{m}G].$$

证明. 第一个断言取

$$\lambda:\mathrm{Hom}(\mathbb{Z},G)\to G,\qquad \varphi\mapsto\varphi(1)$$

就是一个同构. 而对第二个断言, 由于序列 $0\longrightarrow\mathbb{Z}\xrightarrow{m}\mathbb{Z}\longrightarrow\mathbb{Z}/m\mathbb{Z}\longrightarrow 0$ 正合, 那么 $0\longleftarrow\mathrm{Hom}(\mathbb{Z},G)\xleftarrow{\tilde m}\mathrm{Hom}(\mathbb{Z},G)\xleftarrow{\tilde\varphi}\mathrm{Hom}(\mathbb{Z}/m\mathbb{Z},G)\longleftarrow 0$ 也是正合的. 这里 $\tilde\varphi$ 是单同态导出了 $\mathrm{Hom}(\mathbb{Z}/m\mathbb{Z},G)\cong\mathrm{im}\,\varphi$, 而序列正合导出了 $\mathrm{im}\,\tilde\varphi\cong\ker\tilde m$. □

在单纯同调理论中, 我们通过单纯链复形 $\mathcal{C}$ 导出了单纯同调群 $H_p(\mathcal{C})$. 那么 Hom 函子作用之后的 $\mathrm{Hom}(\mathcal{C};G)$, 我们希望它也对偶地导出同调群.

对单纯复形 K, 考虑如下导出的链群:

$$\begin{array}{ccc} & \mathcal{C}(K)=\{(C_p(K),\partial)\} & \longrightarrow H_p(\mathcal{C}(K)) \\ K\nearrow\searrow & \Big\downarrow{\scriptstyle \mathrm{Hom}(\,\cdot\,,G)} & \\ & \mathrm{Hom}(\mathcal{C}(K);G)=\{(C^p(K;G),\delta)\} & \longrightarrow H^p(\mathcal{C}(K)) \end{array}$$

称 $\mathrm{Hom}(C_p(K),G):=C^p(K;G)$ 是 p 维**上链群**, 这里的 $\delta=\mathrm{Hom}(\partial,G)$.

命题 1.11.5. *$\mathrm{Hom}(\mathcal{C}(K),G)$ 仍是一个链复形, 它以 $\delta=\mathrm{Hom}(\partial,G)$ 为边缘算子.*

证明. 考虑

$$\delta:C^p(K;G)\to C^{p+1}(K;G),\qquad c^p\mapsto\delta c^p=\tilde\partial c^p=c^p\delta,$$

这里 $\delta c^p:C_{p+1}(K)\to C_p(K)$. 并且

$$\delta c^p(c_{p+1})=c^p(\partial c_{p+1})$$

可以看作是 c^p 作用于 c_{p+1} 上的值, 于是我们把它写成类似于内积的形式: $\langle\delta c^p,c_{p+1}\rangle=\langle c^p,\partial c_{p+1}\rangle$.

而 $\delta^2=0$ 由 $\partial^2=0$ 立刻得到. □

于是可以在 $\mathrm{Hom}(\mathcal{C}(K),G)$ 上定义上同调. 记

$$\ker\delta_p:=Z^p(K;G),\qquad \mathrm{im}\,\delta_{p+1}:=B^p(K;G),$$

称

$$H^p(K;G)=Z^p(K;G)/B^p(K;G)$$

是 K 的 p 维**上同调群**. (这里称作上同调完全是因为此时习惯把指标写在右上角, *cohomology* 的 *co*–前缀表示代数对偶).

对任何 Abel 群 G, 它有自由部分 (以 $\mathbb{Z}$ 为直和项) 和挠部分 (以 $\mathbb{Z}/m\mathbb{Z}$ 为直和项), 而之前已经讨论过

$$\mathrm{Hom}(\mathbb{Z},G)\cong G,\qquad \mathrm{Hom}(\mathbb{Z}/m\mathbb{Z},G)\cong\ker[G\xrightarrow{m}G].$$

我们需要注意, 由

$$C^p(K;G)\cong\mathrm{Hom}(C_p(K),G)\cong\mathrm{Hom}\Big(\bigoplus_{\sigma\in S_p^+(K)}\mathbb{Z}\langle\sigma\rangle,G\Big)\cong\prod_{\sigma\in S_p^+(K)}\mathrm{Hom}(\mathbb{Z}\langle\sigma\rangle,G)$$

未必是一个自由 Abel 群.

要计算 $H^p(K;G)$, 就归结到计算 δc 的问题, 所以下面处理上链如何表达. 在单纯同调理论中, 有 p 维基本链 $\sigma_\alpha \in S_p^+(K)$, 类似地, 我们希望也有 p 维基本上链.

先考虑 $G=\mathbb{Z}$ 的情形, 此时对 $\sigma_\beta \in S_p^+(K)$, 定义

$$\langle \sigma_\alpha^*, \sigma_\beta \rangle := \begin{cases} 1, & \beta = \alpha \\ 0, & \beta \neq \alpha \end{cases}$$

那么对任何 $c^p \in C^p(K;\mathbb{Z}) =: C^p(K)$, 就有 $c^p = \sum\limits_\alpha n_\alpha \sigma_\alpha^*$ (这里的和未必是有限和). 对一般的 Abel 群 G, 我们考虑 $g_\alpha \sigma_\alpha^*$, 这里 $g_\alpha \in G$. 之前把它写成内积的形式, 我们同样希望这里的 "内积" 保持双线性, 于是可以定义

$$\langle g_\alpha \sigma_\alpha^*, \sigma_\beta \rangle := \begin{cases} g_\alpha, & \beta = \alpha \\ 0, & \beta \neq \alpha \end{cases}$$

那么对任何 $c^p \in C^p(K;\mathbb{Z}) =: C^p(K)$, 就有 $c^p = \sum\limits_\alpha g_\alpha \sigma_\alpha^*$ (同样地, 这里的和未必是有限和).

命题 1.11.6. 对 $c^p = \sum\limits_\alpha g_\alpha \sigma_\alpha^*$, 有 $\delta c^p = \sum\limits_\alpha g_\alpha \delta \sigma_\alpha^*$.

证明. 对 $\tau \in S_{p+1}^+(K)$, 有 $\delta\tau = \sum\limits_{i=0}^{p+1} \varepsilon_i \sigma_i$, 其中 $\varepsilon_i = \pm 1$, 取决于 σ_i 的定向. (这因低维单形的定向未必总是与高维单形定向相同.) 此时

$$\langle \delta c^p, \tau \rangle = \left\langle \sum_\alpha g_\alpha \sigma_\alpha^*, \sum_{i=0}^{p+1} \varepsilon_i \sigma_i \right\rangle = \sum_\alpha \sum_{i=0}^{p+1} g_\alpha \varepsilon_i \langle \sigma_\alpha^*, \sigma_i \rangle = \sum_{i=0}^{p+1} \varepsilon_i g_i.$$

而

$$\left\langle \sum_\alpha g_\alpha \delta\sigma_\alpha^*, \tau \right\rangle = \sum_{i=0}^{p+1} \varepsilon_i g_i.$$

这说明在每一个单形 τ 上 $c^p = \sum\limits_\alpha g_\alpha \sigma_\alpha^*$, 于是命题得证. □

由这一条件, 在几何上, $\delta\sigma_\alpha^*$ 的计算可以简单地遵从下面的规则. 记 τ_i 是所有以 σ_α 为面的单形, 则

$$\delta\sigma_\alpha^* = \sum_i \varepsilon_i \tau_i^*,$$

这里 ε_i 是 σ_α 在 $\partial\tau_i$ 中的系数.

例 1.11.1. 考虑下面的 2 维单纯复形

只需分别计算不同维数的基本上链的边缘即可. 对 2–基本上链

$$\delta\sigma_1^* = \delta\sigma_2^* = 0,$$

对 1–基本上链, 有

$$\delta e_1^* = \delta e_4^* = -\sigma_1^*, \quad \delta e_2^* = \delta e_3^* = -\sigma_2^*, \quad \delta e_5^* = \sigma_1^* - \sigma_2^*.$$

对 0–基本上链, 有

$$\delta v_0^* = e_4^* - e_1^*, \quad \delta v_1^* = e_1^* - e_2^* + e_5^*, \quad \delta v_2^* = e_2^* - e_3^*, \quad \delta v_3^* = e_3^* - e_4^* - e_5^*.$$

考虑 K 关于系数 $\mathbb{Z}$ 的上同调群: 列出 K 的上链复形有 $0 \xleftarrow{\delta_2} \mathbb{Z}\oplus\mathbb{Z} \xleftarrow{\delta_1} \mathbb{Z}^{\oplus 4} \xleftarrow{\delta_0} \mathbb{Z}^{\oplus 4} \longleftarrow 0$ 那么 $\operatorname{im}\delta_0 = 0$ 导出 $H^0(K) \cong \ker\delta_0 = Z^0(K)$, 对任意 $c^0 = \sum_{i=0}^{3} n_i v_i^* \in C^0(K)$, 由

$$\begin{aligned}
\delta c^0 = 0 &\Longleftrightarrow \sum_{i=0}^{3} n_i \delta v_i^* = 0 \\
&\Longleftrightarrow (n_0 - n_3)e_4^* + (n_1 - n_0)e_1^* + (n_2 - n_1)e_2^* + (n_3 - n_2)e_3^* = 0 \\
&\Longleftrightarrow n_0 = n_1 = n_2 = n_3 = n.
\end{aligned}$$

这就说明 $Z^0(K) = \mathbb{Z}\left\langle \sum_{i=0}^{3} v_i^* \right\rangle \cong \mathbb{Z}$, 于是 $H^0(K) \cong \mathbb{Z}$.

再考虑 $p = 1$ 的情形, $Z^1(K) = \ker\delta_1$, 对 $c^1 = \sum_{i=1}^{5} n_i e_i^* \in C^1(K)$, 类似地有

$$\delta c^1 = 0 \Longleftrightarrow \begin{cases} n_1 = n_5 - n_4 \\ n_2 = -n_3 - n_5 \end{cases}.$$

于是对 $c^1 \in Z^1(K)$, 有

$$\begin{aligned}
c^1 &= (n_5 - n_4)e_1^* + (-n_3 - n_5)e_2^* + n_3 e_3^* + n_4 e_4^* + n_5 e_5^* \\
&= n_3(e_3^* - e_2^*) + n_4(e_4^* - e_1^*) + n_5(e_1^* - e_2^* + e_5^*) \\
&= -n_3\delta v_2^* + n_4\delta v_0^* + n_5\delta v_1^*,
\end{aligned}$$

原文未完成片段：以下按源码保留，未补写公式或推导。

```
这即 $ Z^1(K)=B^1(K) $, 于是 $ H^1(K)=Z^1(K)/
```

1.12　单纯同调论

原网页

1.13　几何复形

首先声明我们主要讨论的基底空间 $\mathbb{R}^{\mathbb{N}}$. 记

$$\mathbb{R}^{\mathbb{N}} := \left\{(x_1, x_2, \dots, x_n, \dots) : \text{至多有限个} x_i \text{ 非零}\right\},$$

在其上可以定义**凝聚拓扑**, 即

$$A \subset \mathbb{R}^{\mathbb{N}} \text{ 是闭集} \iff \forall n \in \mathbb{N},\ A \cap \mathbb{R}^n \text{ 为} \mathbb{R}^n \text{ 中闭集}.$$

考虑 $a_0, \dots, a_p$ 是 $\mathbb{R}^{\mathbb{N}}$ 中的 $p+1$ 个点, 称它们**几何独立**, 若 $\{a_1 - a_0, \dots, a_p - a_0\}$ 线性无关. 这确定了一个 p 维线性空间

$$P := \operatorname{span}\{a_1 - a_0, \dots, a_p - a_0\} = \left\{\sum_{i=1}^{p} \lambda_i (a_i - a_0) : \lambda_i \in \mathbb{R}\right\},$$

将其平移得到仿射空间 $a_0 + P$. 此时

$$a_0 + P = \left\{\left(1 - \sum_{i=1}^{p} \lambda_i\right) a_0 + \sum_{i=1}^{p} \lambda_i a_i : \lambda_i \in \mathbb{R}\right\} = \left\{\sum_{i=0}^{p} \lambda_i a_i : \sum_{i=0}^{p} \lambda_i = 1\right\},$$

其中 $\lambda_0 = 1 - \sum\limits_{i=1}^{p} \lambda_i$. 注意到几何独立的 $p+1$ 个点把空间划分成不同的区域, 我们将依此定义单形.

定义 1.13.1 (p 维单形, 真面)**.** *设 $a_0, \dots, a_p$ 是 $\mathbb{R}^{\mathbb{N}}$ 中几何独立, 称*

$$(a_0, \dots, a_p) := \left\{\sum_{i=0}^{p} \lambda_i a_i : \sum_{i=0}^{p} \lambda_i = 1, \lambda_i \geqslant 0\right\}$$

*是一个 $\mathbb{R}^{\mathbb{N}}$ 中的p **维单形**.*

若 $\{i_0, \dots, i_q\} \subset \{0, \dots, p\}$, 则 $a_{i_0}, \dots, a_{i_q}$ 几何独立, 记

$$\tau^q = (a_{i_0}, \dots, a_{i_q}) \prec (a_0, \dots, a_p) = \sigma^p.$$

*若 $q < p$, 称 τ^q 是 σ^p 的一个**真面**.*

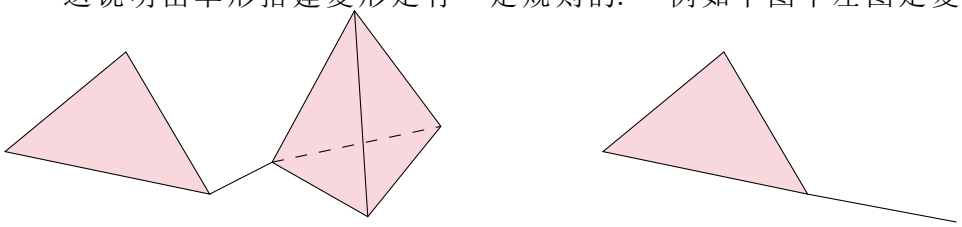
单形是最简单的几何体, 单形的组合构成更复杂的几何体.

定义 1.13.2 (复形)**.** *设 K 是 $\mathbb{R}^{\mathbb{N}}$ 中一些单形的集合, 称 K 是一个 **(单纯) 复形**, 若*

(1) 对任意 $\sigma \in K$, σ 的任何面仍在 K 中;

(2) 对任意 $\sigma,\tau\in K$, $\sigma\cap\tau$ 或者是空集, 或者是它们的公共面.
若 K 中的单形个数有限, 则称 K 是**有限复形**.

这说明由单形搭建复形是有一定规则的:　例如下图中左图是复形而右图不是复形.



定义 1.13.3 (复形的维数)**.** 设 K 是一个复形, 称其**维数**为

$$\dim K:=\sup_{\sigma\in K}\dim\sigma.$$

有限复形必是有限维的, 但反之不真.

对一个单形 $\sigma^p\in\mathbb{R}^{\mathbb{N}}$, 可以从此出发构造复形

$$K_{\sigma^p}:=\{\tau:\tau\prec\sigma^p\}$$

和其边界复形

$$\operatorname{Bd}K_{\sigma^p}=K_{\partial\sigma^p}:=K_{\sigma^p}\setminus\{\sigma\}.$$

更具体一些, 有复形的骨架的概念:

定义 1.13.4 (骨架)**.** 设 K 是一个复形, 称

$$K^{(p)}:=\{\sigma\in K:\dim\sigma\leqslant p\}$$

是 K 的p **维骨架**. 于是 $K^{(p)}$ 也是一个复形, $K^{(p)}\subset K$ 是 K 的一个子复形. 特别地, $K^{(0)}$ 称作是 K 的**顶点集**, 其中的元素称作 K 的**顶点**.

直观上当 $p=1$ 时, 1 维骨架就是直观上看起来的复形的 “骨架”, 它只包含复形的 0 维子复形 (即顶点) 和 1 维子复形 (即各边).

设 K 是 $\mathbb{R}^{\mathbb{N}}$ 的复形, 称

$$|K|=\bigcup_{\sigma\in K}\sigma$$

是 K 的**几何实现**或**伴随多面体**. 此时 $|K|$ 成为 $\mathbb{R}^{\mathbb{N}}$ 中的子集, 赋予拓扑 $\mathcal{T}_{|K|}$ 是 $\mathbb{R}^{\mathbb{N}}$ 上的子空间拓扑.

引理 1.13.1. 设 L 是 K 的子复形, 则 $|L|$ 是 $|K|$ 中的闭集.

证明. 设 $A\subset|L|$ 是闭集, 对任意 $\sigma\in K$, 有

$$\sigma\cap|L|=\bigcup_{\tau\prec\sigma,\tau\in L}\tau,$$

故

$$A\cap\sigma=\bigcup_{\tau\prec\sigma,\tau\in L}(A\cap\tau).$$

图 1.1: $\mathbb{T}^2$ 的三角剖分

而 $A\cap\tau$ 是 τ 中的闭集, 且 $\tau\prec\sigma$, 故 $A\cap\tau$ 是 σ 中的闭集, 进而 $A\cap\sigma$ 也是 σ 中的闭集. 由此可知 A 是 $|K|$ 中的闭集.

特别地, 取 $A=|L|$ 即可证明引理. □

由此可以说明在 $|K|$ 上的拓扑在一般情形下是弱于 Euclid 拓扑的, 即使它可以被嵌入有限维空间. 为此考虑复形

$$K=\{[m,m+1]:m\neq0,\,m\in\mathbb{Z}\}\cup\left\{\left[\frac{1}{n+1},\frac{1}{n}\right]:n\in\mathbb{Z}_+\right\}\cup\mathbb{Z}\cup\left\{\frac{1}{n}:n\in\mathbb{Z}_+\right\}.$$

则 $|K|=\mathbb{R}$, 且 $L=\{1/n:n\in\mathbb{Z}_+\}$ 是 K 的子复形. 由引理 1.13.1 可知 $|L|=L$ 是 $|K|$ 中的闭集, 但在 $|K|$ 的 Euclid 拓扑下 $|L|$ 并不是闭集, 因它有聚点 $0\notin|L|$.

定义 1.13.5 (可三角剖分空间)**.** *称拓扑空间 X* **可三角剖分**, *若存在复形 K 使得 $X\cong|K|$.*

所有的光滑流形都可以被三角剖分, 一个例子是环面 $\mathbb{T}^2$ 可以被下图三角剖分: 但需要注意的是下图的剖分方式并不是三角剖分, 因标注 α 和 β 的两个三角形同时相交于点 A 和线 BC, 它们不是公共面 (因不连通).

1.14　单纯映射

单纯复形范畴可以看作偏序范畴的一种, 考虑由真面诱导的偏序 $\prec$, 那么 $(K,\prec)$ 构成一个偏序集, 因此我们需要保持偏序结构的态射:

定义 1.14.1 (单纯映射)**.** *称 $f:K\to L$ 是一个*单纯映射, *若对任意 $\sigma\in K$ 都有 $f(\sigma)\in L$, f 保持偏序关系, 即*

$$\tau\prec\sigma\Longrightarrow f(\tau)\prec f(\sigma),$$

同时要求 $\dim f(\sigma)\leqslant\dim\sigma$.

这与教材上的定义是等价的. $\dim f(\sigma)\leqslant\dim\sigma$ 保证了 f 将 K 的顶点 $K^{(0)}$ 映到 L 的顶点 $L^{(0)}$. 单形本质上就可以看作是一些顶点的集合 (由习题 1.1 保证). 于是单纯映射可以作为态射, 构成一个范畴 SimComp, 称作是**单纯复形范畴**.

单纯映射 $f:K\to L$ 诱导一个拓扑空间之间的映射 g_f, 它满足

$$g_f:|K|\to|L|,\qquad x=\sum_{i=0}^{p}t_i\sigma_i\mapsto\sum_{i=0}^{p}t_if(\sigma_i),$$

图 1.2: 一种不是三角剖分的 $\mathbb{T}^2$ 的剖分方式

它可以看作是一系列形如 $g_f|_\sigma : \sigma \mapsto |L|$ 拼接起来的映射. 在拓扑空间范畴 Top 中, 我们当然希望 g_f 是连续的, 从而导出一个 SimComp 到 Top 的函子. 为此先证明以下引理:

引理 1.14.1. *$h : |K| \to X$ 连续当且仅当对任意 $\sigma \in K$, 都有 $h|_\sigma : \sigma \to X$ 连续.*

证明. 必要性. 显然.

充分性. 任取 X 上的闭集 C, 对任意 $\sigma \in K$, 只需要证明 $h^{-1}(C) \cap \sigma$ 是 σ 中的闭集即可. 而

$$h^{-1}(C) \cap \sigma = (h|_\sigma)^{-1}(C)$$

是闭的. □

由此得到:

命题 1.14.1. *$F : K \mapsto |K|,\ f \mapsto g_f$ 给出一个 SimComp 到 Top 的函子.*

证明. 只需验证 F 的函子性, 首先 $\mathrm{id}_K : K \to K$ 诱导出 $F(\mathrm{id}_K) = \mathrm{id}_{|K|} : |K| \to |K|$, 于是只需要证明图 $\begin{array}{ccc} |K| & \xrightarrow{g_f} & |L| \\ & \searrow^{g_h g_f} & \downarrow{\scriptstyle g_h} \\ & & |M| \end{array}$ 交换即可. 这因

$$g_{hf}(x) = \sum_{i=0}^{p} t_i (hf)(\sigma_i) = g_h \sum_{i=0}^{p} t_i f(\sigma_i) = g_h g_f(x).$$ □

对复形 K, 其中的单形 $\sigma = (\sigma_0, \ldots, \sigma_p)$ 可以由顶点集 $\{\sigma_0, \ldots, \sigma_p\}$ 描述, 因此 K 在某种程度上可以由 $K^{(0)}$ 描述.

定义 1.14.2 (抽象复形)**.** *设 S 是一个集合, 称 $\mathcal{K} \subset 2^S$ 是一个**抽象复形**, 若*

$$\forall s \in \mathcal{K}\,(t \subset s \Longrightarrow t \in \mathcal{K}).$$

此时的 s 即为抽象单形, $\dim s = |s| - 1$. 特别地, $\dim \varnothing = -1$. 类似可定义抽象复形的维数

$$\dim \mathcal{K} := \sup_{s \in \mathcal{K}} \dim s.$$

对抽象复形, 也可以定义骨架 $\mathcal{K}^{(p)} := \{s \in \mathcal{K} : \dim s \leqslant p\}$.

定义 1.14.3 (几何实现复形)**.** *称 K 是 $\mathcal{K}$ 的**几何实现复形**, 若存在双射 $f : K^{(0)} \to \mathcal{K}^{(0)}$ 使得*

$$(v_0, \ldots, v_p) \in K \Longleftrightarrow \{f(v_0), \ldots, f(v_p)\} \in \mathcal{K}.$$

抽象复形 $\mathcal{K}$ 的所有几何实现复形的伴随多面体都是分段线性同胚的, 因此之后我们不区分抽象复形和几何复形.

定理 1.14.1. *设 $\mathcal{K}^n$ 是 n 维有限抽象复形, 它必可在 $\mathbb{R}^{2n+1}$ 上实现为几何复形.*

证明. 考虑循环曲线

$$\Phi:\mathbb{R}\to\mathbb{R}^{2n+1},\qquad t\mapsto(t,t^2,\dots,t^{2n+1}),$$

断言对于互不相同的 $t_0,\dots,t_{2n+1}$, 有 $\Phi(t_0),\dots,\Phi(t_{2n+1})$ 几何独立. 为此, 考虑

$$\det[\Phi(t_1)-\Phi(t_0),\dots,\Phi(t_{2n+1})-\Phi(t_0)]=\begin{vmatrix} t_1-t_0 & \cdots & t_{2n+1}-t_0\\ \vdots & \ddots & \vdots\\ t_1^{2n+1}-t_0^{2n+1} & \cdots & t_{2n+1}^{2n+1}-t_0^{2n+1}\end{vmatrix}\neq 0$$

即得断言.

令 $\mathcal{K}^{(0)}=\{s_1,\dots,s_N\}\setminus\{\varnothing\}$, 几何复形 K 构造如下:

$$\mathcal{K}^0\ni s_i\longleftrightarrow\Phi(i)\in K^{(0)},$$

则 $s=\{s_{i_1},\dots,s_{i_p}\}\in\mathcal{K}$ 当且仅当 $(\Phi(i_1),\dots,\Phi(i_p))\in K$, 其中 $p\leqslant n$. 下面证明如此构造的 K 是一个几何复形, 定义的第一条显然满足, 只需证明任意两个单形的交或者非空, 或者是公共面即可.

任取 $\sigma^p,\sigma^q\in K$, 设它们有 r 个公共顶点, 则 σ^p 和 σ^q 共有 $p+1+q+1-r$ 个不同的顶点, 而

$$p+q+2-r\leqslant 2n+2-r<2n+2,$$

于是这些顶点几何独立, 它们确定了一个 $p+q+1-r$ 维单形 γ, 满足 $\sigma^p\prec\gamma,\sigma^q\prec\gamma$. 因此由单形的定义可知它们的交或者非空, 或者是公共面. □

推论 1.14.1. *设 K 是有限复形, 且 $\dim K=n$, 则一定存在嵌入 $|K|\hookrightarrow\mathbb{R}^{2n+1}$.*

但是将上一推论的嵌入从 $\mathbb{R}^{2n+1}$ 改为 $\mathbb{R}^{2n}$ 后就不再成立了. 例如 $K^{(n)}_{\sigma^{2n+2}}$ 就不能嵌入 $\mathbb{R}^{2n}$. 最简单的情形, 令 $n=1$, 此时 $K^{(1)}_{\sigma^4}$ 即为下图: 它就是完全图 $K_{5,5}$, 是一类不可平面化的图, 于是它不能嵌入 $\mathbb{R}^2$.

1.15　单纯同调群

对复形的定向无非是对顶点的排序, 而排序本质上只有两种, 即由偶置换确定的同构类. 对 σ 的顶点集 $\{v_0,\dots,v_p\}$, 定义它在偶置换下的同构类为 $[v_0,\dots,v_p]$, 就称它是复形 σ 的一个**定向**.

特别地, 约定 0 维情形下, $[a_0]$ 与 $-[a_0]$ 是两种不同的定向.

定义 1.15.1 (定向复形)**.** *称 K 是一个**定向复形**, 若 $\forall\sigma\in K$, σ 是有定向的. 并记*

$$S_p^+(K):=\{\sigma\in K:\dim\sigma=p\},\qquad S_p(K)=\{\pm\sigma:\sigma\in S_p^+(K)\}.$$

定理 1.15.1. *由 $S_p^+(K)$ 作为生成元可以生成一个自由 Abel 群, 记作*

$$C_p(K):=\left\{\sum_{\sigma\in S_p^+(K)}n_\sigma\sigma\right\},$$

并要求求和为有限和.

图 1.3: $K^{(1)}_{\sigma^4} = K_{5,5}$

图 1.4: 低维复形的定向

证明. 考虑映射 $c_p : S_p(K) \to \mathbb{Z}$, 其中 c_p 仅在 $S_p(K)$ 中有限个元素上取非零值, 且 $\forall \sigma \in S_p(K)$ 都有 $c_p(-\sigma) = -c_p(\sigma)$. 称 c_p 是一个 **p–链**. 取

$$C_p(K) = \{c_p : S_p(K) \to \mathbb{Z}\},$$

在它上面定义加法

$$(c_p + c_p')(\sigma) := c_p(\sigma) + c_p'(\sigma),$$

则 $(C_p(K), +)$ 是一个 Abel 群, 其中的零元 $0 : S_p(K) \to \mathbb{Z}, \sigma \mapsto 0$, c_p 的逆 $(-c_p)(\sigma) := -c_p(\sigma)$.

接下来证明它是由 $S_p^+(K)$ 生成的自由 Abel 群. 我们需要做一点说明, 对 $\sigma \in S_p^+(K)$, 定义

$$c_p^\sigma(\tau) = \begin{cases} \pm 1, & \tau = \pm\sigma \\ 0, & \tau \neq \pm\sigma \end{cases}$$

则 $\left\{c_p^\sigma\right\}_{\sigma \in S_p^+(K)}$ 与 $S_p^+(K)$ 之间可以建立 1–1 对应, 于是可以用 σ 替代 c_p^σ. $\forall c_p \in C_p(K), \forall \sigma \in S_p^+(K)$, 记 $c_p(\sigma) = n_\sigma$, 那么构造的

$$c_p' = \sum_{\sigma \in S_p^+(K)} n_\sigma c_p^\sigma = \sum_{\sigma \in S_p^+(K)} n_\sigma \sigma$$

自然是有限和. 断言 $c_p' = c_p$.

为此, 对任意 $\tau \in S_p(K)$, 考虑

$$c_p'(\tau) = \sum_{\sigma \in S_p^+(K)} n_\sigma c_p^\sigma(\tau) \overset{\tau = \pm\sigma}{=\!=} \pm n_\sigma = c_p(\pm\sigma) = c_p(\tau).$$

再说明唯一性, 令 $c_p = \sum\limits_{\sigma \in S_p^+(K)} n_\sigma \sigma = \sum\limits_{\sigma \in S_p^+(K)} n_\sigma' \sigma$, 则

$$\sum_{\sigma \in S_p^+(K)} (n_\sigma - n_\sigma')\sigma = 0_p.$$

代入 $\tau = \pm\sigma$ 得到

$$\sum_{\sigma \in S_p^+(K)} (n_\sigma - n_\sigma')c_p^\sigma(\tau) = \pm(n_\sigma - n_\sigma') = 0_p(\tau) = 0,$$

即 $n_\sigma = n_\sigma'$, 断言得证. □

考虑以上 2 维复形的边界, 计算 $\partial[v_0, v_1, v_2]$ 得到

$$\partial[v_0, v_1, v_2] = [v_0, v_1] + [v_1, v_2] + [v_2, v_0] = [v_0, v_1] - [v_0, v_2] + [v_1, v_2].$$

并且注意到边界中的每一项符号与缺少的顶点标号有关.

再考虑 3 维复形的边界: 这里所有 2 维面的定向都使用右手法则来确定. 计算 $\partial[v_0, v_1, v_2, v_3]$ 得到

$$\begin{aligned} \partial[v_0, v_1, v_2, v_3] &= [v_0, v_2, v_1] + [v_0, v_1, v_3] + [v_1, v_2, v_3] + [v_0, v_3, v_2] \\ &= -[v_0, v_1, v_2] + [v_0, v_1, v_3] - [v_0, v_2, v_3] + [v_1, v_2, v_3]. \end{aligned}$$

同样地, 注意到边界中的每一项符号都与缺少的顶点标号有关.

536×599 像素. 其他分辨率: 255×240 像素.

061 × 1,000 像素, 文件大小: 24 KB, MIME 类型:

图 1.5: 一个 2 维定向复形的例子

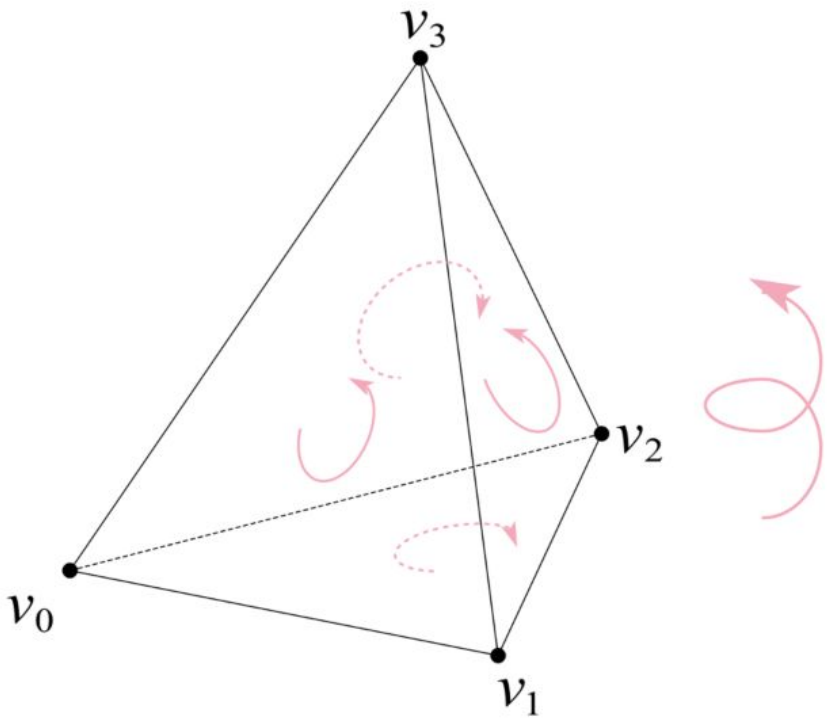


图 1.6: 一个 3 维定向复形的例子

由此可以归纳得到,

$$\partial[v_0,\dots,v_p]=\sum_{k=0}^{p}(-1)^k[v_0,\dots,\hat{v}_k,\dots,v_p],$$

其中 $\hat{v}_k$ 表示缺少这一项.

定义 1.15.2 (边缘算子)**.** 称

$$\partial: C_p(K)\to C_{p-1}(K),\qquad c_p=\sum_{\sigma\in S_p^+(K)}n_\sigma\sigma\mapsto\sum_{\sigma\in S_p^+(K)}n_\sigma\partial\sigma$$

是 $C_p(K)$ 上的**边缘算子**, 有时为了强调也记作 ∂_p. 它是 $C_p(K)$ 到 $C_{p-1}(K)$ 的群同态, 其中 $\partial\sigma$ 如下定义: 对 $\sigma=[v_0,\dots,v_p]$, 有

$$\partial[v_0,\dots,v_p]=\sum_{k=0}^{p}(-1)^k[v_0,\dots,\hat{v}_k,\dots,v_p].$$

命题 1.15.1. $\partial^2=0$. *(严格地讲, 应该写成 $\partial_{p-1}\partial_p=0$)*

证明. 任取 $\sigma=[v_0,\dots,v_p]$, 计算

$$\begin{aligned}\partial^2\sigma&=\partial\left(\sum_{k=0}^{p}(-1)^k[v_0,\dots,\hat{v}_k,\dots,v_p]\right)\\&=\sum_{j<k}(-1)^k(-1)^j[\dots,\hat{v}_j,\dots,\hat{v}_k,\dots]+\sum_{j>k}(-1)^k(-1)^{j-1}[\dots,\hat{v}_k,\dots,\hat{v}_j,\dots]\\&=\sum_{j<k}(-1)^{k+j}[\dots,\hat{v}_j,\dots,\hat{v}_k,\dots]+\sum_{j<k}(-1)^{k+j-1}[\dots,\hat{v}_j,\dots,\hat{v}_k,\dots]\\&=0.\end{aligned}$$

于是命题得证. □

定义 1.15.3 (同调群)**.** 对链复形 $\mathcal{C}(K)=\{(C_p(K),\partial_p\},\quad\xrightarrow{\partial_{p+1}}C_p(K)\xrightarrow{\partial_p}C_{p-1}(K)\xrightarrow{\partial_{p-1}}\cdots\xrightarrow{\partial_2}C_1(K)\to$ 对 $\partial_p: C_p(K)\to C_{p-1}(K)$, 称

(1) $Z_p(K):=\ker\partial_p=\{c_p\in C_p(K):\partial c_p=0\}$ 是 K 的 p **维闭链群**;

(2) $B_p(K):=\operatorname{im}\partial_{p+1}=\{\partial c_{p+1}\in C_p(K):c_{p+1}\in C_{p+1}(K)\}$ 是 K 的 p **维边缘链群**.

满足 $B_p(K)\triangleleft Z_p(K)$, 且 $B_p(K)$ 和 $Z_p(K)$ 仍是自由 *Abel* 群. 称

$$H_p(K)=\ker\partial_p/\operatorname{im}\partial_{p+1}=Z_p(K)/B_p(K)$$

是 K 的p **阶同调群**.

在 $Z_p(K)$ 上可以定义等价关系:

$$\begin{aligned}z_p\sim z_p'&\iff z_p+B_p(K)=z_p'+B_p(K)\\&\iff z_p-z_p'+B_p(K)=0\iff\exists d\in C_{p+1}(K)\,(\partial d=z_p-z_p'),\end{aligned}$$

这一等价关系称作**同调关系**, 确定的等价类 $\bar{z}_p$ 称作**同调类**. 它可以不局限于闭链群, 例如在 $C_p(K)$ 上同样可以定义

$$c_p\sim c_p'\iff\exists d\in C_{p+1}(K)\,(\partial d=c_p-c_p').$$

特别地, 若 $\partial d=c_p$, 则 $c_p\sim 0$.

例 1.15.1. 考虑以下的 2 维复形, 约定每一个 2 维单形的定向都是逆时针.

取

$$z_1 = [v_0, v_1] + [v_1, v_2] + [v_2, v_0] \in C_1(K)$$
$$z_2 = [v_3, v_4] + [v_4, v_5] + [v_5, v_3] \in C_1(K),$$

计算可得

$$\partial z_1 = \partial z_2 = 0,$$

于是 $z_1, z_2 \in Z_1(K)$. 而

$$\partial\left(\sum_{\sigma \in K^{(2)} \setminus K^{(1)}} \sigma\right) = z_1 - z_2,$$

于是由同调关系的定义可知 $z_1 \sim z_2$.

将 ∂_p 限制到 $\operatorname{im}\partial_p$ 上之后可知

$$\partial_p : C_p(K) \twoheadrightarrow \operatorname{im}\partial_p = B_{p-1}(K)$$

是一个满同态. 于是由同态基本定理可知 $C_p(K)/\ker\partial_p \simeq \operatorname{im}\partial_p$.

对复形 $K, v, w \in K^{(0)}$, 则 $v, w \in C_0(K)$. 于是由同调关系的定义可知

$$v \sim w \Longleftrightarrow \exists d \in C_1(K)\,(\partial d = w - v).$$

在直观上, $v \sim w$ 即是存在一列 $\{v_i\}_{i=0}^n$ 满足 $v_0 = v, v_n = w$ 使得 v 与 w 可以经由 $[v_i, v_{i+1}]$ 相连.

于是同调关系可以对 $K^{(0)}$ 分类, 以 $\bar{v} := \left\{w \sim v : w \in K^{(0)}\right\}$ 记 v 所在的等价类, 则

$$K = \bigcup_{\bar{v}\in K^{(0)}/\sim} K_{\bar{v}} = \bigcup_{\bar{v}\in K^{(0)}/\sim} \{(v_0, \ldots, v_p) \in K : \{v_0, \ldots, v_p\} \subset \bar{v}\}.$$

进一步, 其几何实现

$$|K| = \bigcup_{\bar{v}\in K^{(0)}/\sim} |K_{\bar{v}}|,$$

且 $|K_{\bar{v}}|$ 是道路连通的. 此时也称 $K_{\bar{v}}$ 是道路连通的.

定理 1.15.2. *对复形 K, $H_p(K) = \bigoplus_{\bar{v}\in K^{(0)}/\sim} H_p(K_{\bar{v}})$, 即计算同调群只需分别计算各个道路连通分支的同调群即可.*

证明. 首先 $C_p(K) = \bigoplus_{\bar{v}\in K^{(0)}/\sim} C_p(K_{\bar{v}})$, 只需证明

$$Z_p(K) = \bigoplus_{\bar{v}\in K^{(0)}/\sim} Z_p(K_{\bar{v}}), \qquad B_p(K) = \bigoplus_{\bar{v}\in K^{(0)}/\sim} B_p(K_{\bar{v}})$$

即可. 注意到

$$z_p \in Z_p(K) \triangleleft C_p(K) = \bigoplus_{\bar{v}\in K^{(0)}/\sim} C_p(K_{\bar{v}}),$$

这导出 $z_p = \sum_{\bar{v}\in K^{(0)}/\sim} c_p^{\bar{v}}$. 于是

$$0 = \partial z_p = \sum_{\bar{v}\in K^{(0)}/\sim} \partial c_p^{\bar{v}}$$

导出 $\partial c_p^{\bar{v}} = 0$. $B_p(K)$ 的情形类似可证. □

让我们做些计算! 设 X 是一个拓扑空间, $H_0(X)$ 是其 0 阶同调群. 取 $C_0(X)$, 它就是 X 中点的形式和. 考虑边缘函数

$$\partial : C_0(X) \to 0$$

的核, $\ker\partial$ 即为整个空间 $C_0(X)$, 这意味着每一个点都是一个 "环". 而 $[b] - [a] = 0$ 当且仅当存在一个 1 维链连结 a 和 b, 于是这自然导向道路连通空间.

命题 1.15.2. *设 K 是道路连通复形, 则 $H_0(K) = \mathbb{Z}$.*

图 1.7: 可定向的 2 维闭曲面

证明. 设 $v \in K^{(0)}$, 则 $\forall w \in K^{(0)}$, 因 K 是道路连通的, 于是

$$v \sim w \Longleftrightarrow v + B_0(K) = \{v\} = \{w\} = w + B_0(K),$$

则对任意 $\alpha = c_0 + B_0(K) \in H_0(K)$, 都有

$$c_0 + B_0(K) = \sum_{v \in K^{(0)}} n_v v + B_0(K) \overset{\text{固定} v'}{=\!=} nv' + B_0(K) = n\left\{v'\right\},$$

故 $H_0(K) = \mathbb{Z}_{\{v'\}} = \{n\{v'\} : n \in \mathbb{Z}\}$. 断言 $\{v'\}$ 是一个无限阶元.

为此, 令 $n\{v'\} = 0$, 也即 $nv' + B_0(K) = 0$, 则

$$\exists c_1 \in C_1(K)\,(\partial c_1 = nv'),$$

另一方面

$$c_1 = \sum n_{ij}[v_i, v_j], \qquad \partial[v_i, v_j] = v_j - v_i,$$

故 $\partial c_1 = \sum\limits_{v \in K^{(0)}} n_v v$ 中的系数和 $\sum\limits_{v \in K^{(0)}} n_v = 0$, 又由 $\partial c_1 = \sum\limits_{v \in K^{(0)}} n_v v = nv'$ 可知

$$n_v = \begin{cases} n, & v = v' \\ 0, & v \neq v' \end{cases}$$

而系数和为零, 于是只能 $n = 0$, 此即 $H_0(K) = \mathbb{Z}$. □

推论 1.15.1. $|K|$ 是道路连通的当且仅当 $H_0(K) = \mathbb{Z}$.

1.16　2 维闭曲面的同调群

例 1.16.1 (2 维闭曲面的代数表示)**.** 2 维闭曲面一共两类, 一类是可定向的:
其中 $\sharp$ 表示连通和. 它在两个流形中各挖去一个球, 并将球面边界粘贴在一起得到. 而不可定向的一类为

$$\mathbb{RP}^2,\ \mathbb{RP}^2\sharp\mathbb{RP}^2,\ \cdots,\ \mathbb{RP}^2\sharp\cdots\sharp\mathbb{RP}^2,\cdots,$$

其中 *Klein* 瓶实际上就是 $\mathbb{RP}^2\sharp\mathbb{RP}^2$. 由此, 2 维闭曲面的全体可以看作代数系

$$\left\langle \mathbb{S}^2, \mathbb{T}^2, \mathbb{RP}^2; \sharp \right\rangle.$$

图 1.8: 一个 $\mathbb{RP}^2$ 和 $\mathbb{RP}^2$ 作连通和的例子

(当然, 这里需要证明 $\sharp$ 的确是一个代数运算. 首先对于可定向曲面, 亏格 g 就是它的拓扑不变量. 对于不可定向曲面, 无论做了多少次连通和, 它永远可以通过连续变形构成一个定向曲面和一个 Möbius 带的形式.)

为了应用所谓的 push off 方法, 我们约定所有的 2 维单形定向均为逆时针定向. 这有利于我们将复形推到它的边界上进行讨论.

引理 1.16.1. *设 K 是一个闭曲面, 则*

(1) 对 $z_1 \in Z_1(K)$, 存在 $z_1' \in Z_1(\mathrm{Bd}\, K)$ 使得 $z_1 \sim z_1'$;

(2) 对 $d \in Z_2(K)$ 使得 $\partial d \in C_1(\mathrm{Bd}\, K)$, 有 $d = l\left(\sum\limits_{\dim \sigma = 2} \sigma\right)$.

证明. (1) 因 $z_1 \in Z_1(K) \lhd C_1(K)$, $z_1 = \sum n_{ij}[v_i, v_j]$, 若一个非边界的 $[v_i, v_j]$ 使得 $n_{ij} \neq 0$, 则可以取到一个使得 $[v_i, v_j]$ 系数为负的 2 维复形 σ_1:

此时

$$\partial(n_{ij}\sigma_1) = n_{ij}\partial\sigma_1 = -n_{ij}[v_i, v_j] + \cdots,$$

注意到取边界并不影响 $B_1(K)$, 于是

$$z_1 + B_1(K) = z_1 + \partial(n_{ij}\sigma_1) + B_1(K) = \cdots = z_1' + B_1(K)$$

使得 $z_1 \in Z_1(\mathrm{Bd}\, K)$.

(2) 由 $d \in Z_2(K)$ 可知

$$d = \sum_{\dim \sigma = 2} n_\sigma \sigma, \qquad \partial d \in Z_1(\mathrm{Bd}\, K),$$

由此, 考虑非边界上的 1 维边, 如下图所示

于是对粉色边, 应当有 $n_\sigma = n_{\sigma'}$, 再对蓝色边讨论, 有 $n_{\sigma'} = n_{\sigma''}$, 由此进行下去可知 d 中所有的 n_σ 都相同. □

例 1.16.2 ($\mathbb{T}^2$ 的同调群)**.** *考虑 $\mathbb{T}^2$ 的三角剖分*

对 $p < 0$ 和 $p > 2$ 自然有 $H_p(\mathbb{T}^2) = 0$.

因 $\mathbb{T}^2$ 道路连通, 由命题 1.15.2 可知 $H_0(\mathbb{T}^2) = \mathbb{Z}$, 于是只需要计算 $H_1(\mathbb{T}^2)$ 和 $H_2(\mathbb{T}^2)$.

对 $\alpha \in H_1(\mathbb{T}^2)$, 有

$$\alpha = z_1 + B_1(K) = \{z_1\},$$

由引理 *1.16.1* 的 *(1)* 可知存在 $z_1' \in Z_1(\mathrm{Bd}\, K)$ 使得 $z_1 \sim z_1'$. 而

$$\mathrm{Bd}\, K = \{[a,b],[b,c],[c,a],[a,d],[d,e],[e,a]\}.$$

设

$$z_1' = \sum_{i,j} n_{i,j}[i,j],$$

令

$$\begin{aligned} 0 &= \partial z_1' \\ &= (-n_{ab} + n_{ca} - n_{ad} + n_{ea})a + (n_{ab} - b_{bc})b + (n_{bc} - n_{ca})c + (n_{ad} - n_{de})d + (n_{de} - n_{ea})e, \end{aligned}$$

即

$$n_{ab} = n_{bc} = n_{ca} = n, \qquad n_{ad} = n_{de} = n_{ea} = m.$$

为此, 记 $w = [a,b] + [b,c] + [c,a]$, $z = [a,d] + [d,e] + [e,a]$, 则

$$\{z_1\} = \left\{z_1'\right\} = \{nw + mz\} = nw + mz + B_1(K) = n\{w\} + m\{z\},$$

故 $H_1(\mathbb{T}^2) = \mathbb{Z}_{\{w\}} + \mathbb{Z}_{\{z\}}$. 下面证明这一和为直和, 且 $\{w\}$ 和 $\{z\}$ 都是无限阶元.

令 $n\{w\} + m\{z\} = 0$, 则

$$\begin{aligned} nw + mz + B_1(K) = 0 &\Longleftrightarrow \exists d \in C_2(K)\,(\partial d = nw + mz \in Z_1(\mathrm{Bd}\, K)) \\ &\Longrightarrow d = l\Big(\sum_{\dim\sigma=2} \sigma\Big). \end{aligned}$$

其中第二行的推出由引理 *1.16.1* 的 *(2)* 得到. 此时 $\partial\left(\sum\limits_{\dim\sigma=2} \sigma\right) = 0$ *(*在三角剖分图中标注出定向即可看出*)*, 即 $\partial d = 0$, 从而 $nw + mz = 0 \in Z_1(\mathrm{Bd}\, K)$, 且 $n = m = 0$. 故

$$H_1(\mathbb{T}^2) = \mathbb{Z} \oplus \mathbb{Z}.$$

再计算 $H_2(\mathbb{T}^2)$, 注意到 $C_3(\mathbb{T}^2) = 0$, 故 $B_2(\mathbb{T}^2) = 0$, 故 $H_2(\mathbb{T}^2) = Z_2(\mathbb{T}^2)$. 对 $\alpha = z_2 + B_2(K) = z_2 \in Z_2(\mathbb{T}^2)$, 有

$$\begin{aligned} \partial z_2 = 0 &\Longrightarrow \partial z_2 \in Z_1(\mathrm{Bd}\, K) \\ &\Longrightarrow z_2 = l\Big(\sum_{\dim\sigma=2} \sigma\Big) \\ &\Longrightarrow H_2(\mathbb{T}^2) = \mathbb{Z}_{\left(\sum\limits_{\dim\sigma=2} \sigma\right)} \cong \mathbb{Z}. \end{aligned}$$

同样地, 第二行的推出由引理 *1.16.1* 的 *(2)* 得到.

由此, 我们得到环面 $\mathbb{T}^2$ 的同调群如下:

$$H_p(\mathbb{T}^2) = \begin{cases} \mathbb{Z}, & p = 0, 2 \\ \mathbb{Z} \oplus \mathbb{Z}, & p = 1 \\ 0, & \text{其他} \end{cases}$$

例 1.16.3 (Klein 瓶的同调群). *Klein* 瓶 S 的三角剖分如下图所示:

类似地记

$$w=[a,b]+[b,c]+[c,a],\qquad z=[a,d]+[d,e]+[e,a],$$

类似于 $\mathbb{T}^2$ 的计算方法, 对 $\{z_1\}=z_1+B_1(S)=z_1'+B_1(S)$, 其中 $z_1'\in Z_1(\mathrm{Bd}\,S)$, 有

$$\{z_1'\}=mw+nz+B_1(S)=m\{w\}+n\{z\},$$

也即 $H_1(S)=\mathbb{Z}_{\{w\}}+\mathbb{Z}_{\{z\}}$.

令 $m\{w\}+n\{z\}=0$, 则

$$\begin{aligned}mw+nz+B_1(S)=0&\Longleftrightarrow\exists d\in C_2(S)\,(\partial d=mw+nz\in Z_1(S))\\&\Longrightarrow d=l\Big(\sum_{\dim\sigma=2}\sigma\Big),\end{aligned}$$

此时 $\partial d=2lz$, 记 $2lz=mw+nz=0$, 这意味着 $\{z\}$ 是 2 阶元. 从而

$$H_1(S)=\mathbb{Z}\oplus(\mathbb{Z}/2\mathbb{Z}).$$

2 阶同调群的情形, 我们只需要注意到对唯一的 2 维单形 d 有 $\partial d\neq 0$ 即可得到 $H_2(S)=0$.

由此, 我们得到 *Klein* 瓶的同调群如下:

$$H_p(S)=\begin{cases}\mathbb{Z}, & p=0\\ \mathbb{Z}\oplus(\mathbb{Z}/2\mathbb{Z}), & p=1\\ 0, & \text{其他}\end{cases}$$

1.17　约化同调

对单纯复形 K 的链复形 $\xrightarrow{\partial_{p+1}}C_p(K)\xrightarrow{\partial_p}C_{p-1}(K)\xrightarrow{\partial_{p-1}}\cdots\xrightarrow{\partial_2}C_1(K)\xrightarrow{\partial_1}C_0(K)\xrightarrow{\partial_0}$ 它有增广链复形 $\xrightarrow{\partial}C_p(K)\xrightarrow{\partial}\cdots\xrightarrow{\partial}C_1(K)\xrightarrow{\partial}C_0(K)\xrightarrow{\varepsilon}\mathbb{Z}\longrightarrow 0$ 为了使得 $\varepsilon\circ\partial=0$, 只需要将 $v\in C_0(K)$ 映到 $\varepsilon(v)=1$ 即可. 这由

$$\varepsilon\partial[v_1,v_0]=\varepsilon(v_1)-\varepsilon(v_0)=0$$

即可验证. 而对抽象复形 K, 由于 $\{n\emptyset:n\in\mathbb{Z}\}=C_{-1}(K)\cong\mathbb{Z}$ 的存在, 它与几何复形的增广链复形相同.

定义 1.17.1 (约化同调). 称增广链复形导出的同调群 $\tilde{H}_i(K)$ 是**约化同调群**, 当 $i>0$ 时

$$\tilde{H}_i(K)=H_i(K),$$

而 $i=0$ 时 $\tilde{H}_0(K)=\ker\varepsilon/\mathrm{im}\,\partial_1$, 这里 $\ker\varepsilon\leqslant C_0(K)$.

命题 1.17.1. $H_0(K)\cong\tilde{H}_0(K)\oplus\mathbb{Z}$.

证明. 设 $K=\coprod_i K_i$, 其中 K_i 是 K 的道路连通分支. 任取 $v_i\in K_i^{(0)}$, 有

$$H_0(K)\cong\bigoplus_i\mathbb{Z}_{\{v_i\}}\cong\bigoplus_i\mathbb{Z}.$$

考虑约化同调群 $\tilde{H}_0(K)$, 设 $\{z_0\}=z_0+\operatorname{im}\partial_1\in\tilde{H}_0(K)$, 其中 $z_0\in\ker\varepsilon$.

将 $z_0+\operatorname{im}\partial_1$ 在 $C_0(K)$ 中观察得到

$$z_0+\operatorname{im}\partial_1=\sum_i n_iv_i+\operatorname{im}\partial_1,$$

在 $\ker\varepsilon$ 中观察, 得到

$$0=\varepsilon z_0=\sum_i n_i\Longrightarrow n_0=-\sum_{i\neq0}n_i,$$

由此

$$\tilde{H}_0(K)=\bigoplus_{i\neq0}\mathbb{Z}_{\{v_i-v_0\}}.$$

令 $\sum\limits_{i\neq0}n_i\{v_i-v_0\}=0\in\tilde{H}_0(K)$, 即 $\sum\limits_{i\neq0}n_i(v_i-v_0)+\operatorname{im}\partial_1$, 故存在 $d=\sum n_{uv}[u,v]\in C_1(K)$ 使得

$$\sum n_vv=\sum_{i\neq0}n_i(v_i-v_0).$$

这里 $\sum n_v=0$, 因注意到 $\partial n_{uv}[u,v]=n_{uv}v-n_{uv}u$ 中点的系数总是一正一负交替出现, 比较上式两侧系数可知 $n_i=0$. □

定义 1.17.2 (锥). 设 K 是一个复形, $v\in\mathbb{R}^{\mathbb{N}}$. 对任意 $\sigma=(v_0,\dots,v_p)\in K$, 若 $v,v_0,\dots,v_p$ 几何独立, 则它确定一个单形

$$v*\sigma:=(v,v_0,\dots,v_p),$$

它的定向由 $[v,\sigma]:=[v,v_0,\dots,v_p]$ 确定. 由此得到一个复形

$$v*K=K\cup\{v\}\cup\bigcup_{\sigma\in K}v*\sigma,$$

称作在 v 处的**锥**.

从几何上来看, 容易得到 $|v*K|\cong\mathrm{pt}$.

引理 1.17.1. 设 c 是 K 的单形, 则

$$\partial[v,c]=\begin{cases}c-[v,\partial c], & \dim c>0\\ c-\varepsilon(c)v, & \dim c=0\end{cases}$$

证明. 计算 $v*K$ 的约化同调 $\tilde{H}_p(v*K)$, 考虑

$$c_{p+1}=c_1+c_2\in C_{p+1}(K),\qquad c_1\in C_{p+1}(K),\quad c_2=\sum_{\sigma\in K,\dim\sigma=p}n_\sigma(v*\sigma)=\left[v,\sum_{\sigma\in K,\dim\sigma=p}n_\sigma\sigma\right].$$

图 1.9: 一个 2 维复形 K 与点 v 形成的锥

对 $p>0$, 有

$$\begin{aligned}\partial[v,v_0,\dots,v_p]&=[v_0,\dots,v_p]+\sum_{i=0}^{p}(-1)^{i+1}[v,v_0,\dots,\hat{v}_i,\dots,v_p]\\&=[v_0,\dots,v_p]-\sum_{i=0}^{p}(-1)^{i}[v,v_0,\dots,\hat{v}_i,\dots,v_p]\\&=[v_0,\dots,v_p]-\left[v,\sum_{i=0}^{p}[v,v_0,\dots,\hat{v}_i,\dots,v_p]\right]\\&=\sigma-[v,\partial\sigma].\end{aligned}$$

对 $p=0$, 有 $\partial[v,v_0]=v_0-v$. □

定理 1.17.1. 锥 $v*K$ 是零调的, 即 $\tilde{H}_p(v*K)=0$.

证明. 对 $p\leqslant 0$ 和 $p>\dim K$, 显然 $\tilde{H}_p(K)=0$.

对 $1\leqslant p\leqslant \dim K$, 取 $z\in C_p(v*K)$ 满足 $\partial z=0$, 即 $z\in Z_p(v*K)$. 记 $z=c_1+c_2$, 其中 $c_1\in C_p(K)$ 被 K 承载, $c_2=[v,c_2']$ 被 $v*K$ 承载. 计算

$$0=\partial z=\partial c_1+\partial c_2=\partial c_1+\partial[v,c_2']=\begin{cases}\partial c_1+c_2'-[v,\partial c_2'], & p>1\\ \partial c_1+c_2'-\varepsilon(c_2')v, & p=1\end{cases}$$

即 $\partial c_1+c_2'=0$, 且 $[v,\partial c_2']=0$, $\varepsilon(c_2')v=0$, 导出 $c_2'=-\partial c_1$. 故

$$z=c_1+[v,c_2']=c_1+[v,-\partial c_1]=c_1-[v,\partial c_1]=\partial[v,c_1]\in B_p(v*K).$$

从而 $Z_p(v*K)=B_p(v*K)$, 得到

$$\tilde{H}_p(v*K)=\frac{Z_p(v*K)}{B_p(v*K)}=0.$$ □

例 1.17.1 ($\mathbb{B}^n$, $\mathbb{S}^n$ 的同调群)**.** 我们需要承认同调群是拓扑不变量. 考虑 $\mathbb{B}^n\cong|K_{\Delta^n}|=\Delta^n=(v_0,\dots,v_n)$, 将 K_{Δ^n} 看作是 $\left|v_0*K_{(v_1,\dots,v_n)}\right|$, 它是凸的. 于是 $\tilde{H}_p(\mathbb{B}^n)=0$.

再考虑 $\mathbb{S}^n$ 的同调群: 因

$$\mathbb{S}^n=\left|\operatorname{Bd}K_{\Delta^{n+1}}\right|,\quad \mathbb{B}^{n+1}=\left|K_{\Delta^{n+1}}\right|,\quad \operatorname{Bd}K_{\Delta^{n+1}}=K_{\partial\Delta^{n+1}}=K_{\Delta^{n+1}}\setminus\left\{\Delta^{n+1}\right\},$$

故 $C_{n+1}(K_{\Delta^{n+1}})\cong\mathbb{Z}$. 由此得链复形

$$\begin{array}{ccccccccccc}
 & & & & \mathbb{Z} & & & & & & \\
 & & & & \cong\uparrow & & & & & & \\
0 & \longrightarrow & C_{n+1}(K_{\Delta^{n+1}}) & \xrightarrow{\partial} & C_n(K_{\Delta^{n+1}}) & \xrightarrow{\partial} & \cdots & \xrightarrow{\partial} & C_1(K_{\Delta^{n+1}}) & \longrightarrow & 0\\
 & & & & \| & & & & \| & & \\
 & & 0 & \longrightarrow & C_n(K_{\partial\Delta^{n+1}}) & \xrightarrow{\partial} & \cdots & \xrightarrow{\partial} & C_1(K_{\partial\Delta^{n+1}}) & \longrightarrow & 0
\end{array}$$

于是 $p\neq n$ 时有 $\tilde{H}_p(\mathbb{S}^n)=H_p(\mathbb{B}^{n+1})$, 当 $p=n$ 时有

$$H_n(\mathbb{S}^n)=H_n(K_{\partial\Delta^{n+1}})=\ker\partial_n=\operatorname{im}\partial_{n+1}=\left\{l\partial\Delta^{n+1}:l\in\mathbb{Z}\right\}\cong\mathbb{Z},$$

而

$$H_p(\mathbb{B}^{n+1})=\frac{\ker\partial_p}{\operatorname{im}\partial_{p+1}}=0,$$

于是

$$H_p(\mathbb{S}^n)=\begin{cases}\mathbb{Z}, & p=0,n\\ 0, & \text{其他}\end{cases}$$

1.18　单纯映射诱导的链映射

定义 1.18.1 ((诱导) 链映射). 设 $f:K\to L$ 是单纯映射, 则 $f:\sigma\mapsto f(\sigma)$ 诱导

$$f_\sharp:C_p(K)\to C_p(L),\qquad \sigma=[v_0,\dots,v_p]\mapsto f_\sharp(\sigma)=\begin{cases}[f(v_0),\dots,f(v_p)], & f(v_i)\text{ 两两不同}\\ 0, & \text{其他}\end{cases}$$

称 $f_\sharp$ 是 f **诱导的链映射**.

引理 1.18.1. 设 $f_\sharp$ 是 f 诱导的链映射, 则对任何 $p\in\mathbb{Z}$, $\begin{array}{ccc} C_p(K) & \xrightarrow{f_\sharp} & C_p(L)\\ {\scriptstyle\partial}\downarrow & & \downarrow{\scriptstyle\partial}\\ C_{p-1}(K) & \xrightarrow{f_\sharp} & C_{p-1}(L)\end{array}$ 交换.

证明. 即证对任意 $\sigma=[v_0,\dots,v_p]$, 有 $\partial f_\sharp(\sigma)=f_\sharp(\partial\sigma)$. 由定义可知

$$\partial f_\sharp(\sigma)=\begin{cases}\tau=[f(v_0),\dots,f(v_p)], & \dim\tau=p\\ 0, & \dim\tau<p\end{cases}$$

而

$$f_\sharp(\partial\sigma)=f_\sharp\left(\sum_{i=0}^{p}(-1)^i[v_0,\dots,\hat{v}_i,\dots,v_p]\right)=\sum_{i=0}^{p}(-1)^i f_\sharp([v_0,\dots,\hat{v}_i,\dots,v_p]).$$

当 $\dim\tau=p$ 时易知 $\partial f_\sharp(\sigma)=f_\sharp(\partial\sigma)$, 若 $\dim\tau<p$, 分两类情况讨论:

若 $\dim\tau\leqslant p-2$, 则 $\{f(v_i)\}$ 中或者至少有三个相同, 或者至少有两对相同, 这意味着去掉一个顶点之后确定的 $f_\sharp([v_0,\dots,\hat{v}_i,\dots,v_p])=0$ 总是成立的. 此时 $\partial f_\sharp(\sigma)=f_\sharp(\partial\sigma)=0$.

若 $\dim\tau=p-1$, 不妨设 $f(v_0)=f(v_1)$, 此时 $\partial f_\sharp(\sigma)=0$, 而展开 $f_\sharp(\partial\sigma)$ 得到

$$f_\sharp(\partial\sigma)=[f(v_1),f(v_2),\dots,f(v_p)]-[f(v_0),f(v_2),\dots,f(v_p)]=0.$$

于是此时也有 $\partial f_\sharp(\sigma)=f_\sharp(\partial\sigma)=0$. □

命题 1.18.1. 对 $f_\sharp$, 它又可以诱导

$$f_*:H_p(K)\to H_p(L),\qquad z_1+B_p(K)\mapsto f_\sharp(z_1)+B_p(K),$$

且 $f_\sharp(Z_p(K))\leqslant Z_p(L)$, $f_\sharp(B_p(K))\leqslant B_p(L)$.

证明. 对任意 $z_p\in Z_p(K)$, 由

$$\partial f_\sharp(z_p)=f_\sharp(\partial z_p)=f_\sharp(0)=0$$

可知 $f_\sharp(z_p)\in Z_p(L)$; 对任意 $b_p\in B_p(K)$, 由 $\exists d\in C_{p+1}(K)$ 使得 $b_p=\partial d$,

$$f_\sharp(b_p)=f_\sharp(\partial d)=\partial f_\sharp(d)$$

可知 $f_\sharp(b_p)\in B_p(L)$. □

于是 $f\mapsto f_*$ 的确给出了从 SimComp 到 Grp 的一个函子: 只需要验证 $(gf)_\sharp=g_\sharp f_\sharp$ 即可. 对 $\sigma=[v_0,\dots,v_p]$, 记 $\tau=[f(v_0),\dots,f(v_p)]$, 则

$$g_\sharp f_\sharp(\sigma)=\begin{cases}g_\sharp(\tau), & \dim\tau=p\\ 0, & \dim\tau<p\end{cases}=\begin{cases}[gf(v_0),\dots,gf(v_p)], & \dim g_\sharp(\tau)=p\\ 0, & \dim g_\sharp(\tau)<p\end{cases}=(gf)_\sharp(\sigma).$$

推论 1.18.1. *若 K, L 都是单纯复形, $K \cong L$, 则 $H_p(K) \cong H_p(L)$.*

至此, 我们完成了 $\mathsf{Top} \xleftarrow{F} \mathsf{SimComp} \xrightarrow{G} \mathsf{Grp}$ 的刻画.

1.19　同调群的同伦不变性

原网页

1.20　单纯逼近定理

希望对拓扑空间 X, Y, 若 $X \cong Y$, 则有 $H_p(X) \cong H_p(Y)$. 对可以三角剖分的拓扑空间, 存在复形 K 使得 $X = |K|$. 于是希望 $|K| \cong |L|$ 时有 $H_p(|K|) \cong H_p(|L|)$. 一个自然的问题是如何定义 $H_p(|K|)$.

对任何 $|K|$ 的三角剖分 K', K'', 有 $|K'| = |K''|$ 成立, 于是定义 $H_p(|K|) = H_p(K)$ 即可.

考虑 $f \in C(|K|, |L|)$, 它未必是单纯映射. 我们希望找到单纯映射 $g : K \to L$ 使得 $\tilde{g} : |K| \to |L|$ 满足 $f = \tilde{g}$, 即寻找一个单纯映射用来 "代替" f.

定义 1.20.1 (单纯逼近)**.** *设 $f : C(|K|, |L|)$, 称单纯映射 g 是 f 的*单纯逼近*, 若对任意 $x \in |K|$ 都有*

$$\exists! \sigma \in K \, (x \in \operatorname{Int} \sigma) \wedge \exists \tau \in L \, (f(x) \in \operatorname{Int} \tau \wedge g(x) \in \tau).$$

即 $f(x)$ 与 $g(x)$ 落在 L 的同一个单形 τ 中.

命题 1.20.1. *设 g 是 $f \in C(|K|, |L|)$ 的单纯逼近, 仍记它诱导的连续映射为 g, 则 $f \simeq g$. 更具体地, 若 f 可以被单纯映射诱导, 则 $f = g$.*

证明. 由单纯映射的定义可知存在 $\tau \in L$ 使得 $f(x), g(x) \in \tau$, 而单形 τ 是一个凸集. 于是总存在同伦

$$H(x, t) = tf(x) + (1 - t)g(x).$$

任取 $v \in K^{(0)}$, 因 f 被单纯映射诱导, 于是 $f(v) \in L^{(0)}$. 而 g 是 f 的单纯逼近, 于是

$$\forall v \in K^{(0)} \, (f(v) = w = g(v) \in L^{(0)}),$$

这即 $f|_{K^{(0)}} = g|_{K^{(0)}} : K^{(0)} \to L^{(0)}$. 进而 $f = g$, 因从顶点出发可以一直诱导在高维骨架上的相等. □

然而并不是所有的连续映射都存在单纯逼近, 考虑下面的例子: 设 K 的几何实现为 $\mathbb{S}^1$, 而 L 的几何实现是一个平环 $\{z \in \mathbb{C} : 1 \leqslant |z| \leqslant 2\}$. 这因任何 $K \to L$ 的单纯映射必定落在 L 的一个 2–单形中. 下面的问题就是: 在什么条件下存在单纯逼近?

定义 1.20.2 (Star, Link)**.** *设 K 是一个复形, $v \in K^{(0)}$, 定义*

$$\operatorname{St}(v, K) = \bigcup_{\sigma \in K, v \in \sigma} \operatorname{Int} \sigma,$$

图 1.10: 一个 $\mathrm{Int}\,\sigma$ 不是 $|K|$ 中开集的例子

称作 v 在 K 中的 ***star***. 并记

$$\mathrm{Link}\,(v,K) := \overline{\mathrm{St}(v,K)} \setminus \mathrm{St}(v,K) = \bigcup_{v*\tau\in K} \tau,$$

称作 v 在 K 中的 ***link***.

命题 1.20.2. $\mathrm{St}(v,K)$ 是 $|K|$ 中的开集.

证明. 这因

$$|K| \setminus \mathrm{St}(v,K) = \bigcup_{\sigma\in K, v\notin\sigma} \sigma = L$$

是 $|K|$ 中的闭子集, 其中 $L = \{\sigma \in K : v \notin \sigma\}$ 是 K 的子复形. □

但 $\mathrm{Int}\,\sigma$ 在一般的情况下未必是 $|K|$ 中的开集. 而 link 就要更复杂一些, 下图给出了一个简单的 2 维复形中 link 的图示:

图 1.11: 一个 2 维复形中顶点的闭 star, star 和 link

命题 1.20.3. 设 $\{v_0,\dots,v_p\}\subset K^{(0)}$, 则 $\{v_0,\dots,v_p\}$ 落在某一单形中当且仅当 $\bigcap\limits_{i=0}^{p}\mathrm{St}(v_i,K)\neq\emptyset$.

证明. 必要性. 取 $\sigma=(v_0,\dots,v_p)\in K^{(0)}$, 则 $\forall i\in\{0,\dots,p\}$, 有

$$\mathrm{Int}\,\sigma\subset\mathrm{St}(v_i,K)\Longrightarrow\mathrm{Int}\,\sigma\subset\bigcap_{i=0}^{n}\mathrm{St}(v_i,K),$$

而 $\mathrm{Int}\,\sigma\neq\emptyset$.

充分性. 取 $x\in\bigcap\limits_{i=0}^{p}\mathrm{St}(v_i,K)\subset|K|=\bigcup\limits_{\sigma\in K}\mathrm{Int}\,\sigma$. 则存在 $\sigma\in K$ 使得 $x\in\mathrm{Int}\,\sigma$. 由 star 的定义, v_i 是 σ 的顶点, 即 $v_0,\dots,v_p$ 是 σ 中的顶点. □

定义 1.20.3 (Star 条件)**.** 设映射 $f\in C(|K|,|L|)$, 称 f 满足 ***star* 条件**, 若

$$\forall v\in K^{(0)}\,\exists w\in L^{(0)}\,(f(\mathrm{St}(v,K))\subset\mathrm{St}(w,L)).$$

命题 1.20.4 (用 star 条件刻画单纯逼近)**.** $f\in C(|K|,|L|)$ 存在单纯逼近当且仅当它满足 *star* 条件. 特别地, 若 g 是 f 的单纯逼近, 则对任意 $v\in K^{(0)}$, 有

$$f(\mathrm{St}(v,K))\subset\mathrm{St}(g(v),L).$$

证明. 必要性. 设 $g:K\to L$ 是 f 的单纯逼近, 只需证明

$$\forall v\in K^{(0)}\,(f(\mathrm{St}(v,K))\subset\mathrm{St}(g(v),L)).$$

任取 $x\in\mathrm{St}(v,K)$, 根据 star 的定义可知存在 $\sigma\in K$ 使得 $x\in\mathrm{Int}\,\sigma$ 且 $v\in\sigma$. 又因为 g 是 f 的单纯逼近, 故存在 $\tau\in L$ 使得 $f(x)\in\mathrm{Int}\,\tau$ 且 $g(x)\in\tau$. 因 g 是单纯映射, $g(\sigma)=\tau_1\in L$, 且 $g(v)\in g(\sigma)$. 断言 $g(x)\in\mathrm{Int}\,\tau_1$. 注意到对 $\sigma=(v_0,\dots,v_p)$,

$$x=\sum_{i=0}^{p}t_iv_i\Longrightarrow g(x)=\sum_{i=0}^{p}t_ig(v_i)\in\mathrm{Int}\,\tau_1,$$

(即使 $g(v_i)$ 中有相等项也没有关系), 于是断言成立. 因 $g(x)\in\tau$, 故 $\tau_1\subset\tau$. 而 $g(v)$ 是 τ_1 的顶点, 故它也是 τ 的顶点.

此时, $f(x)\in\mathrm{Int}\,\tau\subset\mathrm{St}(g(v),L)$, 也即 $f(\mathrm{St}(v,K))\subset\mathrm{St}(g(v),L)$.

充分性. 假设 f 满足 star 条件, 由这一条件可以定义映射 g

$$g:K^{(0)}\to L^{(0)},\qquad v\mapsto g(v)=w$$

使得 $f(\mathrm{St}(v,K))\subset\mathrm{St}(g(v),L)$. 下面将 g 扩张成一个单纯映射.

任取 $\sigma=(v_0,\dots,v_p)\in K$, 只需证明 $\{g(v_0),\dots,g(v_p)\}$ 落在 L 的同一单形中. 由 1.20.3 知 $\bigcap\limits_{i=0}^{p}\mathrm{St}(v_i,K)\neq\emptyset$. 故

$$\emptyset\neq f\left(\bigcap_{i=0}^{p}\mathrm{St}(v_i,K)\right)\subset\bigcap_{i=0}^{p}f(\mathrm{St}(v_i,K))\subset\bigcap_{i=0}^{p}\mathrm{St}(g(v_i),L).$$

再由 1.20.3 可知 $\{g(v_0),\dots,g(v_p)\}$ 落在 L 的某一单形中, 不妨记作 $\tau=:g(\sigma)$, 故 g 是单纯映射.

再证 g 是 f 的单纯逼近, 首先

$$\forall x \in |K| = \bigcup_{\sigma\in K} \operatorname{Int}\sigma \, \exists! \sigma = (v_0, \ldots, v_p) \in K \, \forall i \, (x \in \operatorname{Int}\sigma \subset \operatorname{St}(v_i, K)).$$

由题设和 star 的定义, 因 $f(x) \in \operatorname{St}(g(v_i), L)$, 有

$$\exists \tau \in L \, (f(x) \in \operatorname{Int}\tau),$$

此时可知 $g(v_i) \in \tau$. 对 $x = \sum_{i=0}^{p} t_i v_i$, 都有 $g(x) = \sum_{i=0}^{p} t_i g(v_i) \in \tau$. 故 g 是 f 的一个单纯逼近. □

单纯逼近不能解决所有问题, 例如之前的例子 中 f 不存在单纯逼近. 此时就需要另外一种工具: 重分.

定义 1.20.4 (重分)**.** *称 K' 是 K 的一个*重分*, 若*

(1) $\forall \tau \in K' \exists \sigma \in K \, (\tau \subset \sigma)$*, 即重分只能分出更小的单形.*

(2) $\forall \sigma \in K \, \exists \tau_1, \ldots, \tau_r \in K' \, (\sigma = \bigcup_{i=1}^{r} \tau_i)$*, 即重分都是有限分割.*

注意到在 $\mathbb{R}^{\mathbb{N}}$ *中* $|K'| = |K|$*, 于是重分实际上只是将 star 变小了.*

定理 1.20.1 (单纯逼近定理)**.** *对任意的* $f \in C(|K|, |L|)$*, 存在 K 的重分 K' 使得* $f \in C(|K'|, |L|)$ *存在单纯逼近* $g : K' \to L$*.*

这是本节的中心定理, 我们只处理 K 是有限复形的情形. 但即使如此, 我们也需要先进行以下的准备工作, 再回来处理这一定理的证明.

引理 1.20.1. *任取* $w \in K^{(0)}$*, 存在* $v \in K$ *使得* $\operatorname{St}(w, K) \subset \operatorname{St}(v, K)$*.*

命题 1.20.5. $\mathrm{id} : |K'| \to |K|$ *存在单纯逼近* $g : K' \to K$*.*

定义 1.20.5 (标准重心重分)**.** *对* $\sigma = (v_0, \ldots, v_p)$*, 其*重心*定义为*

$$\hat{\sigma} = \frac{v_0 + \cdots + v_p}{p+1}.$$

归纳地定义: $p = 0$ 时, 在 $K^{(0)}$ 上 $L^{(0)} = K^{(0)}$; $p = 1$ 时, 在 $K^{(1)}$ 上

$$L^{(1)} = \left\{ \hat{\sigma} * L_\sigma : \sigma \in K^{(1)}, \dim\sigma = 1, |L_\sigma| = |\operatorname{Bd}\sigma| \right\},$$

归纳地, 当 $p = k+1$ 时, 在 $K^{(k+1)}$ 上

$$L^{(k+1)} := \left\{ \hat{\sigma} * L_\sigma : \sigma \in K^{(k+1)}, \dim\sigma = k+1, L_\sigma \subset L^{(k)} \, (|L_\sigma| = |\operatorname{Bd}\sigma|) \right\},$$

图 1.12: 一个 2 维单形构成的复形及其标准重心重分

并称 $\operatorname{sd} K = \bigcup_{p\geqslant 0} L^{(p)}$ 为 K 的**标准重心重分**. 类似地记 $\operatorname{sd}^2 = \operatorname{sd}(\operatorname{sd} K)$.

也可以这样定义标准重心重分:

称 K 中的一列 $\sigma_1 \subset \sigma_2 \subset \cdots \subset \sigma_r$ 是一个***flag***, 它确定单形 $(\hat{\sigma}_1, \ldots, \hat{\sigma}_r)$ (注意到它也可以唯一地确定 *flag*), 定义

$$\operatorname{sd} K := \{(\hat{\sigma}_1, \ldots, \hat{\sigma}_r) : \sigma_1 \subset \cdots \subset \sigma_r\} = \{\sigma_1 \subset \cdots \subset \sigma_r\}.$$

定义 1.20.6 (网径)**.** 当 K 是有限复形时, 定义 K 的**网径**为

$$\operatorname{mesh} K := \max_{\sigma\in K} \operatorname{diam}\sigma = \max_{\sigma\in K}\max_{x,y\in\sigma} \|x-y\|.$$

那么 *star* 自然就有这样的性质: 对任意 $v \in K^{(0)}$ 有 $\operatorname{St}(v, K) \subset B(v, \operatorname{mesh} K)$.

直观上, 标准重心重分减小了网径, 即 $\operatorname{mesh}\operatorname{sd} K < \operatorname{mesh} K$, 下面的引理给出了这一结论的一个估计:

引理 1.20.2. 设 K 是有限复形, 则

$$\operatorname{mesh}(\operatorname{sd} K) \leqslant \frac{n}{n+1}\operatorname{mesh} K,$$

进而 $m \to \infty$ 时 $\operatorname{mesh}(\operatorname{sd}^m K) \to 0$.

证明. 对 $\sigma = (v_0, \ldots, v_p) \in K$, 有 $\operatorname{diam}\sigma = \max_{i,j} \|v_i - v_j\|$. 这因 $\forall x, y \in \sigma$, 记 $y = \sum_{i=0}^{p} t_i v_i$ 后

$$\|x-y\| \leqslant \left\|x - \sum_{i=0}^{p} t_i v_i\right\| = \left\|\sum_{i=0}^{p} t_i (x - v_i)\right\| \leqslant \max_i \|x - v_i\|,$$

类似地 $\|v_i - x\| \leqslant \max_j \|v_i - v_j\|$.

任取 $\tau, \sigma \in \operatorname{sd} K, \tau \subset \sigma$. 则 $\tau \subset \sigma \in K$. 设 $\sigma = (v_0, \ldots, v_p)$, 则存在 $i_0 \in \{0, \ldots, p\}$ 使得

$$\|\hat{\sigma} - \hat{\tau}\| \leqslant \|\hat{\sigma} - v_{i_0}\| = \left\|\frac{1}{p+1}\sum_{i=0}^{p} v_i - v_{i_0}\right\| = \left\|\sum_{i=0}^{p} \frac{v_i - v_{i_0}}{p+1}\right\| \leqslant \frac{p}{p+1}\operatorname{mesh} K.$$

而 $p \leqslant n$ 时总有 $p/(p+1) \leqslant n/(n+1)$. 得证. □

现在我们回到单纯逼近定理 1.20.1 的证明:

定理 1.20.1 的证明. 若 K 是有限复形, 则 $|K|$ 一定是紧度量空间. 由 Lebesgue 覆盖引理可知对任何开覆盖 $\mathcal{A}$, 都存在 Lebesgue 数 $\delta > 0$ 使得

$$\forall x \in |K|\ \exists C \in \mathcal{A}\,(B(x,\delta) \subset C).$$

对 $f : |K| \to |L|$, 考虑 $|L|$ 的开覆盖 $\left\{\mathrm{St}(w,L) : w \in L^{(0)}\right\}$, 存在 K 的开覆盖

$$\mathcal{A} = \left\{f^{-1}(\mathrm{St}(w,L)) : w \in L^{(0)}\right\},$$

由 Lebesgue 覆盖引理, 存在 $\delta > 0$ 使得

$$\forall x \in |K|\ \exists w \in L^{(0)}\,(B(x,\delta) \subset f^{-1}(\mathrm{St}(w,L))).$$

由 1.20.2 可知存在正整数 m 使得 $\mu = \mathrm{mesh}(\mathrm{sd}^m K) < \delta$. 任取 $v \in (\mathrm{sd}^m K)^{(0)}$, 存在 $w \in L^{(0)}$ 使得

$$\mathrm{St}(v, \mathrm{sd}^m K) \subset B(v,\mu) \subset B(v,\delta) \subset f^{-1}(\mathrm{St}(w,L)).$$

也即 $f(\mathrm{St}(v, \mathrm{sd}^m K)) \subset \mathrm{St}(w,L)$. 这即 star 条件, 于是 f 存在单纯逼近 $\varphi : \mathrm{sd}^m K \to L$. □

1.21　链同伦

在上一小节我们知道了对复形进行适当次数的标准重心重分, 就可以得到连续映射的单纯逼近. 还有两个问题需要考虑: 其一是不同的单纯逼近是否诱导相同的同态; 其二是对不同的重分 K', K'', 它们的同调群 $H_p(K')$ 和 $H_p(K'')$ 之间是否存在同态.

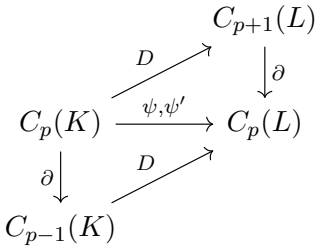
本小节主要来讨论第一个问题:

定义 1.21.1 (链同伦)**.** 设 $\psi, \psi' : C_p(K) \to C_p(L)$ 是链映射, 称 ψ 与 ψ' **链同伦**, 若

$$\exists D : C_p(K) \to C_{p+1}(L)\,(\partial D + D\partial = \psi - \psi'),$$

记作 $\psi \simeq \psi'$.

或者使用交换图的语言, 即图



交换.

定理 1.21.1. 设 $\varphi, \varphi' : K \to L$ 都是 $f : |K| \to |L|$ 的单纯逼近, 则 $\varphi_\sharp \simeq \varphi'_\sharp$.

证明. 任取 $v \in K'^{(0)}$, φ, φ' 是 f 的单纯逼近, 故应当满足 star 条件

$$f(\mathrm{St}(v,K')) \subset \mathrm{St}(\varphi(v),L) \cap \mathrm{St}(\varphi'(v),L).$$

对任意 $x \in K$, 存在单形 $\sigma = (v_0, \ldots, v_p) \in K'$ 使得 $x \in \mathrm{Int}\,\sigma$. 则存在 $\tau \in L$ 使得

$$f(x) \in \mathrm{Int}\,\tau, \qquad \varphi(x), \varphi'(x) \in \tau.$$

对任意 $i \in \{0,\dots,p\}$, 都有

$$f(x) \in \operatorname{Int}\tau \subset f(\operatorname{St}(v_i),K) \subset \operatorname{St}(\varphi(v_i),L) \cap \operatorname{St}(\varphi'(v_i),L),$$

故 $\{\varphi(v_0),\dots,\varphi(v_p),\varphi'(v_0),\dots,\varphi'(v_p)\}$ 落在 L 的同一个单形中. 不妨记以这些顶点为顶点集生成的单形及其面构成的子复形为 $L(\sigma)$. 则

$$\dim L(\sigma) \leqslant 2p+1,$$

且满足 $L(\sigma)$ 非空, $L(\sigma)$ 零调 (这因 $L(\sigma)$ 是单形生成的子复形), 而且对 $s \subset \sigma$, 有 $L(s) \subset L(\sigma)$ (这只需注意到 $L(s)^{(0)} \subset L(\sigma)^{(0)}$). $\varphi_\sharp(\sigma)$ 和 $\varphi'_\sharp(\sigma)$ 被 $L(\sigma)$ 承载.

(下面这一段证明的 *pattern* 非常重要, 这是一类基本技巧, 请务必掌握.) 归纳地证明结论: 当 $p=0$ 时, 对 $v \in K^{(0)}$, 因 $\varphi_\sharp(v)=\varphi(v)$, $\varphi'_\sharp(v)=\varphi'(v)$, 于是自然有

$$\varphi_\sharp(v),\varphi'_\sharp(v) \in C_0(L(v)).$$

因 $L(v)$ 道路连通, 故 $\varphi_\sharp(v)$ 与 $\varphi'_\sharp(v)$ 在同一同调类中, 考虑到 $L(v)$ 零调, 于是

$$\exists d \in C_1(L(v))\,(\partial d = \varphi_\sharp(v) - \varphi'_\sharp(v)),$$

取 $D(v)=d$ 即可. 此时

$$\partial D_0(v) + D_{-1}\partial v = \partial D(v) = \varphi_\sharp(v) - \varphi'_\sharp(v)$$

满足条件.

接下来设对任意 $\leqslant p-1$ 的情形都成立, 那么对 p 的情形, 取 $\sigma \in K'$, $\dim\sigma = p$, 我们希望

$$\partial D\sigma = \varphi_\sharp(\sigma) - \varphi'_\sharp(\sigma) - D\partial\sigma$$

成立, 右侧的每一项都是已经定义过的, 记右侧为 c, 计算

$$\begin{aligned}\partial c &= \partial\varphi_\sharp(\sigma) - \partial\varphi'_\sharp(\sigma) - \partial D(\partial\sigma)\\ &= \partial\varphi_\sharp(\sigma) - \partial\varphi'_\sharp(\sigma) - (\varphi_\sharp(\partial\sigma) - \varphi'_\sharp(\partial\sigma) - D\partial^2\sigma)\\ &= 0,\end{aligned}$$

这因 $\varphi_\sharp$ 和 $\varphi'_\sharp$ 可以与 ∂ 交换. 这说明 $\{c\} \in H_p(L(\sigma))$, 进一步 c 被 $L(\sigma)$ 承载 (因每一项都被 $L(\sigma)$ 承载). 故 $\{c\} \in H_p(L(\sigma)) = 0$, 也即 $c \in Z_p(L(\sigma)) = B_p(L(\sigma))$, 于是

$$\exists d \in C_{p+1}(L(\sigma))\,(\partial d = c),$$

取 $D(\sigma)=d$ 即可.

由此, 我们归纳地构造了 D 满足链同伦的条件, 从而 $\varphi_\sharp \simeq \varphi'_\sharp$. □

推论 1.21.1. 设 $\varphi,\varphi' : K \to L$ 都是 $f : |K| \to |L|$ 的单纯逼近, 则 $\varphi_* = \varphi'_*$.

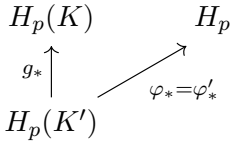
证明. 因对任意 $z_p + B_p(K') \in H_p(K')$, 都有

$$\begin{aligned}\varphi_*(z_p + B_p(K')) &= \varphi_*(z_p) + B_p(L)\\ &= \partial D(z_p) + D(\partial z_p) + \varphi'_*(z_p) + B_p(L)\\ &= \varphi'_*(z_p) + \partial D(z_p) + B_p(L)\\ &= \varphi'_*(z_p) + B_p(L)\end{aligned}$$

成立. 其中第二行的等号因 $\partial z_p = 0$, 第三行的等号因 $D(z_p) \in B_{p+1}(L)$ 导出 $\partial D(z_p) \in B_p(L)$. □

1.22　代数重分定理

本小节主要来讨论第二个问题: 即单纯逼近导出的同态是否依赖于对复形 K 的重分. 设 K' 是 K 的重分, id : $|K'| \to |K|$ 有单纯逼近 $g : K' \to K$. 那么 g 诱导的 g_* 是否是同构?



如果 g_* 是同构, 那么 $H_p(K)$ 到 $H_p(L)$ 的同态就可以给出.

为此考虑这样的一类映射:

定义 1.22.1 (零调承载子)**.** *称 $\Phi : K \to L$, $\sigma \mapsto \Phi(\sigma) \neq \emptyset$ 是一个***零调承载子***, 若对任意 $\sigma \in K$, 都有 $\Phi(\sigma)$ 零调, 且*

$$s \subset \sigma \Longrightarrow \Phi(s) \subset \Phi(\sigma).$$

设 $f : C_p(K) \to C_p(L)$ 是一个链映射, 称 f 被 Φ 承载, 若对任意 $\sigma \in K$, $\dim \sigma = p$ 都有 $f(\sigma)$ 被 $\Phi(\sigma)$ 承载.

命题 1.22.1 (零调承载子定理)**.** *设 $\Phi : K \to L$ 是零调承载子:*

1. *若 $\varphi, \psi : \mathcal{C}(K) \to \mathcal{C}(L)$ 被 Φ 承载, 则存在 D 确定 $\varphi \simeq \psi$;*
2. *存在被 Φ 承载的保持增广的链映射 $\varphi : \mathcal{C}(K) \to \mathcal{C}(L)$.*

证明. (1) 无非就是 1.21.1 中使用相同的 pattern 证明, 只需要注意到 $\Phi(\sigma)$ 也满足相应的条件 (非空, 零调, 保持包含).

(2) 归纳地,

$$\begin{array}{ccccccccccc}
\longrightarrow & C_p(K) & \longrightarrow & \cdots & \longrightarrow & C_1(K) & \xrightarrow{\partial} & C_0(K) & & & \\
 & \downarrow{\scriptstyle \varphi_1} & & & & \searrow{\scriptstyle \varphi_0\partial} & & \downarrow{\scriptstyle \varphi_0} & \overset{\varepsilon}{\searrow} & & \\
 & & & & & & & & \mathbb{Z} & \longrightarrow & 0 \quad \text{当 } p=0 \\
 & & & & & & & & \underset{\varepsilon'}{\nearrow} & & \\
\longrightarrow & C_p(L) & \longrightarrow & \cdots & \longrightarrow & C_1(L) & \xrightarrow{\partial} & C_0(L) & & &
\end{array}$$

时, 取 $v \in K^{(0)}$, $\varepsilon(v) = 1$. 取 w 是 $\Phi(v)$ 的顶点, 并取 $\varphi_0(v) = w$. 则

$$1 = \varepsilon(v) = \varepsilon'(\varphi(v)) = \varepsilon'(w) = 1,$$

成立.

当 $p = 1$ 时, 取 $\sigma = [v_0, v_1]$, $\dim \sigma = 1$. 再取 $c = \varphi_0(\partial\sigma)$, 那么

$$\varepsilon' c = \varepsilon' \varphi_0(\partial\sigma) = \varepsilon(\partial\sigma) = 0,$$

即 c 是增广链中的闭链. 故 $\{c\} \in \tilde{H}_0(\Phi(\sigma)) = 0$, 从而

$$c \in \ker \varepsilon' = \operatorname{im} \partial \Longrightarrow \exists d \in C_1(\Phi(\sigma)) \subset C_1(L)\ (\partial d = c).$$

定义 $\varphi_1(\sigma) = d$ 即可. 由

$$\partial\varphi_1(\sigma) = \partial d = \varphi_0(\partial\sigma)$$

可知图中 $C_1(K), C_0(K), C_1(L), C_0(L)$ 所在的方块交换.

设 $\leqslant p-1$ 的情形都成立, 即对 $\sigma \in K$, $\dim \sigma \leqslant p-1$ 都存在相应的 φ 被 Φ 承载. 取 $\sigma \in K$, $\dim \sigma = p$, 那么考虑

$$\partial\varphi(\partial\sigma) = \varphi\partial^2\sigma = 0,$$

则

$$\varphi(\partial\sigma)\in Z_p(\Phi(\sigma))=B_p(\Phi(\sigma))\Longrightarrow\exists d\in C_{p+1}(\Phi(\sigma))\subset C_{p+1}(L)\,(\partial d=\varphi(\partial\sigma)).$$

定义 $\varphi(\sigma)=d$ 即可.

至此, 我们归纳地构造了 φ. □

现在提出本小节的中心定理:

定理 1.22.1 (代数重分定理)**.** *设 K' 是 K 的一个重分, 则*

*(1) 存在唯一的链映射 $\lambda:C_p(K)\to C_p(K')$ 使得 $\lambda(\sigma)$ 可被 K'_σ 承载, 称 λ 是*重分算子*.*

(2) λ 与 $g_\sharp$ 互为链同伦逆, 由此 g_* 和 λ_* 均为同构.

证明. 考虑

$$\Lambda:K\to K',\qquad \sigma\mapsto\Lambda(\sigma)=K'_\sigma,$$

假设 K'_σ 是零调的. 此时 Λ 是零调承载子, 由 1.22.1 可知存在唯一的 λ 使得 $\lambda(\sigma)\in\mathcal{C}(\Lambda(\sigma))$ 对任意 $\sigma\in K$ 成立. 再考虑

$$\Theta:K'\to K,\qquad \tau\mapsto\Theta(\tau)=K_{\sigma_\tau},$$

其中 $\sigma_\tau\in K$ 是包含 τ 的最低维子复形. 故 K_{σ_τ} 也是锥, 而锥是零调的. 此时 Θ 也是零调承载子, 由 1.22.1 可知存在唯一的 θ 使得 $\theta(\tau)\in\mathcal{C}(\Theta(\tau))$ 对任意 $\tau\in K'$ 成立.

同样地, 可以得到 $\mathrm{id}_{\mathcal{C}(K)}$ 被 Ψ 承载, $\mathrm{id}_{\mathcal{C}(K')}$ 被 Φ 承载.

Claim 1. $\theta\circ\lambda:\mathcal{C}(K)\to\mathcal{C}(K)$ 被 Ψ 承载, $\lambda\circ\theta:\mathcal{C}(K')\to\mathcal{C}(K')$ 被 Φ 承载.

下面证明 Claim 1. 我们只说明 $\theta\circ\lambda$ 被 Ψ 承载, 另一侧类似可证. 任取 $\sigma\in K$, 由 $\lambda(\sigma)$ 被 $\Lambda(\sigma)$ 承载可知 $\lambda(\sigma)$ 可以被 $\Lambda(\sigma)=K'_\sigma$ 中单形 $\tau<\sigma$ 线性表示. 又由 $\theta(\tau)$ 被 $\Theta(\tau)$ 承载可知 $\Theta(\tau)=K_{\sigma_\tau}=K_\sigma=\Psi(\sigma)$. 由此 $\theta\circ\lambda(\sigma)$ 可被 $\Psi(\sigma)=K_\sigma$ 承载.

Claim 2. Θ 承载 $g_\sharp$.

下面证明 Claim 2. 任取 $\tau \in K'$, $w \in \tau$ 是其一个顶点. 由单纯逼近的 star 条件可知

$$\operatorname{Int}\tau \subset \operatorname{id}(\operatorname{St}(w, K)) \subset \operatorname{St}(g(w), K),$$

设 σ 是顶点集含有 $g(w)$ 的单形, 则 $\tau \subset \sigma$. 而 $\sigma_\tau \subset \sigma$, 故 $g_\sharp$ 被 Θ 承载.

由上述讨论, 仍有 $g_\sharp \circ \lambda = \mathrm{id}_{\mathcal{C}(K)}$, $\lambda \circ g_\sharp = \mathrm{id}_{\mathcal{C}(K')}$, 即 λ 与 $g_\sharp$ 互为同伦逆.

接下来说明 λ 的唯一性. 设 λ' 也被 Λ 承载, 那么由 1.22.1 可知 $\lambda' \simeq \lambda$. 即对任意 $\sigma \in K$, $\dim\sigma = p$, 都有

$$\partial D(\sigma) + D(\partial\sigma) = \lambda(\sigma) - \lambda'(\sigma).$$

这里 $D : C_p(K) \to C_{p+1}(K')$, 而 $D(\sigma)$ 被 $\Lambda(\sigma) = K'_\sigma$ 承载, K'_σ 又是一个 p 维复形, 从而只能 $D(\sigma) = 0$. 也即

$$\forall\sigma \in K\,(\lambda(\sigma) - \lambda'(\sigma)) \Longrightarrow \lambda = \lambda'.$$

最后, 我们来说明最开始假设 K'_σ 零调的假设是合理的. 首先考虑特殊情况, 若 $K' = \operatorname{sd} K$, 那么

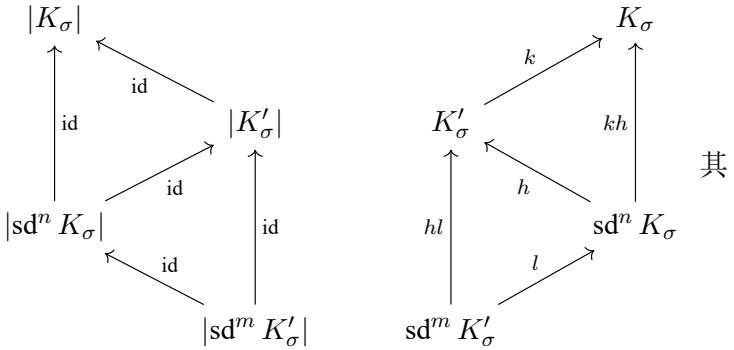
$$\forall\sigma \in K\,(K'_\sigma = \operatorname{sd} K_\sigma = \hat{\sigma} * L_\sigma),$$

这里 $|L_\sigma| = |\operatorname{Bd} K_\sigma|$, 此时定理成立, $H_p(K) \cong H_p(\operatorname{sd} K)$. 若 $K' = \operatorname{sd}^m K$, 此时

$$H_p(K) \cong H_p(\operatorname{sd} K) \cong H_p(\operatorname{sd}^2 K) \cong \ldots \cong H_p(\operatorname{sd}^m K).$$

对任何的 $\sigma \in K$, 注意到 $0 \cong \tilde{H}_p(\operatorname{sd} K_\sigma) \cong \tilde{H}_p(\operatorname{sd}^m K_\sigma)$. 故 $\operatorname{sd}^m K_\sigma$ 零调, 这说明 K' 零调.

再考虑任意的重分 K', 考虑下图:

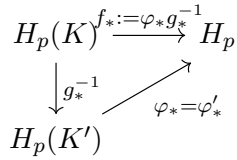


其中右侧映射是左侧 (轴对称对应的) 单纯逼近. 那么由前面的讨论可知 $(kh)_* = k_*h_*$ 是同构, 这说明 h_* 是满同态; 类似地也有 $(hl)_* = h_*l_*$ 是同构, 这说明 h_* 是单同态. 故 h_* 是同构, 从而 k_* 也是同构. 于是得到

$$\tilde{H}_p(K'_\sigma) \cong \tilde{H}_p(K_\sigma) = 0.$$

于是 K'_σ 的确是零调的, 那么命题得证. □

于是 f_* 可以如下图定义:

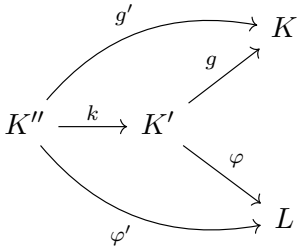


此时一个自然的问题是 f_* 是否与 K 的重分无关. 若另有 K 的重分 K'' 和 $\varphi' : K'' \to L$, 是否有

$$\varphi_* g_*^{-1} = \varphi'_* (g'_*)^{-1}$$

成立?

若 K'' 是 K' 的重分, 考虑到 id : $|K''| \to |K'|$ 存在单纯逼近 $k : K'' \to K'$, 有



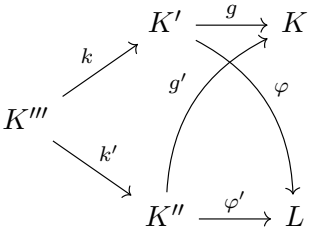
交换, 这即

$$(gk)_\sharp \simeq g'_\sharp, \qquad (\varphi k)_\sharp \simeq \varphi'_\sharp,$$

于是 $(gk)_* = g'_*$, $(\varphi k)_* = \varphi'_*$. 代入得到

$$\varphi'_*(g'_*)^{-1} = \varphi_* k_* (g_* k_*)^{-1} = \varphi_* g_*^{-1}.$$

若 K'' 不是 K' 的重分, 考虑到重分的有限性 (即重分是在 K 的每个单形中增加有限多个点), 那么总存在 K''' 同时是 K' 和 K'' 的重分, 此时



交换, 这即

$$(gk)_* = (g'k')_*, \qquad (\varphi k)_* = (\varphi' k')_*,$$

于是

$$\varphi'_*(g'_*)^{-1} = (\varphi' k')_* (g' k')_*^{-1} = (\varphi k)_* (gk)_*^{-1} = \varphi_* g_*^{-1}.$$

1.23　同调群的同伦不变性

至此, 我们终于完成了以下函子的刻画:

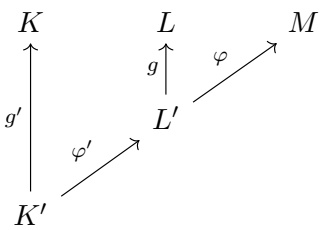
$$\begin{array}{ccc} \mathsf{Top} & & \mathsf{Grp} \\ |K| & \longmapsto & H_p(K) \end{array}$$

定理 1.23.1. 以下 $f : K \to L \longmapsto f_* : H_p(K) \to H_p(L)$ 给出了一个从 Top 到 Grp 的函子.

证明. 为此只需验证 id : $|K| \to |K|$ 被作用变成 $\mathrm{id}_* : H_p(K) \to H_p(K)$, 和对 $f : |K| \to |L|$, $h : |L| \to |M|$, 有 $(hf)_* = h_* f_*$.

$$|K| \xrightarrow{f} |L| \xrightarrow{h} |M|$$

其中第一条是显然的, 这里只证明第二条. 考虑以下的交换图:



则 $h_* = \varphi_* g_*^{-1}$, $f_* = (g\varphi')_* (g'_*)^{-1}$, 计算得到

$$h_* f_* = \varphi_* g_*^{-1} g_* \varphi'_* (g'_*)^{-1}$$
$$(hf)_* = (\varphi\varphi')_* (g'_*)^{-1} = \varphi_* \varphi'_* (g'_*)^{-1}$$

也即 $(hf)_* = h_* f_*$. □

推论 1.23.1. 同调群是同胚不变量.

证明. 对 $|K| \cong |L|$, 考虑 $f : C(|K|, |L|), h \in C(|L|, |K|)$ 使得 $fh = \mathrm{id}_{|L|}, hf = \mathrm{id}_{|K|}$. 由 1.23.1 可知

$$f_* h_* = (fh)_* = (\mathrm{id}_{|L|})_* = \mathrm{id}, \qquad h_* f_* = (hf)_* = (\mathrm{id}_{|K|})_* = \mathrm{id},$$

也即 f_* 和 h_* 是同构, 得到 $H_p(K) \cong H_p(L)$. □

$$\text{homeomorphism } X \cong Y \Longrightarrow \text{homotopy } X \simeq Y$$

我们知道: $\text{homeomorphism invariant} \Longleftarrow \text{homotopy invariant}$ 所以如果可以验证同调群是同伦不变量, 那么从逻辑上讲, 自然就不必说明它是同胚不变量. 幸运的是, 确实如此. 在这里只证明 K 是有限复形的情形:

定理 1.23.2. 设 $f, g \in C(|K|, |L|)$, K 是有限复形. 若 $f \simeq g$, 则 $f_* = g_*$.

证明. 因 K 是有限复形, H 是连接 f 与 g 的同伦, 故 $|K| \times I$ 是紧度量空间. 考虑 $|K| \times I$ 的开覆盖

$$\mathcal{C} = \left\{ H^{-1}(\mathrm{St}(w, L)) : w \in L^{(0)} \right\},$$

由 Lebesgue 覆盖引理可知

$$\exists \delta > 0 \, \forall x \in |K| \times I \, \exists w \in L^{(0)} \, (B(x, \delta) \subset H^{-1}(\mathrm{St}(w, L))).$$

而存在 $m \in \mathbb{Z}_{\geq 1}$ 和 $k > 0$ 使得 $v \in (\mathrm{sd}^m K)^{(0)}$, 且

$$\mathrm{St}(v, \mathrm{sd}^m K) \times \left[\frac{r-1}{k}, \frac{r}{k} \right] \subset B(x, \delta).$$

也即存在 $w \in L^{(0)}$ 使得

$$\mathrm{St}(v, \mathrm{sd}^m K) \times \left[\frac{r-1}{k}, \frac{r}{k} \right] \subset H^{-1}(\mathrm{St}(w, L)).$$

这导出

$$h_{(r-1)/k}(\mathrm{St}(v, \mathrm{sd}^m K)) \subset \mathrm{St}(w, L), \quad h_{r/k}(\mathrm{St}(v, \mathrm{sd}^m K)) \subset \mathrm{St}(w, L).$$

意味着 $h_{(r-1)/k}$ 和 $h_{r/k}$ 可以取到相同的单纯逼近 $\varphi_r : \mathrm{sd}^m K \to L$. 而 $\mathrm{id} : \mathrm{sd}^m K \to K$ 也有相应的单纯逼近, 于是

$$\forall 1 \leqslant r \leqslant k \ (h_{(r-1)/k})_* = (h_{r/k})_*,$$

那么

$$f_* = (h_0)_* = (h_{1/k})_* = \cdots = (h_{(k-1)/k})_* = (h_1)_* = g_*.$$ □

推论 1.23.2 (同调群是同伦不变量)**.** 若 $|K| \simeq |L|$, 则 $H_p(|K|) \cong H_p(|L|)$.

证明. 这即存在 $f \in C(|K|, |L|), g \in C(|L|, |K|)$ 使得

$$fg \simeq \mathrm{id}_{|L|}, \qquad gf \simeq \mathrm{id}_{|K|},$$

由此 f_* 与 g_* 互逆. 即 $H_p(|K|) \cong H_p(|L|)$. □

1.24　应用: 映射度, Brouwer 不动点定理

考虑到 $\mathbb{S}^0$ 退化, 不妨总假设 $n>0$. Brouwer 不动点主要讨论球面上的映射:

定义 1.24.1 (映射度)**.** *设 $f\in C(\mathbb{S}^n,\mathbb{S}^n)$, 它诱导*

$$f_*:H_n(\mathbb{S}^n)\to H_n(\mathbb{S}^n),\quad \alpha\mapsto f_*(\alpha)=d\alpha, d\in\mathbb{Z}.$$

*其中 α 是 $\mathbb{S}^n$ 的生成元. d 称作 f 的***映射度***, 记作 $\deg f$.*

当 $n=1$ 时, 映射度 $\deg f$ 有明显的几何意义. $\deg f$ 即 f 覆盖球面的层数. $\pi_1(\mathbb{S}^1)=[\mathbb{S}^1,\mathbb{S}^1]\to\mathbb{Z}$, $f\mapsto\deg f$ 给出了 $\mathbb{S}^1$ 的基本群.

命题 1.24.1. *映射度具有以下基本性质: 以下设 $f,g:\mathbb{S}^n\to\mathbb{S}^n$,*

1. *若 $f\simeq g$, 则 $\deg f=\deg g$;*
2. *$\deg(gf)=\deg f\cdot\deg g$;*
3. *$\deg\mathrm{id}=1$. 若 c 是常值映射, 则 $\deg c=0$.*

证明. (2) 因

$$\deg(gf)\alpha=(gf)_*\alpha=g_*(\deg f\alpha)=\deg f\cdot g_*\alpha=\deg f\cdot\deg g\,\alpha.$$

而 (1), (3) 是显然的. □

引理 1.24.1. *设 $F\in C(\mathbb{D}^{n+1},\mathbb{S}^n)$, 则 $\deg(F|_{\partial\mathbb{D}^{n+1}})=0$.*

$$\mathbb{S}^n\xhookrightarrow{\ \iota\ }\mathbb{D}^{n+1}\xrightarrow{\ F\ }\mathbb{S}^n$$

证明. 这因 且 $F|_{\partial\mathbb{D}^{n+1}}=F\circ\iota$. 由 $H_n(\mathbb{D}^{n+1})=0$ 可知

$$H_n(\mathbb{S}^n)\xhookrightarrow{\ \iota_*\ }H_n(\mathbb{D}^{n+1})\xrightarrow{\ F_*\ }H_n(\mathbb{S}^n)$$

$(F|_{\partial\mathbb{D}^{n+1}})_*=0$. □

引理 1.24.2. *不存在这样的 $r\in C(\mathbb{D}^{n+1},\mathbb{S}^n)$ 使得 $r|_{\partial\mathbb{D}^{n+1}}=\mathrm{id}_{\mathbb{S}^n}$.*

证明. 由 1.24.1 可知若存在这样的 r, 有 $\deg(r|_{\partial\mathbb{D}^{n+1}})=0$. 再由 1.24.1 可知 $\deg(\mathrm{id}_{\mathbb{S}^n})=1$, 矛盾. □

定理 1.24.1 (Brouwer 不动点定理)**.** *任何的 $f\in C(\mathbb{D}^n,\mathbb{D}^n)$ 存在不动点.*

证明. 用反证法, 任取 $x\in\mathbb{D}^n$, 都有 $f(x)\neq x$. 取

$$h:\mathbb{D}^n\to\mathbb{S}^{n+1},\qquad x\mapsto\frac{x-f(x)}{\|x-f(x)\|}.$$

由 1.24.1 可知 $\deg(h|_{\partial\mathbb{D}^n})=0$, 作同伦

$$H:\mathbb{S}^{n-1}\times I\to\mathbb{S}^{n-1},\qquad (x,t)\mapsto\frac{x-tf(x)}{\|x-tf(x)\|}.$$

这一同伦的定义是合理的, 因 $x\neq f(x)$ 可知 $t=1$ 时 $x-f(x)\neq0$, 而 $0\leqslant t<1$ 时若有 $x=tf(x)$, 则

$$1=\|x\|=\|tf(x)\|=t\,\|f(x)\|\leqslant t<1,$$

矛盾. 此时

$$H(x,0)=\mathrm{id}_{\mathbb{S}^{n-1}}(x),\qquad H(x,1)=h|_{\partial\mathbb{D}^n}(x),$$

即 $\mathrm{id}_{\mathbb{S}^{n-1}}\simeq h|_{\partial\mathbb{D}^n}$, 而它们的映射度不相等, 矛盾. □

Brouwer 不动点定理在 $n=1$ 时即为介值定理, 在 $n=2$ 时使用分析的方法已经比较繁琐, 但还是可以借助基本群来解决. 以上给出了 $n \geqslant 1$ 时使用同调论的解决方法.

接下来考虑从球面到球面的映射, $f \in C(\mathbb{S}^n, \mathbb{S}^n)$ 可以不存在任何不动点, 最常见的取法就是取**对径映射** $r: x \mapsto -x$.

命题 1.24.2. $\deg r = (-1)^{n+1}$.

证明. 考虑 $r_i: \mathbb{S}^n \to \mathbb{S}^n$, 它只将第 i 个分量反转, 即

$$(x_1, \ldots, x_i, \ldots, x_{n+1}) \mapsto (x_1, \ldots, -x_i, \ldots, x_{n+1}).$$

则 $r = r_{n+1} r_n \cdots r_1$. 于是 $\deg r = \prod_{i=1}^{n+1} \deg r_i$. 由

$$\begin{array}{ccc} (x_1, \ldots, x_i, \ldots, x_{n+1}) & & \mathbb{S}^n & \xrightarrow{r_i} & \mathbb{S}^n & & (x_1, \ldots, -x_i, \ldots \\ \downarrow & & \downarrow{\scriptstyle h_{i,n+1}} & & \downarrow{\scriptstyle h_{i,n+1}} & & \downarrow \\ (x_1, \ldots, x_{n+1}, \ldots, x_i) & & \mathbb{S}^n & \xrightarrow{r_{n+1}} & \mathbb{S}^n & & (x_1, \ldots, -x_{n+1} \end{array}$$

有 $r_{n+1} = h_{i,n+1} r_i h_{i,n+1}^{-1}$, 从而

$$\deg r_{n+1} = \deg h_{i,n+1} \deg r_i \deg h_{i,n+1}^{-1} = \deg r_i.$$

故 $\deg r = (\deg r_{n+1})^{n+1}$.

考虑 $\mathbb{S}^n$ 的如下三角剖分: 它由 $\mathbb{R}^{n+1}$ 上典范正交基的顶点 $\{\pm e_1, \pm e_2, \ldots, \pm e_{n+1}\}$ 确定. 记 $e_{-i} = -e_i$, 得到复形 K

$$|K| = \left\{ (x_1, \ldots, x_{n+1}) \in \mathbb{R}^{n+1} : \sum_{i=1}^{n+1} |x_i| = 1 \right\}.$$

考虑中心辐射 $\varphi : |K| \to \mathbb{S}^n$, 有

$$\begin{array}{ccc} \mathbb{S}^n & \xrightarrow{\quad r_{n+1} \quad} & \mathbb{S}^n \\ \uparrow{\scriptstyle \varphi} & & \uparrow{\scriptstyle \varphi} \\ |K| & \xrightarrow{\quad \tilde{r}_{n+1} \quad} & |K| \end{array}$$

那么对 K 中的 n 维单形, 给定如下的定向: (约定 $i_j = \pm j$)

$$(x_1, \ldots, x_n, x_{n+1}) \longmapsto (x_1, \ldots, x_n, -x_{n+1})$$

$$\pm[e_{i_1}, e_{i_2}, \ldots, e_{i_{n+1}}] \begin{cases} \text{取}+ \Longleftrightarrow \{i_k\} \text{ 中有偶数个负数} \\ \text{取}- \Longleftrightarrow \{i_k\} \text{ 中有奇数个负数} \end{cases}$$

令 $z_n = \sum_{\sigma \in \mathbb{S}^n_+(K)} \sigma$ 可知 $\partial z_n = 0$. 这里 $\mathbb{S}^n_+(K)$ 是 K 中正定向 n 维单形全体. 此时 $z_n \in Z_n(K) = H_n(\mathbb{S}^n)$, 那么

$$(\tilde{r}_{n+1})_\sharp : C_n(K) \to C_n(K), \qquad \pm[e_1, \ldots, e_{n+1}] \mapsto \mp[e_1, \ldots, e_{n+1}].$$

故 $(\tilde{r}_{n+1})_\sharp(z_n) = -z_n$, 由此 $\deg \tilde{r}_{n+1} = -1$ 得到 $\deg r_{n+1} = -1$. □

命题 1.24.3. *设 $h \in C(\mathbb{S}^n, \mathbb{S}^n)$, 若 $\deg h \neq (-1)^{n+1}$, 则 h 必有不动点.*

证明. 用反证法, 设 $\forall x \in \mathbb{S}^n\,(h(x) \neq x)$. 取

$$H : \mathbb{S}^n \times I \to \mathbb{S}^n, \qquad (x, t) \mapsto \frac{(1-t)h(x) + t(-x)}{\|(1-t)h(x) + t(-x)\|}.$$

类似地, 若有 $(1-t)h(x) - tx = 0$, 则

$$\|(1-t)h(x)\| = t\,\|x\| \Longrightarrow 1 - t = t \Longrightarrow t = \frac{1}{2}.$$

而将 $t = 1/2$ 代入之后有 $1/2(h(x) - x) \neq 0$, 于是 H 是合理定义的. 此时由

$$H(x, 0) = h(x), \qquad H(x, 1) = r(x)$$

可知 $h \simeq r$, 而 $\deg h \neq \deg r = (-1)^{n+1}$, 矛盾. □

推论 1.24.1. *取 $f \in C(\mathbb{S}^n, \mathbb{S}^n)$, $\deg f \neq 1$, 则存在 $x_0 \in \mathbb{S}^n$ 使得 $f(x_0) = -x_0$.*

证明. 考虑 $rf : \mathbb{S}^n \to \mathbb{S}^n$, 有 $\deg(rf) \neq (-1)^{n+1}$. 由 1.24.3 可知 rf 存在不动点 x_0, 那么

$$rf(x_0) = x_0 \Longrightarrow -f(x_0) = x_0,$$

于是命题得证. □

下面把这一问题应用于所谓的 “毛球问题”:

定义 1.24.2 (向量场)**.** *称 $v : X \to \mathbb{R}^n$ 是一个 X 上的*向量场 *(当 $n = 1$ 时称为*数量场*). 特别地, $v : \mathbb{S}^n \to \mathbb{R}^{n+1}$ 称作一个切向量场, 若对任意 $x \in \mathbb{S}^n$ 都有 $\langle x, v(x) \rangle = 0$.*

推论 1.24.2 (Brouwer–Poincaré)**.** *$\mathbb{S}^n$ 上存在非零切向量场 (向量场非零指它在任何一点都非零.) 当且仅当 n 为奇数.*

证明. 充分性. 因 $n=2k-1$, 直接取

$$v:\mathbb{S}^n\to\mathbb{S}^n,\qquad (x_1,x_2,\dots,x_{2k-1},x_{2k})\mapsto(-x_2,x_1,\dots,-x_{2k},x_{2k-1})$$

是一个非零切向量场即可.

必要性. 若 $\mathbb{S}^n$ 存在非零切向量场 v, 考虑它诱导的映射

$$h:\mathbb{S}^n\to\mathbb{S}^n,\qquad x\mapsto\frac{v(x)}{\|v(x)\|},$$

因 v 是切向量场, 故 h 也是切向量场, 于是

$$\forall x\in\mathbb{S}^n\,(h(x)\neq x\wedge h(x)=-x).$$

由 1.24.3 可知 $\deg h=(-1)^{n+1}$, 又由 1.24.1 可知 $\deg h=1$, 于是 n 是奇数. □

1.25　应用: Euler 示性数, Lefschetz 不动点定理

设 K 是有限复形, 但 $H_p(K)=\ker\partial_p/\operatorname{im}\partial_{p+1}$ 未必是自由的, 但一定是有限生成 Abel 群. 而有限生成 Abel 群 G 总有分解

$$G\cong\mathbb{Z}\oplus\mathbb{Z}\oplus\cdots\oplus\mathbb{Z}\oplus(\mathbb{Z}/t_1\mathbb{Z})\oplus(\mathbb{Z}/t_2\mathbb{Z})\oplus\cdots\oplus(\mathbb{Z}/t_k\mathbb{Z})=F(G)\oplus T(G),$$

其中 $F(G)$ 是 G 的自由部分, 它有秩 r, $T(G)$ 是 G 的挠部分, 满足 $t_1\mid t_2\mid\cdots\mid t_k$, 则 $H_i(K)$ 中自由部分的秩 $\beta_1(K):=\operatorname{rank}F(H_i(K))$ 称作复形 K 的第 i 个 **Betti 数**.

我们把 Brouwer 定理再次推广, 不仅仅考虑球面之间的映射, 而考虑一般的复形 K, 它的几何实现到自身的映射 $f\in C(|K|,|K|)$ 的不动点. 为此, 我们先回顾一点点初等线性代数的内容:

定义 1.25.1 (线性变换的迹)**.** 设 $\varphi:V\to V$ 是线性变换, $\{e_1,\dots,e_n\}$ 是 V 的基, φ 在 $\{e_i\}$ 下的矩阵为 A. 因迹对基变换是不变的, 于是可以定义

$$\operatorname{tr}\varphi:=\operatorname{tr}A$$

是线性变换 φ 的**迹**.

引理 1.25.1. 设 W 是 V 的 φ–不变子空间, 即 $\varphi(W)\subset W$, 则 φ 可以诱导

$$\varphi_*:V/W\to V/W,\qquad x+W\mapsto\varphi(x)+W,$$

此时有

$$\operatorname{tr}\varphi=\operatorname{tr}(\varphi|_W)+\operatorname{tr}\varphi_*.$$

证明. 设 W 有一组基 $\{\xi_1,\dots,\xi_r\}$, V 的基 $\{\xi_1,\dots,\xi_r,\xi_{r+1},\dots,\xi_n\}$ 由 W 的基扩张而来, 则 V/W 以 $\{\xi_{r+1}+W,\dots,\xi_n+W\}$ 为一组基. 设 $\varphi,\varphi|_W$ 和 φ_* 在对应基下的矩阵分别为 A,A_W,A_*, 则

$$A=\begin{bmatrix}A_W & *\\ & A_*\end{bmatrix},$$

这导出 $\operatorname{tr}\varphi=\operatorname{tr}(\varphi|_W)+\operatorname{tr}\varphi_*$. □

设 G 是有限生成的自由 Abel 群, $\varphi : G \to G$ 是线性映射. 也可以定义 $\operatorname{tr}\varphi$: 考虑 G 的 φ–不变子群 H, 那么 φ 可以诱导商群上的线性映射

$$\varphi_* : G/H \to G/H,$$

当 G/H 也是自由群时, 仍然有 $\operatorname{tr}\varphi = \operatorname{tr}(\varphi|_H) + \operatorname{tr}\varphi_*$.

为了记号方便, 我们约定将链群、闭链群、边缘链群中的复形记号省略掉, 并且记

$$H_p = Z_p/B_p = T_p \oplus F_p,$$

这里 T_p 是 H_p 的挠部分, F_p 是 H_p 的自由部分. 那么 Z_p 可以分解成两部分, 一部分被 B_p 模掉之后产生挠部分, 把这一部分记作

$$W_p := \{c_p \in C_p : \exists \ell \in \mathbb{Z} \setminus \{0\}\ (\ell c_p \in B_p)\}.$$

另一部分被 B_p 模掉产生自由部分.

定理 1.25.1 (Hopf 迹定理)**.** *设 $\varphi : \mathcal{C}(K) \to \mathcal{C}(K)$ 是链映射, K 是有限复形, 那么*

$$\sum_p (-1)^p \operatorname{tr}\varphi|_{C_p} = \sum_p (-1)^p \operatorname{tr}(\varphi_*, H_p/T_p),$$

这里 φ_* 是合理定义的. (因形如 $\mathbb{Z}/m\mathbb{Z} \to \mathbb{Z}$ 的同态必为零同态, 于是一定 $\varphi(T_p) \leqslant T_p$.)

证明. 根据前面约定的记号, 容易得到

$$B_p \leqslant W_p \leqslant Z_p \leqslant C_p.$$

首先考虑 $Z_p \leqslant C_p$. 由 $C_p \overset{\partial_p}{\twoheadrightarrow} B_{p-1} \leqslant C_{p-1}$ 且 $Z_p = \ker\partial_p$, 由第一同构定理得到

$$C_p/Z_p \cong B_{p-1}.$$

因 B_{p-1} 自由, 故 C_p/Z_p 自由. 由 1.25.1 可知

$$\operatorname{tr}(\varphi|_{C_p}) = \operatorname{tr}(\varphi|_{Z_p}) + \operatorname{tr}(\varphi_*, C_p/Z_p) = \operatorname{tr}(\varphi|_{Z_p}) + \operatorname{tr}(\varphi_*, B_{p-1}).$$

$$Z_p \xrightarrow{\pi_{B_p}} H_p \xrightarrow{\pi_{T_p}} F_p, \qquad Z_p \xrightarrow{\ j\ } F_p$$

再考虑 $W_p \leqslant Z_p$. 由 则对任何 $z_p \in W_p$, 有

$$z_p \longmapsto \{z_p\} = z_p + B_p \longmapsto \{z_p\} + T_p$$

$j(z_p) = \{z_p\} + T_p$. 由 W_p 的定义可知存在 $\ell \in \mathbb{Z} \setminus \{0\}$ 使得 $\ell z_p \in B_p$. 那么

$$\ell\{z_p\} + T_p = \{\ell z_p\} + T_p = 0 + T_p,$$

即 $W_p \leqslant \ker j$. 反之, 任取 $z_p \in \ker j$, 由

$$j(z_p) = \{z_p\} + T_p = 0 \Longrightarrow \exists \ell \in \mathbb{Z} \setminus \{0\}\ (\ell\{z_p\} = 0) \Longrightarrow z_p \in W_p$$

知 $\ker j \subset W_p$. 于是 $W_p = \ker j$, 由第一同构定理可知

$$Z_p/W_p \cong H_p/T_p = F_p.$$

由 F_p 自由可知 Z_p/W_p 也自由, 由 1.25.1 可知

$$\operatorname{tr}(\varphi|_{Z_p}) = \operatorname{tr}(\varphi|_{W_p}) + \operatorname{tr}(\varphi_*, Z_p/W_p) = \operatorname{tr}(\varphi|_{W_p}) + \operatorname{tr}(\varphi_*, F_p).$$

最后, 考虑 $B_p \leqslant W_p$. 我们断言 $\operatorname{rank} B_p = \operatorname{rank} W_p$ 且 $\operatorname{tr}(\varphi|_{B_p}) = \operatorname{tr}(\varphi|_{W_p})$. 用反证法, 否则 $m = \operatorname{rank} B_p < \operatorname{rank} W_p = r$.

设 $\{\alpha_1, \dots, \alpha_r\}$ 是 W_p 的一组基, 不妨设 $n_i \in \mathbb{Z}_{\geq 1}$, $\{n_1\alpha_1, \dots, n_m\alpha_m\}$ 是 B_p 的一组基. 那么由 $r > m$ 可知存在 $n_{m+1} \in \mathbb{Z}_{\geq 1}$ 使得 $n_{m+1}\alpha_{m+1} \in B_p$ (这由 W_p 的定义可得), 矛盾, 于是 $r = m$.

设 $\varphi|_{W_p}$ 在 $\{\alpha_1, \dots, \alpha_r\}$ 下的矩阵为 $A = [a_{ij}]$, $\varphi|_{B_p}$ 在 $\{n_1\alpha_1, \dots, n_r\alpha_r\}$ 下的矩阵为 $B = [b_{ij}]$. 则

$$\varphi|_{W_p}(\alpha_j) = \sum_{i=1}^{r} a_{ij}\alpha_i \Longrightarrow \varphi|_{W_p}(n_j\alpha_j) = \sum_{i=1}^{r} a_{ij}n_j\alpha_i.$$

且

$$\varphi|_{B_p}(n_j\alpha_j) = \sum_{i=1}^{r} b_{ij}n_i\alpha_i.$$

而 $B_p \leqslant W_p$, 于是 $\sum\limits_{i=1}^{r} a_{ij}n_j\alpha_i = \sum\limits_{i=1}^{r} b_{ij}n_i\alpha_i$, 这说明对任意 i, j 成立

$$a_{ij}n_j = b_{ij}n_i.$$

特别地, 当 $i = j$ 时就有 $a_{ii} = b_{ii}$. 那么

$$\operatorname{tr}(\varphi|_{W_p}) = \operatorname{tr} A = \sum_{i=1}^{r} a_{ii} = \sum_{i=1}^{r} b_{ii} = \operatorname{tr} B = \operatorname{tr}(\varphi|_{B_p}).$$

于是:

$$\begin{aligned}\operatorname{tr}(\varphi|_{C_p}) &= \operatorname{tr}(\varphi|_{Z_p}) + \operatorname{tr}(\varphi|_{B_{p-1}}) \\ &= \operatorname{tr}(\varphi|_{W_p}) + \operatorname{tr}(\varphi_*, F_p) + \operatorname{tr}(\varphi|_{B_{p-1}}) \\ &= \operatorname{tr}(\varphi|_{B_p}) + \operatorname{tr}(\varphi|_{B_{p-1}}) + \operatorname{tr}(\varphi_*, F_p).\end{aligned}$$

因 K 是有限复形, 于是下面的交错和是有限和, 得到

$$\begin{aligned}\sum_{p\geqslant 0}(-1)^p \operatorname{tr}(\varphi|_{C_p}) &= \sum_{p\geqslant 0}(-1)^p \operatorname{tr}(\varphi|_{B_p}) - (-1)^{p-1} \operatorname{tr}(\varphi|_{B_{p-1}}) + (-1)^p \operatorname{tr}(\varphi_*, F_p) \\ &= \sum_{p\geqslant 0}(-1)^p \operatorname{tr}(\varphi_*, F_p).\end{aligned}$$

于是定理成立. □

我们再进一步讨论一下这些子群: 由 $Z_p/W_p \cong H_p/T_p$, 若记 $Z_p = W_p \oplus A$, 那么由 $W_p/B_p \cong T_p$,

$$\frac{Z_p}{B_p} \cong \frac{W_p \oplus A}{B_p} \cong \frac{W_p}{B_p} \oplus A \cong T_p \oplus A,$$

这说明 $A \cong F_p$.

定义 1.25.2 (Lefschetz 数)**.** 设 K 是有限复形, $h:|K|\to|K|$ 是连续映射, 称

$$\Lambda(h):=\sum_p(-1)^p\operatorname{tr}(h_*,H_p/T_p)$$

是 h 的 ***Lefschetz* 数**.

由定义可知若 $h\simeq h'$, 则 $\Lambda(h)=\Lambda(h')$, 这因 $h_*=h'_*$; 同时 $\Lambda(h)$ 与 $|K|$ 的三角剖分无关, 因为之前已经说明过同调群不会随着重分改变.

特别地, 若允许同调群中的系数在数域中取值, 那么 C_p 和 H_p 都是向量空间, 其维数即为复形 K 的 p 阶 Betti 数, 也即 $\dim_{\mathbb{F}}H_p=\beta_p(K)$.

考虑 $\mathrm{id}:|K|\to|K|$ 的 Lefschetz 数:

$$\Lambda(\mathrm{id})=\sum_{p\geqslant0}(-1)^p\operatorname{tr}(\mathrm{id}_*,H_p/T_p)=\sum_{p\geqslant0}(-1)^p\beta_p(K).$$

设 K 的 p 维单形个数为 f_p, 取 $f(K)=(f_0,\dots,f_n)$, 那么

$$\operatorname{rank}C_p(K)=f_p=\beta_p(K).$$

称

$$\chi(K)=\sum_{p\geqslant0}(-1)^p\operatorname{rank}C_p(K)$$

是 K 的 **Euler 示性数**.

定理 1.25.2 (Euler–Poincaré)**.** *Euler* 示性数是拓扑不变量, 它与 $|K|$ 的三角剖分无关, 且

$$\chi(|K|)=\sum_{p\geqslant0}(-1)^p\beta_p(K).$$

一些典型的曲面 Euler 示性数可以计算出来:

$$\chi(\mathbb{T}^2)=0,\quad\chi(S)=0,\quad\chi(\mathbb{RP}^2)=1,\quad\chi(\mathbb{S}^n)=\begin{cases}2, & n=2k\\ 0, & n=2k+1\end{cases}.$$

定理 1.25.3 (Lefschetz 不动点定理)**.** 设 K 是有限复形, $h\in C(|K|,|K|)$, 若 $\Lambda(h)\neq0$, 则 h 必有不动点.

证明. 用反证法, 设 $\forall x\in|K|\ (h(x)\neq x)$, 考虑连续函数

$$f:|K|\to\mathbb{R},\qquad x\mapsto\|x-h(x)\|,$$

因 $|K|$ 是紧度量空间, 它取到最小值 ε. 不妨设 $\operatorname{mesh}K<\varepsilon/2$ (否则取足够多次数的标准重心重分).

记 $h:|K|\to|K|$ 得单纯逼近为 $\varphi:\mathrm{sd}^mK\to K$, 则对任意 $x\in K$, $h(x)$ 和 $\varphi(x)$ 都落在 K 的同一个单形中, 有

$$\|h(x)-\varphi(x)\|\leqslant\operatorname{mesh}K<\frac{\varepsilon}{2},$$

故

$$\|x-\varphi(x)\|\geqslant\|x-h(x)\|-\|h(x)-\varphi(x)\|>\frac{\varepsilon}{2}.$$

由代数重分定理: $C_p(K) \xrightarrow{\exists!\lambda} C_p(\mathrm{sd}^m K) \xrightarrow{\varphi_\sharp} C_p(K)$ 则 $h_* = (\varphi_\sharp\lambda)_* = \xi_*$. 故

$$\Lambda(h) = \sum_{p\geqslant 0}(-1)^p \operatorname{tr}(h_* : H_p/T_p) = \sum_{p\geqslant 0}(-1)^p \operatorname{tr}(\xi_*, C_p).$$

我们断言 $\operatorname{tr}(\xi_*, C_p) = 0$. 这蕴含着 ξ_p 在 $S_p^+(K)$ (K 的定向 p 维单形全体, 这是 $C_p(K)$ 的一组基) 下的矩阵 A 满足 $\operatorname{tr} A = 0$. 这意味着 $\xi_p(\sigma)$ 中不能含有 σ. 因 $\lambda(\sigma)$ 被 $\mathrm{sd}^m K_\sigma$ 承载, 若

$$\xi_p(\sigma) = \varphi_\sharp\lambda(\sigma) = a\sigma + \cdots, \quad a \neq 0,$$

在 $\lambda(\sigma)$ 的展开式中应当存在 p 维单形 $\tau \in \mathrm{sd}^m\ K_\sigma$ 使得

$$\varphi_\sharp(\tau) = a'\sigma + \cdots, \quad a' \neq 0.$$

因对任意 $x \in \tau$ 都有 $\varphi(x) \in \sigma$, 又 $\tau \subset \sigma$, 故 $x \in \sigma$, 从而

$$\frac{\varepsilon}{2} < \|x - \varphi(x)\| \leqslant \operatorname{mesh} K < \frac{\varepsilon}{2},$$

矛盾. □

推论 1.25.1. $\mathbb{S}^n$ 上的对径映射满足 $\deg r = (-1)^{n+1}$.

证明. 由 1.25.3 知 $\Lambda(r) = 1 + (-1)^n \deg r = 0$. □

推论 1.25.2. 设 K 是零调的, 则任意 $f \in C(|K|, |K|)$ 都存在不动点.

证明. 设 $f_* : H_0(K) \to H_0(K)$, 它有单纯逼近 $\varphi : K' \to K$, 那么图

$$\begin{array}{ccccc} C_0(K) & & \xrightarrow{\lambda} & & C_0(K') \xrightarrow{\varphi_\sharp} C_0(K) \\ & \searrow^{\varepsilon} & & \swarrow_{\varepsilon'} & \\ & & \mathbb{Z} & & \end{array}$$

交换.

对 $v \in K^{(0)}$, $K'_v = \{v\}$, 有 $\lambda(v) = \ell v$. 由

$$1 = \varepsilon(v) = \varepsilon'\lambda(v) = \ell\varepsilon'(v) = \ell$$

可知 $\lambda(v) = v$, 于是

$$\varphi_\sharp(\lambda(v)) = \varphi_\sharp(v) = \varphi(v)$$

与 v 在同一同调类中, 这即 $f_*(\{v\}) = \{\varphi_*(\lambda(v))\} = \{v\}$, 即 $f_* = \mathrm{id}_*$. 故 $\Lambda(f) = \Lambda(\mathrm{id}) = 1 \neq 0$. 由 1.25.3 得证. □

推论 1.25.3. 设 $\chi(|K|) \neq 0$, 则对任意同伦于 $\mathrm{id}_{|K|}$ 的连续映射 $f \in C(|K|, |K|)$, f 存在不动点.

证明. 因 $\Lambda(f) = \Lambda(\mathrm{id}) = \chi(|K|) \neq 0$. □

1.26　奇异同调论

原网页

1.27　相对同调

对子复形 $K_0 \subset K$, 它们的链复形 $\mathcal{C}(K_0)$ 与 $\mathcal{C}(K)$ 满足

$$C_p(K_0) \leqslant C_p(K), \quad S_p^+(K_0) \subset S_p^+(K).$$

于是 $C_p(K)/C_p(K_0)$ 仍是自由 Abel 群.

定义 1.27.1 (相对链群, 相对链复形, 相对同调群)**.** *设 K_0 是 K 的子复形, 称*

$$C_p(K, K_0) = C_p(K)/C_p(K_0)$$

*是 K 模 K_0 的***相对链群***. 它是自由 Abel 群, 其中的元素为 $\bar{c}_p = c_p + C_p(K_0)$. 自然地, 有边缘同态*

$$\partial : C_p(K, K_0) \to C_{p-1}(K, K_0), \quad \bar{c}_p \mapsto \partial \bar{c}_p := \partial c_p + C_{p-1}(K_0).$$

*且 $\partial^2 = 0$. 由此, 可以定义***相对链复形***

$$\mathcal{C}(K, K_0) := \{C_p(K, K_0), \partial\}$$

*和***相对同调群***

$$H_p(K, K_0) := \ker \partial_p / \operatorname{im} \partial_{p+1}.$$

出于对记号的继承, 我们仍然记

$$\ker \partial_p = Z_p(K, K_0), \qquad \operatorname{im} \partial_{p+1} = B_p(K, K_0).$$

在相对同调的意义下,

$$\bar{z}_p \in Z_p(K, K_0) \iff \partial \bar{z}_p = 0 \iff \partial z_p \in C_{p-1}(K_0),$$

即 $\bar{z}_p = z_p + C_p(K_0)$ 是相对闭链群中的元素当且仅当 ∂z_p 被 K_0 承载. 而

$$\bar{b}_p \in B_p(K, K_0) \iff \exists c_{p+1} \in C_{p+1}(K)\,(b_p - \partial \bar{c}_{p+1} \in C_p(K_0)).$$

直接计算相对同调群往往是复杂的, 我们举一个简单的例子和一个复杂的例子.

例 1.27.1. *设 σ 是一个 n 维单形, $K = K_\sigma$, $K_0 = K_\sigma \setminus \{\sigma\}$, 那么对任何的 $p < n$ 都有*

$$C_p(K) = C_p(K_0) \Longrightarrow C_p(K, K_0) = C_p(K)/C_p(K_0) = 0.$$

而 $p = n$ 时, $C_n(K) = \mathbb{Z}$, $C_n(K_0) = 0$, 有 $C_n(K, K_0) = \mathbb{Z}$. 于是

$$H_p(K, K_0) = \frac{Z_p(K, K_0)}{B_p(K, K_0)} = \begin{cases} 0, & p \neq n \\ \mathbb{Z}, & p = n \end{cases}.$$

(这可能是可以找到的最简单的例子了.)

例 1.27.2. 考虑下面的复形 K, 其几何实现 $|K| \cong \mathbb{D}^2$. 取 $K_0 = \mathrm{Bd}\, K$. 首先考虑最简单的 $p = 2$ 的情形:

$$H_2(K, K_0) = C_2(K, K_0) = Z_2(K, K_0) = C_2(K),$$

取 $\bar{c}_2 = \sum_{\dim \sigma = 2} m_i \sigma_i \in Z_2(K, K_0)$, 那么

$$\partial c_2 \in C_1(K_0) \Longleftrightarrow m_1 = m_2 = m_3 = m_4,$$

于是 $H_2(K, K_0) = Z_2(K, K_0) \cong \mathbb{Z}$.

对于 $p = 1$ 的情形, 情况就复杂了起来. 我们考虑

$$\{\bar{c}_1\} \in H_1(K, K_0) \Longleftrightarrow \partial \bar{c}_1 = 0 \Longleftrightarrow \partial c_1 \in C_0(K_0),$$

另一方面, 若 $c_1 \sim c_1'$ 使得 c_1' 被下面的复形承载, 那么

$$\begin{aligned} \exists d \in C_2(K)\,(\partial d = c_1 - c_1') &\Longrightarrow \partial \bar{d} = \overline{\partial d} = c_1 - c_1' + C_1(K_0) \\ &\Longrightarrow \partial^2 \bar{d} = \partial c_1 - \partial c_1' + C_0(K_0) \\ &\Longrightarrow \partial^2 \bar{d} = 0 = -\partial c_1' + C_0(K_0) \\ &\Longrightarrow \partial c_1' \in C_0(K_0). \end{aligned}$$

故 c_1' 也被 K_0 承载. 则

$$B_1(K, K_0) \ni \partial d = c_1 - c_1' + C_1(K_0) = c_1 + C_1(K_0) = \bar{c}_1 \in Z_1(K, K_0),$$

于是 $H_1(K,K_0)=Z_1(K,K_0)/B_1(K,K_0)=0$.

对 $p=0$ 的情形, 因各 v_i 都有 $v_i\in C_0(K_0)$ 成立, 于是

$$\{\bar{c}_0\}\in H_0(K,K_0)\Longrightarrow \bar{c}_0=\ell v+C_0(K_0)\Longrightarrow \partial e=v-v_0,$$

则

$$B_0(K,K_0)\ni \partial(\ell\bar{e})=\ell(\partial\bar{e})=\ell v-\ell v_0+C_0(K_0)=\ell v+C_0(K_0)=\bar{c}_0\in Z_0(K,K_0),$$

故 $H_0(K,K_0)=Z_0(K,K_0)/B_0(K,K_0)=0$.

但幸运的是, 下面的引理使得我们计算相对同调群时无需每次都要走一次那么复杂的流程:

定理 1.27.1 (切除定理)**.** 设 K_0 是 K 的子复形, 若开集 $U\subset|K_0|$ 使得

$$|K_0|\setminus U=|L_0|,\qquad |K|\setminus U=|L|,$$

这里 L_0 是 K_0 的子复形, L 是 K 的子复形, 则

$$H_p(K,K_0)\cong H_p(L,L_0).$$

$$\begin{array}{ccccc} C_p(L) & \xhookrightarrow{\iota} & C_p(K) & \xrightarrow{\pi} & C_p(K,K_0) \\ & \searrow^{\varphi} & & \nearrow & \end{array}$$

证明. 因 L 是 K 的子复形, 于是 由 $\sigma\in K\setminus K_0$ 有 $\{\sigma_i\}$

$$c\longmapsto c\longmapsto c+C_p(K_0)$$

是 $C_p(K,K_0)$ 的基, 而 $\sigma_i\in L$, 故 φ 是满的. 又因为 $\ker\varphi=C_p(L_0)$, 由第一同构定理可知 φ 诱导同构

$$C_p(L)/C_p(L_0)\cong C_p(K,K_0).$$

只需要 ∂ 也被这一同构保持即可. 通过直接计算可以验证下图

$$\begin{array}{ccc} c+C_p(L_0) & \longmapsto & \\ \downarrow & & \\ \partial c+C_{p-1}(L_0) & \longmapsto & \end{array}\qquad \begin{array}{ccc} C_p(L,L_0) & \xrightarrow{\bar{\varphi}} & C \\ \downarrow{\scriptstyle\partial} & & \\ C_{p-1}(L,L_0) & \xrightarrow{\bar{\varphi}} & C_{p-1} \end{array}$$

交换, 从而 $H_p(L,L_0)\cong H_p(K,K_0)$. □

设 K_0 是 K 的子复形, 那么 $K_0\overset{\iota}{\hookrightarrow}K$ 诱导 $H_p(K_0)\overset{\iota_*}{\hookrightarrow}H_p(K)$. 而 K 又可以看作复形对 $(K,\emptyset)$, 于是 $(K,\emptyset)\overset{j}{\hookrightarrow}(K,K_0)$ 又可以诱导 $H_p(K)\overset{j_*}{\hookrightarrow}H_p(K,K_0)$. 于是我们得到一个序列

$$H_p(K_0)\xhookrightarrow{\iota_*}H_p(K)\xhookrightarrow{j_*}H_p(K,K_0)\xrightarrow{\partial_*}H_{p-1}(K_0)$$

在继续讨论它之前, 我们需要一点点

$$\{\bar{c}_p\}\longmapsto\{\partial\bar{c}_p\}$$

其他的基础知识.

1.28　正合序列

之前我们讨论的是 R–模上的正合列, 这里只需要讨论 $\mathbb{Z}$–模, 或者说 Abel 群的正合列即可.

定义 1.28.1 (正合序列)**.** 设 (A_i) 是一列 *Abel* 群, (φ_i) 是一列群同态, 若序列 $\cdots \longrightarrow A_{i-1} \xrightarrow{\varphi_{i-1}} A_i \xrightarrow{\varphi_i} A_{i+1}$ 满足 $\operatorname{im}\varphi_{i-1} = \ker\varphi_i$, 则称序列在 A_i 处**正合**. 若它处处正合, 则称序列是一个**正合序列**.

命题 1.28.1. 正合列具有下面基本的性质. 这些性质验证起来都比较容易:

(1) 若序列 $A_1 \xrightarrow{\varphi} A_2 \to 0$ 正合, 则 φ 是满同态;

(2) 若序列 $0 \to A_1 \xrightarrow{\psi} A_2$ 正合, 则 ψ 是单同态;

(3) 若序列 $0 \to A_1 \xrightarrow{\varphi} A_2 \xrightarrow{\psi} A_3 \to 0$ 正合, 则 $\psi\varphi = 0$.

(4) 短正合列 $A_1 \xrightarrow{\varphi} A_2 \longrightarrow A_3 \longrightarrow A_4 \xrightarrow{\psi} A_5$ 诱导的序列 $0 \longrightarrow \operatorname{coker}\varphi \longrightarrow A_3 \longrightarrow \ker$ 也是正合的.

定义 1.28.2 (分裂的正合列)**.** 称短正合列 $0 \longrightarrow A_1 \xrightarrow{\varphi} A_2 \xrightarrow{\psi} A_3 \longrightarrow 0$ 是**分裂**的, 若 $A_2 = \operatorname{im}\varphi \oplus B$. 此时 $\psi|_B : B \to A_3$ 是同构.

证明. $\psi|_B$ 是满同态容易得到, 下证它是单的. 设 $b_1, b_2 \in B$ 使得 $\psi(b_1) = \psi(b_2)$, 则

$$\begin{aligned}\psi(b_1) - \psi(b_2) = \psi(b_1 - b_2) = 0 &\Longrightarrow b_1 - b_2 \in \ker\psi = \operatorname{im}\varphi \\ &\Longrightarrow b_1 - b_2 \in \operatorname{im}\varphi \cap B \\ &\Longrightarrow b_1 - b_2 = 0.\end{aligned}$$

从而 $\psi|_B$ 是同构. □

此时对于分裂的正合列, 有图

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A_1 & \xrightarrow{\varphi} & A_2 & \xrightarrow{\psi} & A_3 & \longrightarrow & 0 \\ & & \| & & \downarrow{\scriptstyle\theta} & & \| & & \\ 0 & \longrightarrow & A_1 & \xhookrightarrow{\iota} & A_1 \oplus A_3 & \xrightarrow{\pi} & A_3 & \longrightarrow & 0 \end{array}$$

交换, 这即群扩张.

命题 1.28.2 (分裂正合列的判定)**.** 设 $0 \longrightarrow A_1 \xrightarrow{\varphi} A_2 \xrightarrow{\psi} A_3 \longrightarrow 0$ 是短正合列, 下列叙述等价:

(1) 序列是分裂的;

(2) 存在 $p : A_2 \to A_1$ 使得 $p\varphi = \mathrm{id}_{A_1}$;

(3) 存在 $j : A_3 \to A_2$ 使得 $\psi j = \mathrm{id}_{A_3}$.

证明. (1) ⇒ (2, 3) : 只需注意到正合列等价于 $0 \longrightarrow A_1 \underset{p}{\overset{\varphi}{\rightleftarrows}} A_2 \underset{j}{\overset{\psi}{\rightleftarrows}} A_3 \longrightarrow 0$ 取 $p : A_1 \oplus A_3 \twoheadrightarrow A_1$ 是投影, 而 $j : A_3 \hookrightarrow A_1 \oplus A_3$ 是嵌入即可.

(2) ⇒ (1) : 因对任意 $x \in A_2$, 有

$$x = \varphi p(x) + (x - \varphi p(x)) \in \operatorname{im}\varphi + \ker p,$$

故只需验证这是直和. 取 $x \in \operatorname{im}\varphi \cap \ker p$, 有

$$x \in \operatorname{im}\varphi \Longrightarrow \exists y \in A_1\,(x = \varphi(y)),$$

而

$$x \in \ker p \Longrightarrow p(x) = p(\varphi(y)) = y = 0 \Longrightarrow \varphi(y) = 0$$

这导出 $x = 0$.

(3) $\Rightarrow$ (1) : 类似于上面, 对任意 $x \in A_2$, 有

$$x = (x - j\psi(x)) + j\psi(x) \in \ker\psi + j(A_3)$$

只需验证这是直和. 取 $x \in \ker\psi \cap j(A_3)$, 有

$$x \in j(A_3) \Longrightarrow \exists y \in A_3(x = j(y)),$$

而

$$x \in \ker\psi \Longrightarrow y = \psi j(y) = \psi(x) = 0 \Longrightarrow j(y) = 0,$$

这导出 $x = 0$. □

推论 1.28.1. *若 A_3 是自由 Abel 群, 则短正合列 $0 \longrightarrow A_1 \xrightarrow{\varphi} A_2 \xrightarrow{\psi} A_3 \longrightarrow 0$ 分裂.*

证明. 取 A_3 的一组基 $\{e_i\}$, 则 ψ 是满同态, $\psi^{-1} \neq \emptyset$. 取

$$j : A_3 \to A_2, \qquad e_i \mapsto a_i \in \psi^{-1}(e_i),$$

则 $\psi_j(e_i) = \psi(a_i) = e_i$, 这导出 $\psi j = \mathrm{id}_{A_3}$. □

定义 1.28.3 (链复形的短正合列)**.** *用 0 记零链, 设 $\varphi : \mathcal{C} \to \mathcal{D}$ 和 $\psi : \mathcal{D} \to \mathcal{E}$ 都是链映射, 称 $0 \longrightarrow \mathcal{C} \xrightarrow{\varphi} \mathcal{D} \xrightarrow{\psi} \mathcal{E} \longrightarrow 0$ 正合, 若对任何的 $p \in \mathbb{Z}$ 都有 $0 \longrightarrow C_p \xrightarrow{\varphi} D_p \xrightarrow{\psi} E_p \longrightarrow 0$ 正合.*

下面的内容是有关正合列的三个重要结论, 分别是 zig–zag 引理, 五引理和蛇引理. 首先说明的就是 zig–zag 引理, 因为它的存在, 我们可以想到从短正合列出发构造长正合列. 而又因为正合列的性质, 我们可以从此出发构造相对同调群.

引理 1.28.1 (Zig–zag 引理)**.** *设 $0 \longrightarrow \mathcal{C} \xrightarrow{\varphi} \mathcal{D} \xrightarrow{\psi} \mathcal{E} \longrightarrow 0$ 正合, 则有长正合列*

$$\cdots \longrightarrow H_p(\mathcal{C}) \xrightarrow{\varphi_*} H_p(\mathcal{D}) \xrightarrow{\psi_*} H_p(\mathcal{E}) \xrightarrow{\partial_*} H_{p-1}(\mathcal{C}) \longrightarrow \cdots$$

证明. 先明确 ∂_* 的定义: 对 $\{e_p\} \in H_p(\mathcal{E})$, 为了确定 $\partial_*(\{e_p\})$, 我们使用以下的 “ 图追踪法” 来说明解决的思路. *这种方法非常重要, 后面会反复使用.*

$$\begin{array}{ccccccccc} 0 & \longrightarrow & C_p & \xrightarrow{\varphi} & D_p & \xrightarrow{\psi} & E_p & \longrightarrow & 0 \\ & & \partial_C\downarrow & & \partial_D\downarrow & & \partial_E\downarrow & & \\ 0 & \longrightarrow & C_{p-1} & \xrightarrow{\varphi} & D_{p-1} & \xrightarrow{\psi} & E_{p-1} & \longrightarrow & 0 \end{array}$$

因 ψ 是满同态, 于是存在 $d_p \in D_p$ 使得 $\psi(d_p) = e_p$; 对 d_p 做边缘运算: 由 $e_p \in Z_p(\mathcal{E})$ 可知

$$\psi(\partial_D d_p) = \psi\partial_D d_p = \partial_E(\psi(d_p)) = \partial_E e_p = 0,$$

于是 $\partial_D d_p \in \ker\psi = \mathrm{im}\,\varphi$. 于是存在 $c_{p-1} \in C_{p-1}$ 使得 $\varphi(c_{p-1}) = \partial_D d_p$, 还需要验证 $c_{p-1} \in Z_{p-1}(\mathcal{C})$. 计算

$$\varphi(\partial_C c_{p-1}) = \partial_D\varphi(c_{p-1}) = \partial_D(\partial_D d_p) = \partial_D^2 d_p = 0,$$

由 φ 是单同态可知 $\partial_C c_{p-1}=0$, 于是可以定义 $\partial_*(\{e_p\})=\{c_{p-1}\}$.

如此定义的 ∂_* 是良定义的. 为此, 另取 $e'_p\in\{e_p\}$, 并且记类似得到的为 $c'_{p-1}\in Z_{p-1}(\mathcal{C})$, 需证 $\{c_{p-1}\}=\left\{c'_{p-1}\right\}$. 由 $e_p\sim e'_p$ 可知存在 $e_{p+1}\in E_{p+1}$ 使得 $\partial_E e_{p+1}=e_p-e'_p$. 继续使用图追踪法

$$\begin{array}{ccccccccc}
0 & \longrightarrow & C_p & \xrightarrow{\varphi} & D_p & \xrightarrow{\psi} & E_p & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle \partial_C} & & \downarrow{\scriptstyle \partial_D} & & \downarrow{\scriptstyle \partial_E} & & \\
0 & \longrightarrow & C_{p-1} & \xrightarrow{\varphi} & D_{p-1} & \xrightarrow{\psi} & E_{p-1} & \longrightarrow & 0
\end{array}$$

考虑

$$\begin{aligned}
\psi(d_p-d'_p-\partial_D d_{p+1}) &= e_p-e'_p-\psi(\partial_D d_{p+1}) \\
&= e_p-e'_p-\partial_E(\psi(d_{p+1}))=e_p-e'_p-\partial_E e_{p+1}=0,
\end{aligned}$$

即 $d_p-d'_p-\partial_D d_{p+1}\in\ker\psi=\operatorname{im}\varphi$, 故存在 $c_p\in C_p$ 使得 $\varphi(c_p)=d_p-d'_p-\partial_D d_{p+1}$, 计算

$$\varphi(\partial_C c_p)=\partial_D(\varphi(c_p))=\partial_D(d_p-d'_p-\partial_D d_{p+1})=\partial_D d_p-\partial_D d'_p=\varphi(c_{p-1}-c'_{p-1}),$$

因 φ 是单同态, $\partial_C c_p=c_{p-1}-c'_{p-1}$, 即 $c_{p-1}\sim c'_{p-1}$.

下面验证序列在 $H_p(\mathcal{E})$ 处的正合性, 即 $\ker\partial_*=\operatorname{im}\psi_*$.

首先验证 $\ker\partial_*\subset\operatorname{im}\psi_*$: 任取 $\{e_p\}\in\ker\partial_*$, 即 $\{c_{p-1}\}=\partial_*(\{e_p\})=0$. 取 $d_p\in D_p$ 使得 $\psi(d_p)=e_p$. 再取 $c_p\in C_p$ 使得 $\partial_C c_p=c_{p-1}$.

$$\begin{array}{ccccccccc}
0 & \longrightarrow & C_p & \xrightarrow{\varphi} & D_p & \xrightarrow{\psi} & E_p & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle \partial_C} & & \downarrow{\scriptstyle \partial_D} & & \downarrow{\scriptstyle \partial_E} & & \\
0 & \longrightarrow & C_{p-1} & \xrightarrow{\varphi} & D_{p-1} & \xrightarrow{\psi} & E_{p-1} & \longrightarrow & 0
\end{array}$$

考虑 $d_p-\varphi(c_p)$, 计算

$$\partial_D(d_p-\varphi(c_p))=\partial_D d_p-\partial_D\varphi(c_p)=\partial_D d_p-\varphi(\partial_C c_p)=\partial_D d_p-\varphi(c_{p-1})=0,$$

进而

$$\psi_*(\{d_p-\varphi(c_p)\})=\{\psi(d_p)-\psi\varphi(c_p)\}=\{\psi(d_p)\}=\{e_p\}.$$

即 $\{e_p\}\in\operatorname{im}\psi_*$.

再验证 $\operatorname{im}\psi_*\subset\ker\partial_*$: 任取 $\{e_p\}\in\operatorname{im}\psi_*$, 则存在 $\left\{d'_p\right\}\in H_p(\mathcal{D})$ 使得

$$\left\{\psi(d'_p)\right\}=\psi_*(\left\{d'_p\right\})=\{e_p\}.$$

因 $\partial_D d'_p=0$, 有 $e_p\sim\psi(d'_p)=:e'_p$.

$$\begin{array}{ccccccccc}
0 & \longrightarrow & C_p & \xrightarrow{\varphi} & D_p & \xrightarrow{\psi} & E_p & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle \partial_C} & & \downarrow{\scriptstyle \partial_D} & & \downarrow{\scriptstyle \partial_E} & & \\
0 & \longrightarrow & C_{p-1} & \xrightarrow{\varphi} & D_{p-1} & \xrightarrow{\psi} & E_{p-1} & \longrightarrow & 0
\end{array}$$

故 $\partial_*(\{e_p\})=\partial_*(\left\{e'_p\right\})=0$.

序列在 $H_p(\mathcal{D})$ 和在 $H_{p-1}(\mathcal{C})$ 处的正合性类似可证. □

定理 1.28.1. 设图

$$\begin{array}{ccccccccc}
0 & \longrightarrow & \mathcal{C} & \xrightarrow{\varphi} & \mathcal{D} & \xrightarrow{\psi} & \mathcal{E} & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \beta} & & \downarrow{\scriptstyle \gamma} & & \\
0 & \longrightarrow & \mathcal{C}' & \xrightarrow{\varphi} & \mathcal{D}' & \xrightarrow{\psi} & \mathcal{E}' & \longrightarrow & 0
\end{array}$$

交换, 那么图

$$\begin{array}{ccccccc}
\longrightarrow & H_p(\mathcal{C}) & \xrightarrow{\varphi_*} & H_p(\mathcal{D}) & \xrightarrow{\psi_*} & H_p & \\
 & \downarrow{\scriptstyle \alpha_*} & & \downarrow{\scriptstyle \beta_*} & & \downarrow{\scriptstyle \gamma_*} & \\
\longrightarrow & H_p(\mathcal{C}') & \xrightarrow{\varphi'_*} & H_p(\mathcal{D}') & \xrightarrow{\psi'_*} & H_p &
\end{array}$$

也交换.

300×376 像素. 其他分辨率: 320×151 像素.

证明. 考虑中间三个方块的交换性. 前两个方块的交换性是简单的, 以第一个为例. 取 $\{c_p\} \in H_p(\mathcal{C})$, 有

$$\beta_*\varphi_*(\{c_p\}) = \{\beta\varphi(c_p)\} = \{\varphi'\alpha(c_p)\} = \varphi'_*\alpha_*(\{c_p\}),$$

故第一个方块交换.

主要讨论第三个方块的交换性, 取 $\{e_p\} \in H_p(\mathcal{E})$. 则存在 c_{p-1} 和 d_p 使得 $\varphi(c_{p-1}) = \partial_D d_p$, $\psi(d_p) = e_p$. 需要验证 $\alpha_*\partial_*(\{e_p\}) = \partial'_*\gamma_*(\{e_p\})$. 由

$$\alpha_*\partial_*(\{e_p\}) = \alpha_*(\{c_{p-1}\}) = \{\alpha(c_{p-1})\},$$

和

$$\partial_*\gamma_*(\{e_p\}) = \partial'_*(\{\gamma(e_p)\}).$$

本质上只需要验证 $\partial'_*(\{\gamma(e_p)\}) = \{\alpha(c_{p-1})\}$ 即可. 注意到淡蓝色标出的方块都是交换的, 由

$$\psi'\beta(d_p) = \gamma\psi(d_p) = \gamma(e_p)$$

和

$$\varphi'\alpha(c_{p-1}) = \beta\varphi(c_{p-1}) = \beta(\partial_D d_p) = \partial'_D\beta(d_p).$$

从而有 $\partial'_*(\{\gamma(e_p)\}) = \{\alpha(c_{p-1})\}$. □

引理 1.28.2 (五引理)**.** 设 A_i, B_j 是 *Abel* 群, 水平的序列都是正合的.

$$\begin{array}{ccccccccc} A_1 & \xrightarrow{\alpha_1} & A_2 & \xrightarrow{\alpha_2} & A_3 & \xrightarrow{\alpha_3} & A_4 & \xrightarrow{\alpha_4} & A_5 \\ \downarrow{\scriptstyle f_1} & & \downarrow{\scriptstyle f_2} & & \downarrow{\scriptstyle f_3} & & \downarrow{\scriptstyle f_4} & & \downarrow{\scriptstyle f_5} \\ B_1 & \xrightarrow{\beta_1} & B_2 & \xrightarrow{\beta_2} & B_3 & \xrightarrow{\beta_3} & B_4 & \xrightarrow{\beta_4} & B_5 \end{array}$$

那么

(1) 若 f_1 是满的, f_2, f_4 是单的, 则 f_3 是单的;

(2) 若 f_5 是单的, f_2, f_4 是满的, 则 f_3 是满的;

(3) 若 f_1 是满的, f_5 是单的, f_2, f_4 是同构, 则 f_3 是同构.

证明. (1) 使用图追踪法: $\begin{array}{ccccccccc} A_1 & \xrightarrow{\alpha_1} & A_2 & \xrightarrow{\alpha_2} & A_3 & \xrightarrow{\alpha_3} & A_4 & \xrightarrow{\alpha_4} & A_5 \\ \downarrow{\scriptstyle f_1} & & \downarrow{\scriptstyle f_2} & & \downarrow{\scriptstyle f_3} & & \downarrow{\scriptstyle f_4} & & \downarrow{\scriptstyle f_5} \\ B_1 & \xrightarrow{\beta_1} & B_2 & \xrightarrow{\beta_2} & B_3 & \xrightarrow{\beta_3} & B_4 & \xrightarrow{\beta_4} & B_5 \end{array}$ 那么取 $x \in \ker f_3$, 由 f_4 是单同态可知 $\alpha_3(x) = 0$, 而 w 和 $f_1(v)$ 的存在性由正合性得到. 这里 v 的存在性因 f_1 是满同态.

由图表的交换性, 有 $f_2(w)=f_2(\alpha_1(v))$, 由 f_2 是单同态, 有 $w=\alpha_1(v)$. 从而

$$x=\alpha_2(w)=\alpha_2\alpha_1(v)=0,$$

其中 $\alpha_2\alpha_1=0$ 由正合性得到, 即 f_3 是单的.

(2) 使用图追踪法: $\begin{array}{ccccccccc} A_1 & \xrightarrow{\alpha_1} & A_2 & \xrightarrow{\alpha_2} & A_3 & \xrightarrow{\alpha_3} & A_4 & \xrightarrow{\alpha_4} & A_5 \\ \downarrow{\scriptstyle f_1} & & \downarrow{\scriptstyle f_2} & & \downarrow{\scriptstyle f_3} & & \downarrow{\scriptstyle f_4} & & \downarrow{\scriptstyle f_5} \\ B_1 & \xrightarrow{\beta_1} & B_2 & \xrightarrow{\beta_2} & B_3 & \xrightarrow{\beta_3} & B_4 & \xrightarrow{\beta_4} & B_5 \end{array}$ 任取 $y\in B_3$, $w\in A_4$ 的存在性由 f_4 是满同态得到. 由正合性可知 $\beta_4\beta_3=0$, 而 f_5 是单同态, 于是

$$w\in\ker\alpha_4=\operatorname{im}\alpha_3,$$

那么 x 的存在性也得到了. 由图表的交换性可得

$$\beta_3f_3(x)=f_4\alpha_3(x)=\beta_3(y),$$

那么 $y-f_3(x)\in\ker\beta_3=\operatorname{im}\beta_2$, 故 z 的存在性得到. 而 $v\in X_2$ 的存在性由 f_2 是满同态得到. 此时

$$f_3(\alpha_2(v)+x)=f_3\alpha_2(v)+f_3(x)=y-f_3(x)+f_3(x)=y,$$

即 $y\in\operatorname{im}f_3$.

(3) 是 (1) 和 (2) 的直接推论. □

引理 1.28.3 (蛇引理)**.** 设下图中水平序列均是正合的, α, β, γ 是同态,

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C & \longrightarrow & 0 \\ & & \alpha\downarrow & & \beta\downarrow & & \gamma\downarrow & & \\ 0 & \longrightarrow & D & \longrightarrow & E & \longrightarrow & F & \longrightarrow & 0 \end{array}$$

则序列 $0\longrightarrow\ker\alpha\longrightarrow\ker\beta\longrightarrow\ker\gamma\longrightarrow\operatorname{coker}\alpha\longrightarrow\operatorname{coker}\beta\longrightarrow\operatorname{coker}\gamma\longrightarrow 0$ 是正合的.

证明. 我们考虑下面的交换图

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \ker\alpha & \longrightarrow & \ker\beta & \longrightarrow & \ker\gamma & & \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & A & \xrightarrow{\varphi} & B & \xrightarrow{\psi} & C & \longrightarrow & 0 \\ & & \alpha\downarrow & & \beta\downarrow & & \gamma\downarrow & & \\ 0 & \longrightarrow & D & \xrightarrow{\varphi'} & E & \xrightarrow{\psi'} & F & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ & & \operatorname{coker}\alpha & \longrightarrow & \operatorname{coker}\beta & \longrightarrow & \operatorname{coker}\gamma & \longrightarrow & 0 \end{array}$$

对它做图追踪. 注意到所有水平方向的序列都是正合的, 所有竖直方向的序列也是正合的, 这直接导出所求证的序列在 $\ker\beta$ 和 $\operatorname{coker}\beta$ 处正合.

下面的证明主要分三步.

(1) 首先, 我们构造 $\delta:\ker\gamma\to\operatorname{coker}\alpha$. 取 $c\in\ker\gamma$, 因 $B\to C\to 0$ 正合, 故 $B\to C$ 是满同态, 即

$$\exists b\in B\,(\psi(b)=c).$$

再设 $e=\beta(b)$, 由图表的交换性由 $\psi'(e)=0$, 故 $e\in\ker\psi'=\operatorname{im}\varphi'$. 那么存在 $d\in D$ 使得 $\varphi'(d)=e$, 并记 d 在 $D\to\operatorname{coker}\alpha$ 下的像为 $\bar{d}$. 于是可以定义 $\delta(c)=d$.

(2) 再验证序列在 $\ker\gamma$ 和 $\operatorname{coker}\alpha$ 处的正合性. 首先考虑 $\ker\gamma$ 处, 序列 $\ker\beta \longrightarrow \ker\gamma \xrightarrow{\delta} \operatorname{coker}\alpha$ 的正合性需要验证 $\ker\delta = \operatorname{im}[\ker\beta \to \ker\gamma]$. 若 $c \in \ker\gamma$ 使得 $\delta(c) = \bar{d} = 0$, 由 δ 的构造过程可知存在一个 $b' \in B$ 使得 $b - b' \in \ker\beta$. 而 $A \to B \to C$ 正合, 故 $c \in \operatorname{im}[\ker\beta \to \ker\gamma]$. 于是 $\ker\gamma$ 处是正合的.

再考虑 $\operatorname{coker}\alpha$ 处, 序列 $\ker\gamma \xrightarrow{\delta} \operatorname{coker}\alpha \longrightarrow \operatorname{coker}\beta$ 的正合性需要验证 $\ker[\operatorname{coker}\alpha \to \operatorname{coker}\beta] = \operatorname{im}\delta$. 若 $d \in D$ 在 $D \to \operatorname{coker}\alpha$ 下的像为 $\bar{d}$, 则 $\bar{d} \in \ker[\operatorname{coker}\alpha \to \operatorname{coker}\beta]$ 当且仅当存在 $b \in B$ 使得 $\beta(b) = \varphi'(d)$. 由图表的交换性可知

$$\gamma\psi(b) = \beta\psi'(\varphi(d)) = 0.$$

于是 $\psi(b) \in \ker\gamma$. 由 δ 的构造可知

$$\bar{d} = 0 \Longleftrightarrow \exists c \in \ker\gamma\,(\delta(c) = \bar{d}).$$

于是 $d \in \operatorname{im}\delta$, 故序列在 $\operatorname{coker}\alpha$ 处正合.

(3) 因出入 0 的同态必为零同态, 故序列在 $\ker\alpha$ 和 $\operatorname{coker}\gamma$ 处的正合性显然. □

一个更简单的办法是直接将竖列看作是链复形, 只有 ∂_1 分别是 α, β, γ, 其他的边缘算子均为零. 此时注意到蛇引理所求证的正合序列无非是 zig–zag 引理导出的长正合列, 而证明中构造的 δ 无非是 zig–zag 引理中的 ∂_*.

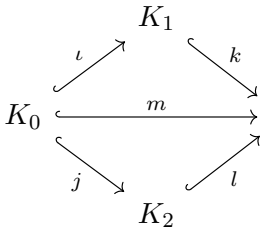
1.29　Mayer–Vietoris 序列

考虑连续映射 $f : (|K|, |K_0|) \to (|L|, |L_0|)$, 它可以诱导长正合列的同态

$$\begin{array}{ccccc} \longrightarrow & H_p(|K_0|) & \longrightarrow & H_p(|K|) & \longrightarrow \\ & \downarrow{\scriptstyle f_*} & & \downarrow{\scriptstyle f_*} & \\ \longrightarrow & H_p(|L_0|) & \longrightarrow & H_p(|L|) & \longrightarrow \end{array}$$

类似于 van Kampen 定理的想法, 对 $K = K_1 \cup K_2$, 且 $K_0 = K_1 \cap K_2$ 是单纯复形, 那么

$$0 \longrightarrow \mathcal{C}(K_0) \xhookrightarrow{\varphi} \mathcal{C}(K_1) \oplus \mathcal{C}(K_2) \xrightarrow{\psi} \mathcal{C}(K) \longrightarrow 0$$



考虑链复形的嵌入

$$c \longmapsto (\iota_\sharp(c), j_\sharp(c))$$

$$(c_1, -c_2) \longmapsto c_1 + c_2$$

那么 φ 是单射, ψ 是满射, 那么 φ, ψ 是链映射.

只是一个给自己的小提示, 在链复形中一定要记得验证 φ, ψ 与 ∂ 的交换性, 即图

$$\begin{array}{ccc} C_p(K_0) & \xhookrightarrow{\varphi} & C_p(K_1) \oplus C_p(K_2) \\ \downarrow{\scriptstyle \partial} & & \downarrow{\scriptstyle \partial} \\ C_{p-1}(K_0) & \xhookrightarrow{\varphi} & C_{p-1}(K_1) \oplus C_{p-1}(K_2) \end{array}$$

下面验证序列的正合性. 即验证 $\ker\psi = \operatorname{im}\varphi$. 这里 $\operatorname{im}\varphi \subset \ker\psi$ 是显然的, 这因 $\psi\varphi = 0$. 下面验证 $\ker\psi \subset \operatorname{im}\varphi$: 取 $c = (c_1, c_2) \in \ker\psi$, 则

$$\psi(c) = c_1 + c_2 = 0 \Longrightarrow c_1 = -c_2 \in \mathcal{C}(K_1) \cap \mathcal{C}(K_2) = \mathcal{C}(K_1 \cap K_2).$$

记 $c_1 = -c_2 = c' \in \mathcal{C}(K_1 \cap K_2)$, 则

$$\varphi(c') = (c', -c') = (c_1, c_2) = c,$$

故 $c \in \operatorname{im}\varphi$.

定义 1.29.1 (Mayer–Vietoris 序列)**.** 设 K 是复形, K_1, K_2 是 K 的子复形, 满足 $K = K_1 \cup K_2$. 记 $K_0 = K_1 \cap K_2$, 则存在长正合序列 $\cdots \longrightarrow H_p(K_0) \longrightarrow H_p(K_1) \oplus H_p(K_2) \longrightarrow H_p(K) \longrightarrow H_{p-1}(K_0) \longrightarrow$ 称作是 (K_1, K_2) 的 ***Mayer–Vietoris* 序列**.

证明. 在上述讨论中已经说明了序列是正合的, 于是它导出同调正合序列. □

Mayer–Vietoris 序列说明了两个复形的一点并导出的同调群恰为它们的直和: 取 $X = A \cup B$, 其中 $A \cap B = \{v\}$, 那么在 $p > 0$ 时列出 Mayer–Vietoris 序列 $\cdots \longrightarrow \tilde{H}_p(v) \longrightarrow \tilde{H}_p(A) \oplus \tilde{H}_p(B) \longrightarrow \tilde{H}_p(X)$ 于是 $\tilde{H}_p(X) \cong \tilde{H}_p(A) \oplus \tilde{H}_p(B)$.

至此, 我们可以完全使用公理化的方式来描述同调论: 下面我们约定提到的空间都是可以三角剖分的拓扑空间. 考虑下面的一组公理:

1. $\mathrm{id} : X \to X$ 应当诱导出恒等映射 id_*.
2. 若 $X \xrightarrow{f} Y \xrightarrow{g} Z$, 则 $(gf)_* = g_* f_*$.
3. 对 $f : (X, A) \to (Y, B)$, 图 $\begin{array}{ccc} H_p(X, A) & \xrightarrow{f_*} & H_p(Y, B) \\ \downarrow{\scriptstyle \partial_*} & & \downarrow{\scriptstyle \partial_*} \\ H_{p-1}(A) & \xrightarrow{(f|_A)_*} & H_{p-1}(B) \end{array}$ 交换.
4. 对复形对 (X, A), 序列 $\cdots \longrightarrow H_p(A) \xrightarrow{\iota_*} H_p(X) \xrightarrow{j_*} H_p(X, A) \xrightarrow{\partial_*} H_{p-1}(A) \longrightarrow \cdots$ 是正合的, 其中 $\iota : X \to A$ 和 $j : (X, \emptyset) \to (X, A)$ 是嵌入.
5. (同伦公理) 若 $f \simeq g : X \to Y$, 则 $f_* = g_*$.
6. (切除公理) 给定复形对 (X, A), 设开集 U 满足 $\bar{U} \subset \operatorname{Int} A$, 则 $H_p(X, A) \cong H_p(X \setminus U, A \setminus U)$.
7. (维数公理) $\tilde{H}_p(\mathrm{pt}) = 0$.
8. (紧支撑公理) 若 $\alpha \in H_p(X, A)$, 则存在 X_0, A_0 是紧的, 同态 $\iota : (X_0, A_0) \to (X, A)$, 使得 $\alpha \in \operatorname{im}[\iota_* : H_p(X_0, A_0) \to H_p(X, A)]$.

在这里如果去掉维数公理, 则称前六条公理构成的同调为**广义同调**. 上述公理称作是单纯同调的 **Eilenberg–Steenrod 公理**.

1.30　奇异链复形

奇异同调导出的思路类似于在对基本群的讨论中, 将道路视作映射 $\sigma : I \to X$, 接下来我们将单形视作映射 $T : \Delta_p \to X$.

在空间 $\mathbb{R}^{\mathbb{N}}$ 上, 有几何独立的点

$$\begin{aligned} \varepsilon_0 &= (0, 0, \dots, 0, \dots) \\ \varepsilon_1 &= (1, 0, \dots, 0, \dots) \\ &\vdots \\ \varepsilon_p &= (0, 0, \dots, 1, \dots) \end{aligned}$$

则 $\{\varepsilon_0, \dots, \varepsilon_p\}$ 确定一个单形 Δ_p.

图 1.13: Δ_3 的所有 2 维面

定义 1.30.1 (奇异单形)**.** *记号同上面的约定, 称连续映射 $T:\Delta_p\to X$ 是一个 p 维*奇异单形*, 此时 p 维奇异单形全体即为 $C(\Delta_p,X)$. 并记以 $C(\Delta_p,X)$ 中元素为生成元产生的自由 Abel 群为 $S_p(X)$, 称作 X 的 p 维*奇异链群*.*

对 $c:C(\Delta_p,X)\to\mathbb{Z}, T\mapsto c(T)$, 它至少应当满足 $c(-T)=-c(T)$, 且仅对 $C(\Delta_p,X)$ 中有限个元素使得 $c(T)\neq 0$.

但将单形看作映射带来的问题是, 如何刻画 T 的 p 维面. 此时已经没有足够好的几何直观, 为此, 我们考虑比较简单的奇异单形:

定义 1.30.2 (线性奇异单形)**.** *设 $a_0,\dots,a_p$ 是 $\mathbb{R}^{\mathbb{N}}$ 中的任意 $p+1$ 个点, 这 $p+1$ 个点未必几何独立. 存在仿射变换*

$$\ell:\Delta_p\to\mathbb{R}^{\mathbb{N}},\qquad \varepsilon_i\mapsto a_i\,(i=0,1,\dots,p),$$

则对 $x=(x_1,x_2,\dots,x_p,0,\dots)$, 有

$$\ell(x)=a_0+\sum_{i=1}^{p}(a_i-a_0)x_i,$$

*记这一仿射变换 ℓ 为 $\ell(a_0,\dots,a_p)$, 并称作 $\mathbb{R}^{\mathbb{N}}$ 中的*线性奇异单形*.*

特别地, 若 $\varepsilon_i=a_i$, 那么此时 $\ell(\varepsilon_0,\dots,\varepsilon_p):\Delta_p\hookrightarrow\mathbb{R}^{\mathbb{N}}$ 是一个到上的同胚, 此时 $\ell(\varepsilon_0,\dots,\varepsilon_p)$ 就是嵌入同胚. 称

$$\ell(\varepsilon_0,\dots,\hat{\varepsilon}_i,\dots,\varepsilon_p):\Delta_{p-1}\hookrightarrow\mathbb{R}^{\mathbb{N}}$$

是 $\ell(\varepsilon_0,\dots,\varepsilon_p)$ 的第 i 个 $p-1$ 维面. 为了方便, 将这一线性奇异单形记作 $\ell_{i,p}$.

定义 1.30.3 (奇异单形的边界)**.** 对 $T_p : \Delta_p \to X$, 称

$$T_p \circ \ell_{i,p} : \Delta_{p-1} \to X$$

是 T_p 的第 i 个 $p-1$ 维面, 故可定义

$$\partial T_p := \sum_{i=0}^{p} (-1)^i T_p \circ \ell_{i,p}$$

是 T_p 的**边界**.

连续映射 $f : X \to Y$ 可以诱导链映射

$$f_\sharp : S_p(X) \to S_p(Y), \qquad T \mapsto f_\sharp(T) := f \circ T.$$

命题 1.30.1. 设 $f \in C(X, Y)$, 则 $f_\sharp \partial = \partial f_\sharp$.

证明. 直接计算

$$f_\sharp \partial T = f_\sharp \left(\sum_{i=0}^{p} (-1)^i T \circ \ell_{i,p} \right) = \sum_{i=0}^{p} (-1)^i f \circ (T \circ \ell_{i,p})$$

和

$$\partial f_\sharp(T) = \partial(f \circ T) = \sum_{i=0}^{p} (-1)^i (f \circ T) \circ \ell_{i,p},$$

得到 $f_\sharp \partial = \partial f_\sharp$. □

定理 1.30.1. $\partial^2 = 0$.

证明. 因 $T \in C(\Delta_p, X)$ 可以诱导

$$T_\sharp : S_p(\Delta_p) \to S_p(X), \qquad \ell(\varepsilon_0, \ldots, \varepsilon_p) \mapsto T \circ \ell(\varepsilon_0, \ldots, \varepsilon_p) = T.$$

则对 $T \in C(\Delta_p, X)$, 计算

$$\begin{aligned}
\partial^2 T &= \partial^2 T_\sharp \ell(\varepsilon_0, \ldots, \varepsilon_p) \\
&= T_\sharp \partial^2 \ell(\varepsilon_0, \ldots, \varepsilon_p) \\
&= T_\sharp \partial \left(\sum_{i=0}^{p} (-1)^i \ell(\varepsilon_0, \ldots, \varepsilon_p) \ell_{i,p} \right) \\
&= T_\sharp \partial \left(\sum_{i=0}^{p} (-1)^i \ell_{i,p} \right) \\
&= 0.
\end{aligned}$$

其中倒数第二个等号因 $\ell(\varepsilon_0, \ldots, \varepsilon_p)$ 实际上就是 id. □

由此我们构造了自由 Abel 群 $S_p(X)$ 和其上的边缘算子 ∂, 并且它们的性质和在单纯同调中的链复形满足的条件是相同的, 于是可以类似地定义链复形如下:

定义 1.30.4 (奇异链复形)**.** 称 $\mathcal{S}(X) := \{(S_p(X), \partial)\}$ 是 X 的**奇异链复形**, 由此可以定义同调群

$$H_p(\mathcal{S}(X)) = H_p(X) = \ker \partial_p / \operatorname{im} \partial_{p+1},$$

并且记号与单纯同调中保持相同, 仍有 $Z_p(X)$ 和 $B_p(X)$.

类似地, $f \in C(X, Y)$ 不仅可以诱导奇异链复形之间的链映射 $f_\sharp : \mathcal{S}(X) \to \mathcal{S}(Y)$, 还可以诱导奇异同调群的同态 $f_* : H_p(X) \to H_p(Y)$.

定理 1.30.2. $H_* : X \mapsto H_p(X),\ f \mapsto f_*$ 是一个从 Top 到 Grp 的函子.

证明. 只需要说明两点, 首先是恒等映射 $\mathrm{id}_X : X \to X$ 诱导出的 $(\mathrm{id}_X)_*$ 应当仍是恒等映射, 这是显然的. 只需要证明若 $X \xrightarrow{f} Y \xrightarrow{g} Z$, 成立 $(gf)_* = g_* f_*$ 即可. 为此, 只需要在链层面证明: 这因

$$(gf)_\sharp(T) = (gf)T = g(fT) = g_\sharp(fT) = g_\sharp f_\sharp(T),$$

于是协变性自然成立. □

推论 1.30.1. 奇异同调群是拓扑不变量.

证明. 设 $X \cong Y$, 则存在 $f \in C(X, Y)$ 和 $g \in C(Y, X)$ 使得 $gf = \mathrm{id}_X$, $fg = \mathrm{id}_Y$, 那么此时

$$g_* f_* = (gf)_* = (\mathrm{id}_X)_*, \qquad f_* g_* = (fg)_* = (\mathrm{id}_Y)_*,$$

即 f_* 和 g_* 都是同构, 于是 $H_p(X) \cong H_p(Y)$. □

奇异链复形也可以做增广 $\cdots \xrightarrow{\partial_{p+1}} S_p(X) \xrightarrow{\partial_p} \cdots \xrightarrow{\partial_2} S_1(X) \xrightarrow{\partial_1} S_0(X) \xrightarrow{\varepsilon} \mathbb{Z} \longrightarrow 0$ 其中 $\varepsilon : S_0(X) \to \mathbb{Z}$ 将 $T \in C(\Delta_0, X)$ 映成 $\varepsilon(T) = 1$. 由此诱导的同调群称作**约化同调群**, 同样地当 $p > 0$ 时 $\tilde{H}_p(X) = H_p(X)$.

定理 1.30.3. $H_0(X)$ 是自由 *Abel* 群, 且 $H_0(X) \cong \tilde{H}_0(X) \oplus \mathbb{Z}$.

证明. 设 $X = \coprod_{i \in \alpha} X_i$, 其中 X_i 是 X 的道路连通分支. 取 $T_i \in C(\Delta_0, X_i)$, 断言 $\{T_i\}_{i \in \alpha}$ 恰好可以成为使得 $H_0(X)$ 成为自由 Abel 群的生成元系.

对任意 $T \in C(\Delta_0, X)$, 存在 $i \in \alpha$ 使得 $T_i(\Delta_0) \in X_i$, 则存在道路

$$\sigma : I \to X_i, \qquad \sigma(0) = T(\Delta_0), \quad \sigma(1) = T_i(\Delta_0),$$

视 σ 是一个 $C(\Delta_1, X)$ 中的元素, 即 1 维奇异单形, 记 $\Delta_1 = (\varepsilon_0, \varepsilon_1) = (0, 1)$, 则

$$\partial\sigma = \sigma\ell_{0,1} - \sigma\ell_{1,1} = T_i - T.$$

这说明 $T \sim T_i$, 即道路连通分支 X_i 中任何两个 0 维单形都在同一个同调类中, 那么对任意 $c_0 \in S_0(X)$, 它都与 $\sum_{i \in \alpha} n_i T_i$ 同调. 也即 $H_0(X)$ 由 $\{T_i\}_{i \in \alpha}$ 生成.

要证 $H_0(X)$ 是自由的, 还需要 $\{T_i\}_{i \in \alpha}$ 无关且任何 T_i 都是无限阶的. 由此令

$$\left\{\sum_{i \in \alpha} n_i T_i\right\} = 0 \Longrightarrow \sum_{i \in \alpha} n_i T_i \in B_0(X) \Longrightarrow \exists d \in S_1(X) \left(\partial d = \sum_{i \in \alpha} n_i T_i\right).$$

又 $d=\sum\limits_{i\in\alpha}d_i$, 其中 $d_i\in S_1(X_i)$, 那么

$$\partial d=\sum_{i\in\alpha}\partial d_i=\sum_{i\in\alpha}n_iT_i\Longrightarrow\forall i\in\alpha\,(\partial d_i=n_iT_i),$$

于是 $\varepsilon\partial d_i=\varepsilon(n_iT_i)=n_i$. 而 $\varepsilon\partial=0$, 这导出 $n_i=0$ 对任意 $i\in\alpha$ 都成立, 这说明 $\left\{\sum\limits_{i\in\alpha}n_iT_i\right\}=0\Longrightarrow n_i=0$, 于是 $H_0(X)$ 是自由 Abel 群. □

例 1.30.1. 考虑单点拓扑空间 pt, 因

$$C(\Delta_0,\mathrm{pt})=\{T:\Delta_p\to\mathrm{pt},\ x\mapsto\mathrm{pt}\}$$

是单点集, 故 $S_p(\mathrm{pt})\cong\mathbb{Z}$, 于是 $\mathcal{S}(\mathrm{pt})$ 具有以下的形式: $\cdots\xrightarrow{\partial}S_p(\mathrm{pt})\xrightarrow{\partial}S_{p-1}(\mathrm{pt})\xrightarrow{\partial}\cdots\xrightarrow{\partial}S_1(\mathrm{pt})\to$ 且

$$\partial_pT_p=\sum_{i=0}^{p}(-1)^iT_p\ell_{i,p}=\sum_{i=0}^{p}(-1)^iT_{p-1}=\begin{cases}0, & n=2k+1\\ T_{p-1}, & n=2k\end{cases}$$

这源于 $S_{p-1}(\mathrm{pt})$ 中只有一个元素 T_{p-1}, 故

$$Z_{2k+1}(\mathrm{pt})=B_{2k+1}(\mathrm{pt})=S_p(\mathrm{pt})\Longrightarrow\tilde{H}_{2k+1}(\mathrm{pt})=0,$$

而 $p\neq0$ 时有

$$Z_{2k}(\mathrm{pt})=B_{2k}(\mathrm{pt})=0\Longrightarrow\tilde{H}_{2k}(\mathrm{pt})=0,$$

当 $p=0$ 时

$$Z_0(\mathrm{pt})=S_0(\mathrm{pt}),\ B_0(\mathrm{pt})=0\Longrightarrow\tilde{H}_0(\mathrm{pt})=0.$$

这说明对任意 $p\in\mathbb{Z}$ 都有 $\tilde{H}_p(\mathrm{pt})=0$. (需要注意的是和单纯链复形不同, 单点拓扑空间的奇异链复形向 $p>0$ 的方向可以全部都是不平凡的.)

设 (X,A) 是一个复形对, 注意到 $C(\Delta_p,A)$ 是 $S_p(A)$ 的生成元集, $C(\Delta_p,X)$ 是 $S_p(X)$ 的生成元集, 那么 $C(\Delta_p,A)\subset C(\Delta_p,X)$ 就导出 $S_p(X)/S_p(A)$ 是自由的. 称 $S_p(X,A)$ 是 (X,A) 的**相对链群**, 其生成元系为 $\{T+S_p(A):T\in C(\Delta_p,X)\setminus C(\Delta_p,A)\}$. 此时可以自然地定义边缘算子 ∂, 相对链复形 $\mathcal{S}(X,A)$ 由此诱导出相对同调群 $H_p(X,A)$.

由此, 有短正合序列 $0\longrightarrow S_p(A)\hookrightarrow S_p(X)\longrightarrow S_p(X,A)\longrightarrow0$ 由 zig–zag 引理导出长正合序列 $\cdots\longrightarrow H_p(A)\longrightarrow H_p(X)\longrightarrow H_p(X,A)\longrightarrow H_{p-1}(A)\longrightarrow\cdots$

1.31　奇异同调的公理

奇异同调同样有下面的公理:

1. $\mathrm{id}_X:X\to X$ 诱导 $(\mathrm{id}_X)_*$ 是恒等映射;
2. $X\xrightarrow{f}Y\xrightarrow{g}Z$ 导出 $(gf)_*=g_*f_*$;
3. 对 $f:(X,A)\to(Y,B)$, 图 $\begin{array}{ccc}H_p(X,A) & \xrightarrow{f_*} & H_p(Y,B)\\ \downarrow\partial_* & & \downarrow\partial_*\\ H_{p-1}(A) & \xrightarrow{(f|_A)_*} & H_{p-1}(B)\end{array}$ 交换;

4. 正合公理: 设 (X,A) 是复形对, 有长正合列 $\cdots \longrightarrow H_p(A) \xrightarrow{\iota_*} H_p(X) \xrightarrow{\pi_*} H_p(X,A) \xrightarrow{\partial_*} H_{p-1}(A$
5. 同伦公理: 若 $f \simeq g : X \to Y$, 则 $f_* = g_*$;
6. 切除公理: 设 (X,A) 是复形对, 若 U 使得 $\bar{U} \subset \operatorname{Int} A$, 则 $H_p(X,A) \cong H_p(X \setminus U, A \setminus U)$; (注意, 这里没有对 U 是开集的限制了)
7. 维数公理: $\forall p \in \mathbb{Z}\,(\tilde{H}_p(\mathrm{pt}) = 0)$.

与单纯同调类似, 奇异同调的公理也都是可以证明的. 其中 (1) 和 (2) 即奇异同调群是拓扑不变量, (3) 是自然的, (4) 由 zig–zag 引理可证. (6) 的证明与单纯同调如出一辙, (7) 即 1.30.1 的结论.

下面我们着重证明 (5), 即奇异同调群的同伦不变性. 在叙述和证明这一结论之前, 先来简单地列一下思路: $f \simeq g$ 即意味着存在 $H : X \times I \to Y$ 使得 $H(\cdot,0) = f$ 且 $H(\cdot,1) = g$, 于是我们可以将 X 加厚: 若将 i, j 代数化得到 $i_\sharp, j_\sharp$, 只需要证明 $i_\sharp$ 与 $j_\sharp$ 链同伦, 那么

$$g_\sharp = H_\sharp j_\sharp, \qquad f_\sharp = H_\sharp i_\sharp.$$

如果可以找到 $D_X : S_p(X) \to S_{p+1}(X \times I)$ 使得 $\partial D_X + D_X \partial = j_\sharp - i_\sharp$, 那么它复合上 H 诱导的链映射 $H_\sharp : S_{p+1}(X \times I) \to S_p(Y)$ 就使得 $D := H_\sharp D_X : S_p(X) \to S_p(Y)$ 是连接 $g_\sharp$ 和 $f_\sharp$ 的链同伦. 这因

$$\partial D + D\partial = \partial H_\sharp D_X + H_\sharp D_X \partial = H_\sharp(\partial D_X + D_X \partial) = H_\sharp(j_\sharp - i_\sharp) = g_\sharp - f_\sharp.$$

引理 1.31.1. $i_\sharp \simeq j_\sharp$.

证明. 我们需要构造这样的 $D_X : S_p(X) \to S_{p+1}(X \times I)$ 使得

$$\partial D_X + D_X \partial = j_\sharp - i_\sharp,$$

且图
$$\begin{array}{ccc} S_p(X) & \xrightarrow{D_X} & S_{p+1}(X \times I) \\ \downarrow{\scriptstyle f_\sharp} & & \downarrow{\scriptstyle (f \times \mathrm{id}_I)_\sharp} \\ S_p(Y) & \xrightarrow{D_Y} & S_{p+1}(Y \times I) \end{array}$$
交换.

当 $p = 0$ 时, 观察 $D_X : S_0(X) \to S_1(X \times I)$, 取 $T \in C(\Delta_0, X)$, 考虑 $T(\Delta_0) = x_0$

$$D_X(T) : \Delta_1 \to X, \qquad (t, 0, \dots) = (x_0, t),$$

其中 $0 \leqslant t \leqslant 1$. 此时

$$\partial D_X(T) = D_X(T) \circ \ell[\hat{\varepsilon}_0, \varepsilon_1] - D_X(T) \circ \ell[\varepsilon_0, \hat{\varepsilon}_1] = j_\sharp(T) - i_\sharp(T).$$

这因

$$D_X(T) \circ \ell[\hat{\varepsilon}_0, \varepsilon_1](\Delta_0) = D_X(T)(1, 0, \dots) = (x_0, 1) = j_\sharp(T)(\Delta_0),$$

类似地有 $t=0$ 的情况. 再验证 D_X 的函子性, 对任意 $T:\Delta_0\in X$ 和任意 $(t,0,\dots)\in\Delta_1$, 有

$$D_Y f_\sharp(T)(t,0,\dots)=D_X(f\circ T)(t,0,\dots)=(f(x_0),t)$$

和

$$(f\times \mathrm{id}_I)_\sharp D_X(T)(t,0,\dots)=(f\times \mathrm{id}_I)_\sharp(x_0,t)=(f(x_0),t)$$

可知图交换.

下面假设 $<p$ 的情形都成立, 考虑 p 维链群上的 D_X: 首先考虑特殊情形, 当 $X=\Delta_p$ 时, 我们需要定义一个这样的 D_{Δ_p}, 如果它存在, 应当有

$$\partial D_{\Delta_p}(\mathrm{id}_{\Delta_p})=\hat{\jmath}_\sharp(\mathrm{id}_{\Delta_p})-\hat{\imath}(\mathrm{id}_{\Delta_p})-D_{\Delta_p}(\partial\mathrm{id}_{\Delta_p}),$$

而后三项都有定义 (注意到 $\partial\mathrm{id}_{\Delta_p}\in S_{p-1}(\Delta_p)$), 又因 Δ_p 零调, 于是 $Z_p(\Delta_p)=B_p(\Delta_p)$. 因此只需 $\partial D_{\Delta_p}(\mathrm{id}_{\Delta_p})=c_p\in Z_p(\Delta_p)$ 即可.

直接计算验证即可:

$$\begin{aligned}\partial c_p&=\partial(\hat{\jmath}_\sharp(\mathrm{id}_{\Delta_p})-\hat{\imath}_\sharp(\mathrm{id}_{\Delta_p})-D_{\Delta_p}(\partial\mathrm{id}_{\Delta_p}))\\&=\partial\hat{\jmath}(\mathrm{id}_{\Delta_p})-\partial\hat{\imath}(\mathrm{id}_{\Delta_p})-(\hat{\jmath}_\sharp(\partial\mathrm{id}_{\Delta_p})-\hat{\imath}_\sharp(\partial\mathrm{id}_{\Delta_p})-D_{\Delta_p}\partial(\partial\mathrm{id}_{\Delta_p}))\\&=0.\end{aligned}$$

于是 $c_p\in Z_p(\Delta_p)=B_p(\Delta_p)$, 则存在 $d\in S_{p+1}(\Delta_p)$ 使得 $\partial d=c_p$, 取 $D_{\Delta_p}(\mathrm{id}_{\Delta_p})=d$ 即可.

现在任取 X 是一个拓扑空间, 定义

$$D_X:S_p(X)\to S_{p+1}(X\times I),\qquad T:\Delta_p\to X\mapsto D_X(T):=(T\times\mathrm{id}_I)_\sharp D_{\Delta_p}(\mathrm{id}_{\Delta_p}).$$

下面只需验证 $\partial D_X=D_X\partial=j_\sharp-i_\sharp$ 即可. 对任意 $T\in S_p(X)$,

$$\begin{aligned}\partial D_X(T)&=\partial((T\times\mathrm{id}_I)_\sharp D_{\Delta_p}(\mathrm{id}_{\Delta_p}))\\&=(T\times\mathrm{id}_I)_\sharp\partial D_{\Delta_p}(\mathrm{id}_{\Delta_p})\\&=(T\times\mathrm{id}_I)_\sharp(\hat{\jmath}(\mathrm{id}_{\Delta_p})-\hat{\imath}(\mathrm{id}_{\Delta_p})-D_{\Delta_p}(\partial\mathrm{id}_{\Delta_p}))\\&=j_\sharp(T)-i_\sharp(T)-(T\times\mathrm{id}_I)_\sharp D_{\Delta_p}(\partial\mathrm{id}_{\Delta_p})\\&=j_\sharp(T)-i_\sharp(T)-D_XT_\sharp(\partial\mathrm{id}_{\Delta-p})\\&=j_\sharp(T)-i_\sharp(T)-D_X\partial T_\sharp(\mathrm{id}_{\Delta_p})\\&=j_\sharp(T)-i_\sharp(T)-D_X\partial(T).\end{aligned}$$

再验证 D_X 在 S_p 上的函子性. 对任意 $T:\Delta_p\to X$ 和 $f:X\to Y$, 有

$$D_Y f_\sharp(T)=D_Y(f\circ T)=((f\circ T)\times \mathrm{id}_I)_\sharp D_{\Delta_p}(\mathrm{id}_{\Delta_p})$$

和

$$(f\times \mathrm{id}_I)_\sharp D_X(T)=(f\times \mathrm{id}_I)_\sharp\circ(T\times \mathrm{id}_I)_\sharp D_{\Delta_p}(\mathrm{id}_{\Delta_p})=((f\circ T)\times \mathrm{id}_I)_\sharp D_{\Delta_p}(\mathrm{id}_{\Delta_p}),$$

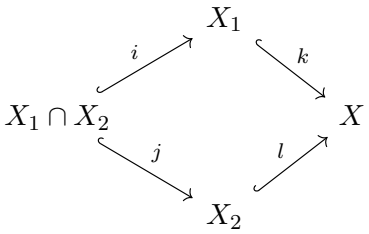
于是在 S_p 上图表交换.

由归纳法可知命题成立. □

定理 1.31.1. *若 $f\simeq g$, 则 $f_*=g_*$.*

这一定理实际上就是 (5), 其证明由引理 1.31.1 和它上面的讨论得到.

对奇异链, 一般来说 $\mathcal{S}(X_1)\oplus\mathcal{S}(X_2)\to\mathcal{S}(X)$ 并不是满的, 因为可能出现下图这样的状况: 但对



它仍然可以诱导短正合列

$$0\longrightarrow \mathcal{S}(X_1\cap X_2)\longrightarrow \mathcal{S}(X_1)\oplus\mathcal{S}(X_2)\longrightarrow \mathcal{S}($$

$$c\longmapsto (i_\sharp(c),-j_\sharp(c))$$

$$(c_1,c_2)\longmapsto$$

由 zig–zag 引理可知它诱导长正合列 $\longrightarrow H_p(X_1\cap X_2)\longrightarrow H_p(X_1)\oplus H_p(X_2)\longrightarrow H_p(\mathcal{S}(X_1)+\mathcal{S}(X_2))$

我们当然希望 $H_p(\mathcal{S}(X_1)+\mathcal{S}(X_2))\cong H_p(X)$, 这样上面的序列就是 Mayer–Vietoris 序列.

定义 1.31.1 (切除对)**.** *设 $X=X_1\cup X_2$, 若*

$$\mathcal{S}(X_1)+\mathcal{S}(X_2)\hookrightarrow\mathcal{S}(X),$$

诱导同调群的同构, 也即 $H_p(\mathcal{S}(X_1)+\mathcal{S}(X_2))\cong H_p(X)$, 则称 (X_1,X_2) 是 X 的一个**切除对**.

定理 1.31.2. 若 (X_1,X_2) 是 X 的切除对, 则存在长正合列 $\longrightarrow H_p(X_1\cap X_2)\longrightarrow H_p(X_1)\oplus H_p(X_2)\longrightarrow H$ 此即切除对 (X_1,X_2) 的 ***Mayer–Vietoris* 序列**.

但直接从定义出发构造切除对太过困难, 下面提供一个简单的构造切除对的方法.

引理 1.31.2. (X_1,X_2) 是 X 的切除对当且仕当 $(X_1,X_1\cap X_2)\hookrightarrow(X,X_2)$ 诱导同调群的同构. *(*此即 *Munkres* 书中习题 *26-3).*

证明. 我们有
$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathcal{S}(X_2) & \hookrightarrow & \mathcal{S}(X) & \twoheadrightarrow & \mathcal{S}(X,X_2) & \longrightarrow & 0 \\ & & \| & & \uparrow & & \uparrow & & \\ 0 & \longrightarrow & \mathcal{S}(X_2) & \hookrightarrow & \mathcal{S}(X_1)+\mathcal{S}(X_2) & \twoheadrightarrow & \mathcal{S}(X_1,X_1\cap X_2) & \longrightarrow & 0 \end{array}$$
其中上下两行都是正合的. 下方的序列正合因取 $\pi:\mathcal{S}(X_1)+\mathcal{S}(X_2)\to\mathcal{S}(X_1,X_1\cap X_2)$ 使得 $\ker\pi=\mathcal{S}(X_2)$.

它们会诱导长正合列
$$\begin{array}{ccccccccc} \longrightarrow & H_{p+1}(X,X_2) & \longrightarrow & H_p(X_2) & \longrightarrow & H_p(X) & \longrightarrow & H_p(X,X_2) & \longrightarrow \\ & \uparrow & & \| & & \uparrow & & \uparrow & \\ \longrightarrow & H_{p+1}(X_1,X_0) & \longrightarrow & H_p(X_2) & \longrightarrow & H_p(\mathcal{S}(X_1)+\mathcal{S}(X_2)) & \longrightarrow & H_p(X_1,X_0) & \longrightarrow \end{array}$$
其中 $X_0=X_1\cap X_2$. 由五引理得证. □

命题 1.31.1. 若 $\{\operatorname{Int}X_1,\operatorname{Int}X_2\}$ 是 X 的开覆盖, 则 (X_1,X_2) 是 X 的切除对.

证明. 不失一般性, 不妨 X_1 和 X_2 都是开集, 那么取 $U=X_2\setminus(X_1\cap X_2)$, 则
$$\bar{U}=\overline{X_2\setminus(X_1\cap X_2)}\subset X\setminus X_1\subset X_2=\operatorname{Int}X_2.$$
由切除公理, 有
$$H_p(X,X_2)\cong H_p(X\setminus U,X_2\setminus U)=H_p(X_1,X_1\cap X_2),$$
得证. □

例 1.31.1 (球面的一点并)**.** 设 $X=\mathbb{S}^m\vee\mathbb{S}^n$, $m,n>0$, 取 $X_1=\mathbb{S}^m$, $X_2=\mathbb{S}^n$, 则 (X_1,X_2) 是 X 的切除对. 我们有 *Mayer–Vietoris* 序列
$$\to H_p(\mathrm{pt})\to H_p(X_1)\oplus H_p(X_2)\to H_p(X)\to H_{p-1}(\mathrm{pt})\to$$
因 X 道路连通, 故 $H_0(X)\cong\mathbb{Z}$. 当 $p>0$ 时, 由 *Mayer–Vietoris* 序列可知
$$H_p(X)\cong\begin{cases}\mathbb{Z}\oplus\mathbb{Z}, & p=m=n\\ 0, & p\neq m\end{cases}\qquad H_p(X)\cong\begin{cases}\mathbb{Z}, & p=m\\ \mathbb{Z}, & p=n\\ 0, & p\neq m,\ p\neq n.\end{cases}$$

例 1.31.2 (锥和悬挂 (双锥))**.** 考虑拓扑空间 X, 它的**锥**和**悬挂** *(*或**双锥***)* 可以分别看作以下的商空间*:*
$$w*X:=(X\times I)/(X\times\{1\})$$
和
$$S(X):=(X\times[-1,1])/(X\times\{1\}\cup X\times\{-1\}).$$
将悬挂看作是对空间的一元运算, 那么映射 $S:X\mapsto S(X)$ 可以对自身复合, 归纳地定义 $S^{n+1}(X):=S(S^n(X))$.

锥总是零调的, 于是讨论锥的同调群没什么价值. 但双锥一般情况下并不是可缩的, 于是我们还是需要探讨一下 $S(X)$ 和 X 的同调群之间的关系.

定理 1.31.3. $H_p(S(X)) \cong \tilde{H}_{p-1}(X)$.

证明. 设 $S(X)$ 的两个悬挂点为 v, w, 记

$$X_1 = S(X) \setminus \{w\}, \qquad X_2 = S(X) \setminus \{v\},$$

则 X_1, X_2 是开集, 且 (X_1, X_2) 是 X 的切除对. 考虑投影 π, 有

$$\pi(X \times (-1, 1]) = X_1, \qquad \pi(X \times [-1, 1)) = X_2.$$

考虑切除对 (X_1, X_2) 的 Mayer–Vietoris 序列,

$$\to H_p(X_1) \oplus H_p(X_2) \to H_p(S(X)) \to H_{p-1}(X_1 \cap X_2) \to H_{p-1}(X_1) \oplus H_{p-1}(X_2) \to$$

而 X_1, X_2 都可缩, 于是

$$\tilde{H}_p(X_1) = \tilde{H}_p(X_2) = 0.$$

对 $X_1 \cap X_2$, 注意到投影 π 限制到 $X \times (-1, 1)$ 上是一个双射, 且 $X \times \{0\}$ 是一个形变收缩核, 那么 $X_1 \cap X_2 \simeq X$. 于是命题得证. □

1.32　应用: 拓扑流形和局部同调群

下面总假设 X 是一个 Hausdorff 空间, $x \in X$, A 是 x 的邻域, $U = X \setminus A$. 那么

$$\bar{U} = \overline{X \setminus A} \subset X \setminus x = \operatorname{Int}(X \setminus x),$$

由切除公理可知

$$H_p(X, X \setminus x) \cong H_p(A, A \setminus x),$$

这称作 x 处的**局部同调**.

例 1.32.1. 对 $X=\mathbb{R}^n$, 有

$$H_p(\mathbb{R}^n,\mathbb{R}^n\setminus\{0\})\cong H_p(\mathbb{B}^n,\mathbb{B}^n\setminus\{0\})\cong H_p(\mathbb{B}^n,\mathbb{S}^{n-1})=\begin{cases}\mathbb{Z}, & p=n\\ 0, & p\neq n.\end{cases}$$

由此可导出对任何 $m\neq n$, 都有 $\mathbb{R}^m$ 和 $\mathbb{R}^n$ 不同胚. 但在点集拓扑中, 要处理这样的问题是十分困难的.

注：由 $\mathbb{R}^n$ 可缩，得到

$$H_p(\mathbb{R}^n,\mathbb{R}^n\setminus\{0\})\cong\tilde{H}_{p-1}(\mathbb{R}^n\setminus\{0\})\cong\tilde{H}_{p-1}(\mathbb{S}^{n-1}).$$

定义 1.32.1 (拓扑流形)**.** 称 M 是一个**拓扑流形**, 若 M 是 *Hausdorff* 的, 且对任意 $x\in M$, 存在 $U\in\mathcal{N}(x)$ 使得存在同胚 $\varphi:U\to\mathbb{R}^m$. 若在局部 $M\cong\mathbb{R}^m$, 则称 M 是**无边流形**, 若在局部 $M\cong\mathbb{H}^m$, 则称 M 是**带边流形**.

因此, 局部同调群可以用来确定一个流形的维数. 对无边流形 M, 有

$$H_p(M,M\setminus x)\cong H_p(\mathbb{R}^m,\mathbb{R}^m\setminus\{0\})\cong\begin{cases}\mathbb{Z}, & p=m\\ 0, & p\neq m.\end{cases}$$

对带边流形 M, 若 $x\in\operatorname{Int}M$, 其局部同调与无边流形类似, 只是此时只有 $p=m-1$ 时 $H_p(M,M\setminus x)\cong\mathbb{Z}$; 若 $x\in\operatorname{Bd}M$, 有

$$H_p(M,M\setminus x)\cong H_p(\mathbb{H}^m,\mathbb{H}^m\setminus\{0\})\cong H_p(\mathbb{B}^m_+,\mathbb{B}^m_+\setminus\{0\})=0.$$

一些有关拓扑流形的同调群的结论列举如下:

定理 1.32.1. 设 M 是可三角剖分的带边流形, 若存在复形 K 使得 $h:|K|\to M$ 是一个同构, 则 $h^{-1}(\operatorname{Bd}M)$ 是 K 的子复形的几何实现.

定理 1.32.2. 设 K 是 M 的一个三角剖分, $v\in K^{(0)}$, 那么对任意 $p\in\mathbb{Z}$ 成立

$$H_p(M,M\setminus h(v))\cong H_p(\overline{\operatorname{St}}\,v,\operatorname{Link}v).$$

1.33　附录

原网页

comment|The author is indebted to 芝麻油 for his great help to and my teaching assistant Jacircen for her careful examination of the solution below. My thanks go also to SHS for offering me the opportunity to have several conversations with professor Luzhi when I was in senior one, without which I would have never taken this course.

4.3 (a) $\mathbb{Z}/m\oplus\mathbb{Z}/n$ contains an element (n,m) of order mn, so $\mathbb{Z}/m\oplus\mathbb{Z}/n\simeq\mathbb{Z}/mn$.

(b) $G\simeq\mathbb{Z}/2\oplus\mathbb{Z}/9\oplus\mathbb{Z}/4\oplus\mathbb{Z}/9$.

(c) $3,6=2\cdot3,36=4\cdot9$.

(d) $30=3\cdot2\cdot5,180=9\cdot4\cdot5$.

6.4 (a) $H_1(X_n) = \mathbb{Z}^{2n}$, $H_2(X_n) = \mathbb{Z}$.
(b) $H_1(Y_n) = \mathbb{Z}^{n-1} \oplus \mathbb{Z}/2$, $H_2(Y_n) = 0$.

9.6 Using Exercise 39.2, $H_i(K, K_0) \simeq \tilde{H}_i(K/K_0) = \tilde{H}_i(\sigma/\operatorname{Bd}\sigma) \simeq \tilde{H}_i(S^2) = \begin{cases} \mathbb{Z}, & n = 2 \\ 0, & \text{else} \end{cases}$.

10.3 $H_i(S;\mathbb{Z}/3) = \begin{cases} \mathbb{Z}/3, & i = 0 \\ \mathbb{Z}/3, & i = 1 \\ 0, & i = 2 \end{cases}$. $H_i(S;\mathbb{Z}/4) = \begin{cases} \mathbb{Z}/4, & i = 0 \\ \mathbb{Z}/4 \oplus \mathbb{Z}/2, & i = 1 \\ \mathbb{Z}/2, & i = 2 \end{cases}$.

14.1 $f(a) = A', N'$, $f(b) = B'$, $f(c) = C'$, $f(d) = C'$, $f(e) = E'$, $f(f) = E', F'$, $f(g) = F', G'$, $f(h) = G', H'$, $f(i) = I'$, $f(j) = J'$, $f(k) = K', L'$, $f(l) = L'$, $f(m) = L', M'$, $f(n) = N'$, $f(o) = O'$, altogether $2^6 = 64$ simplicial approximations.

21.2 (a) Note that the ball $|z| \le c \simeq$ an interior point 0, and so is image $h(|z| \le c) \simeq h(0)$ in $\mathbb{R}^2 - 0$. In particualr, the boundary $h(S_c)$ collapses into a point $h(0)$. It follows that $h_*\{S_c\} = h_*\{0\} = 0$.
(b) Set $F(z,t) = z^n + t(a_{n-1}z^{n-1} + \cdots + a_0)$. The image of F never meets 0 if c is sufficiently large since $|F(z,t)| \ge c^n - t(|a_{n-1}|c^{n-1} + \cdots + |a_0|) > 0$. This is a homotopy between $F(-,1) = h$ and $F(-,0) = k$.
(c) $h_* = k_*\colon H_1(S_c) \to H_1(\mathbb{R}^2 - 0)$ by (b), yet $h_* = 0, k_* = n$.

22.4 (a) $F(-,t)\colon M \to M$ for a fixed $t \in (-\epsilon, \epsilon)$ is a map with no fixed point, and $\simeq$ id as $-$ varies from t to 0. Then $\chi(M) = \Lambda(\mathrm{id}) = 0$.
(b) Exactly those with $\chi(M) = 0$, i.e. the torus or the Klein bottle.

24.7 The long exact sequence we shall use is

$$\begin{array}{ccccccccc} 0 & \longrightarrow & H_2(K;G) & \longrightarrow & H_2(K;G') & \longrightarrow & H_2(K;G'') & & \\ & & & \overset{\beta_*}{\swarrow} & & & & & \\ & & H_1(K;G) & \longrightarrow & H_1(K;G') & \longrightarrow & H_1(K;G'') & & \\ & & & \overset{\beta_*}{\swarrow} & & & & & \\ & & H_0(K;G) & \longrightarrow & H_0(K;G') & \longrightarrow & H_0(K;G'') & \longrightarrow & 0 \end{array}$$

(a) For $|K| = P^2$ and $0 \to \mathbb{Z} \to \mathbb{Z} \to \mathbb{Z}/2 \to 0$, we have

$$\begin{array}{ccccccccc} 0 & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & \mathbb{Z}/2 & & \\ & & & & & \overset{1}{\swarrow} & & & \\ & & \mathbb{Z}/2 & \longrightarrow & \mathbb{Z}/2 & \longrightarrow & \mathbb{Z}/2 & & \\ & & & & & \overset{0}{\swarrow} & & & \\ & & \mathbb{Z} & \longrightarrow & \mathbb{Z} & \longrightarrow & \mathbb{Z}/2 & \longrightarrow & 0. \end{array}$$

For $|K| = P^2$ and $0 \to \mathbb{Z}/2 \to \mathbb{Z}/4 \to \mathbb{Z}/2 \to 0$, we have

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathbb{Z}/2 & \longrightarrow & \mathbb{Z}/2 & \longrightarrow & \mathbb{Z}/2 & & \\ & & & & & \overset{1}{\swarrow} & & & \\ & & \mathbb{Z}/2 & \longrightarrow & \mathbb{Z}/2 & \longrightarrow & \mathbb{Z}/2 & & \\ & & & & & \overset{0}{\swarrow} & & & \\ & & \mathbb{Z}/2 & \longrightarrow & \mathbb{Z}/4 & \longrightarrow & \mathbb{Z}/2 & \longrightarrow & 0. \end{array}$$

(b) For $|K| =$ Klein and $0 \to \mathbb{Z} \to \mathbb{Z} \to \mathbb{Z}/2 \to 0$, we have

$$\begin{array}{ccccccc}
0 & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & \mathbb{Z}/2 \\
 & & & \swarrow_{(0,1)} & & & \\
 & & \mathbb{Z}\oplus\mathbb{Z}/2 & \longrightarrow & \mathbb{Z}\oplus\mathbb{Z}/2 & \longrightarrow & \mathbb{Z}/2\oplus\mathbb{Z} \\
 & & & \swarrow_{0} & & & \\
 & & \mathbb{Z} & \longrightarrow & \mathbb{Z} & \longrightarrow & \mathbb{Z}/2
\end{array}$$

For $|K| =$ Klein and $0 \to \mathbb{Z}/2 \to \mathbb{Z}/4 \to \mathbb{Z}/2 \to 0$, we have

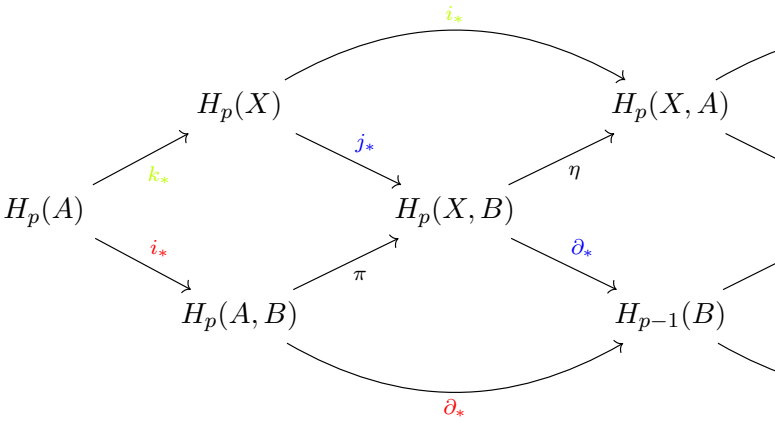
$$\begin{array}{ccccccc}
0 & \longrightarrow & \mathbb{Z}/2 & \longrightarrow & \mathbb{Z}/2 & \longrightarrow & \\
 & & & \swarrow_{(0,1)} & & & \\
 & & \mathbb{Z}/2\oplus\mathbb{Z}/2 & \longrightarrow & \mathbb{Z}/4\oplus\mathbb{Z}/2 & \longrightarrow & \\
 & & & \swarrow_{0} & & & \\
 & & \mathbb{Z}/2 & \longrightarrow & \mathbb{Z}/4 & &
\end{array}$$

25.3 S^{n+1} is the suspension of S^n, and let $Sf\colon (w * S^n)/S^n \to (w * S^n)/S^n$ induced by f be the suspension of f. The naturality of the boundary maps in the long exact sequence of the pair $(w * S^n, S^n)$ gives the commutative diagram, so $\deg Sf = \deg f$.

$$\begin{array}{ccc}
\tilde{H}_{n+1}(S^{n+1}) & \xrightarrow[\simeq]{\partial} & \tilde{H}_n(S^n) \\
\downarrow{\scriptstyle Sf_*} & & \downarrow{\scriptstyle f_*} \\
\tilde{H}_{n+1}(S^{n+1}) & \xrightarrow[\simeq]{\partial} & \tilde{H}_n(S^n)
\end{array}$$

The conclusion follows from induction and $k\colon S^1 \to S^1, k(z) = z^n$ has degree n.

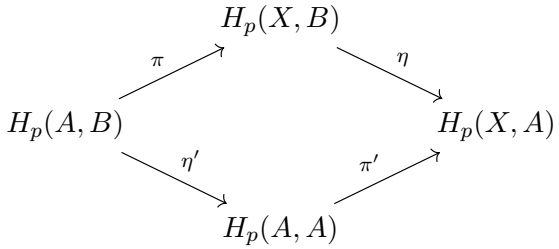
26.2 Let us take a look at a grotesque diagram below.



The three sequences whose maps are of the same color are exact. From the exactness of

$$H_p(A) \xrightarrow{i_*} H_p(A) \xrightarrow{\pi_*} H_p(A,A) \xrightarrow{\partial_*} H_{p-1}(A)$$

we deduce that $H_p(A,A) = 0$, and due to the commutativity (as all four maps are mere inclusions) of



we get $\eta \circ \pi = 0$. Then apply the braid lemma in Exercise 1(b), we obtain the exactness of the sequence with Greek alphabets above the arrows in the above diagram.

26.3 [Hatcher] Section 2.2 Exercise 38, p.159.

29.1 (b) By (a), $H_p(\text{tsc}) = \begin{cases} \mathbb{Z}^2, & p = 0 \\ 0, & p > 0 \end{cases}$.

31.3 $j\colon (X, A) \to (X, X)$ is a homotopy equivalence, so $H_p(X, A) = H_p(X, X) = 0$.

31.5 [Hatcher] Theorem 2.10, p.111.

$$\begin{aligned}\partial D_{\Delta_p} i_p = &\sum_{j\le i}(-1)^i(-1)^j l((\epsilon_0,0),\dots,\widehat{(\epsilon_j,0)},\dots,(\epsilon_i,0),(\epsilon_i,1),\dots,(\epsilon_p,1))\\ &+\sum_{j\ge i}(-1)^i(-1)^{j+1} l((\epsilon_0,0),\dots,(\epsilon_i,0),(\epsilon_i,1),\dots,\widehat{(\epsilon_j,1)},\dots,(\epsilon_p,1)).\end{aligned}$$

The terms with $i = j$ in the two sums cancel except for $l(\widehat{(\epsilon_0,0)},(\epsilon_0,1),\dots,(\epsilon_p,1))$, which is $j \circ i_p = j_\#(i_p)$, and $-l((\epsilon_0,0),\dots,(\epsilon_p,0),\widehat{(\epsilon_p,1)})$, which is $-i \circ i_p = -i_\#(i_p)$. The terms with $i \ne j$ are exactly $-D_{\Delta_p}\partial i_p$ since

$$\begin{aligned}D_{\Delta_p}\partial i_p = &\sum_{i<j}(-1)^i(-1)^j l((\epsilon_0,0),\dots,(\epsilon_i,0),(\epsilon_i,1),\dots,\widehat{(\epsilon_j,1)},\dots,(\epsilon_p,1))\\ &+\sum_{j>i}(-1)^{i-1}(-1)^j l((\epsilon_0,0),\dots,\widehat{(\epsilon_j,0)},\dots,(\epsilon_i,0),(\epsilon_i,1),\dots,(\epsilon_p,1)).\end{aligned}$$

33.1 Let X_1 be the arc minus the ends $(0,-1)$ and $(1,\sin 1)$, and X_2 be X minus a point of X_1. X_1, X_2 is an open covering of X, yielding an exact sequence

$$H_1(X_1)\oplus H_1(X_2) \longrightarrow H_1(X) \xrightarrow{\partial} H_0(X_1\cap X_2) \xrightarrow{i} H_0(X_1)\oplus H_0(X_2)$$

All path components of X_1 and X_2 are contractible, so $H_1(X_1)\oplus H_1(X_2) = 0$, and consequently $H_1(X) \simeq \operatorname{im}\partial = \ker i = 0$. And $H_0(X) = \mathbb{Z}$ since X is contractible.

33.3 S^n is the sphere with equator S^{n-1} and latitude $[-1,1]$, identifying $S^{n-1}\times 1$ to a north pole and $S^{n-1}\times(-1)$ to a south pole. Hence $H_p(S^n) = \begin{cases} \mathbb{Z}, & p = 0, n \\ 0, & \text{else} \end{cases}$.

35.4 (a) As $\overline{\text{St}}\, v$ is contractible and all of its homology groups vanish except for $p = 0$, for $p > 0$ $H_p(\text{Lk}\, v) \simeq H_p(\overline{\text{St}}\, v - v) \simeq H_{p+1}(\overline{\text{St}}\, v, \overline{\text{St}}\, v - v) = \begin{cases} \mathbb{Z}, & p = n-1 \\ 0, & \text{else} \end{cases}$ when $x \in \text{Int}\, M$ or 0 when $x \in \text{Bd}\, M$.

(b) Suppose that there is a simplex σ of dimension less than m that is not a face of any simplex of dimension m. Then $\text{Lk}\,\hat\sigma = \text{Bd}\,\sigma$ is a sphere of dimension less than $m-1$, contradicting (a) for $v = \hat\sigma$ and $K \cup \text{sd}\,\sigma$ in place of K.

(c) s cannot be a face of three simplices of maximal dimension, for otherwise the interiors of two of the simplices would overlap. If there are two m-simplices with s being a face, then $\hat s$ is an interior point of K. Therefore s is a face of precisely two m-simplices iff $\hat s$ is an interior point for any neighborhood of $\hat s$ in K, iff $h(s) \in \text{Int}\, M$; on the contrary s is a face of precisely one m-simplex iff $h(s) \in \text{Bd}\, M$.

so s is a face of precisely two m-simplices iff $h(s) \in \text{Bd}\, M$.

35.5 $S^3 \simeq \mathrm{Bd}(B^2 \times B^2) = (B^2 \times \mathrm{Bd}\, B^2) \cup (\mathrm{Bd}\, B^2 \times B^2) \simeq T_1 \cup T_2$, and $T_1 \cap T_2 \simeq \mathrm{Bd}\, B^2 \times \mathrm{Bd}\, B^2 =$

$$\begin{array}{ccccccccc} 0 & \longrightarrow & H_3(T) & \longrightarrow & H_3(T_1) \oplus H_3(T_2) & \longrightarrow & H_3(S^3) & \longrightarrow & H_2(T) \longrightarrow \\ & & \| & & \| & & \| & & \| \\ & & 0 & & 0 & & \mathbb{Z} & \xrightarrow{1} & \mathbb{Z} \end{array}$$

$S^1 \times S^1 =$ a hollow torus T, so

$$\begin{array}{ccccccccc} \longrightarrow & H_1(T) & \longrightarrow & H_1(T_1) \oplus H_1(T_2) & \longrightarrow & H_1(S^3) & \longrightarrow & H_0(T) & \longrightarrow \\ & \| & & \| & & \| & & \| & \\ & \mathbb{Z} \oplus \mathbb{Z} & \xrightarrow{(1,-1)} & \mathbb{Z} \oplus \mathbb{Z} & & 0 & & \mathbb{Z} & \end{array}$$

38.1 [Rotman] Lemma 8.18, Theorem 8.19, pp.201–2.

Lemma. For a cell $e \in X$, the closure $\bar{e}$ lies in a finite CW subcomplex.

Proof. We proceed by induction on $p = \dim e$; the statement is obviously true when $p = 0$. If $p > 0$, then Theorem 38.2(a) gives $\bar{e} - e = g(\mathrm{Bd}\, B) \subset X^{p-1}$. By "closure–finiteness", $\bar{e}$ meets only finitely many cells other than e, say, $e_1, \ldots, e_m$, and we have just seen that $\dim e_i \leq p-1$ for all i. By induction, there is a finite CW subcomplex X_i containing $\bar{e}_i$, for $i = 1, \ldots, m$, and each X_i is closed. But $\bar{e} \subset e \cup X_1 \cup \cdots \cup X_m$, so that this union of finitely many cells is closed and hence is a finite CW subcomplex. □

(a) For each cell e with $A \cap e \neq \varnothing$, choose a point $a_e \in A \cap e$, and let E be the set compromised of all such a_e. For each cell e, the lemma says that there is a finite CW subcomplex X_e containing $\bar{e}$. Therefore $E \cap \bar{e} \subset E \cap X_e$ is a finite set and hence is closed in $\bar{e}$. Since X has the weak topology, E is closed in X; indeed the same argument shows that every subset of E is closed in X, hence E is discrete. But E is also compact, being a closed subset of A. Thus E is finite, so that A meets only finitely many cells e, say, $e_1, \ldots, e_m$.

(b) By the lemma again, there are finite CW subcomplexes X_i with $\bar{e}_i \subset X_i$ for $i = 1, \ldots, m$. It follows that A lies in the finite CW subcomplex $\bigcup_{1 \leq i \leq m} X_i$.

38.4 (a) This is a special case of the proposition, having nothing to do with CW complexes, that $X \times Y$ is compactly generated if X is compactly generated Hausdorff and Y is locally compact. For the proof, cf. [Hatcher] Theorem A.15, p.531.

(b) Let $\{f_\alpha \colon B_\alpha \to X\}_\alpha$ be the family of the characteristic maps of X, $\{g_\beta \colon B_\beta \to Y\}_\beta$ be the family of the characteristic maps of X. Then $\{(f_\alpha, g_\beta) \colon B_\alpha \times B_\beta \to X \times Y\}_{\alpha,\beta}$ is a CW complex structure on $X \times Y$. Observe that every compact set A of $X \times Y$ lies in the product of its projections onto X and Y, and the two projections are compact and hence lie in finite subcomplexes of X and Y according to Exercise 38.1, so A lies in the image of a finite union of products $B_\alpha \times B_\beta$. Hence the prodoct CW structure generates $X \times_C Y$.

38.5 [Lu&Wei] Theorem 1.7, p.80.

Let X be a regular cell complex. Let $T(X)$ denote the abstract simplicial complex with vertices corresponding to the cells of X and n-simplices members of the set $\{(e_0, e_1, \ldots, e_n) \mid$ each e_i is a cell, $e_0 \subsetneq e_1 \subsetneq \cdots \subsetneq e_n\}$. Let $|T(X)|$ be the geometric realization of this complex. We define a homeomorphism $\Psi \colon |T(X)| \to X$ such that if e is a cell of X, $\Psi^{-1}(e)$ is a subcomplex of $|T(X)|$.

The map Ψ is constructed by first noticing that $T(X) = \bigcup_n T(X^n)$ and inductively constructing $\Psi^n\colon |T(X^n)| \to X^n$ so that Ψ^n defines a triangulation of each cell of dimension less than $n+1$ which contains the triangulated boundary of the cell as a subcomplex. First observe that if Ψ^{n-1} is defined and e^n is an n-cell, then $\mathrm{Bd}\, e^n$ is a subcomplex of X^{n-1}, and $(\Psi^{n-1})^{-1}(\mathrm{Bd}\, e^n)$ is a subcomplex of $|T(X^n)|$ which is homeomorphic to the $(n-1)$-sphere. The usual conical construction enables us to extend Ψ^{n-1} over $|T(X^n)|$.

Geometrically, we are performing a generalized barycentric subdivision of the complex X. The new vertices correspond to the "barycenters" of the cells of X. The new cells correspond to the cones over the new $(n-1)$-cells of X with vertices at the "barycenters" of the n-cells.

39.2 [Hatcher] Proposition 2.22, p.124.

Let V be a neighborhood of A in X that deformation retracts onto A. We have a commutative diagram
$$\begin{array}{ccccc} H_p(X,A) & \longrightarrow & H_p(X,V) & \longleftarrow & H_p(X-A,V-A) \\ \downarrow{\scriptstyle q_*} & & \downarrow{\scriptstyle q_*} & & \downarrow{\scriptstyle q_*} \\ H_p(X/A,A/A) & \longrightarrow & H_p(X/A,V/A) & \longleftarrow & H_p(X/A-A/A,V/A-A/A) \end{array}$$
The upper left horizontal map is an isomorphism since in the long exact sequence of the triple (X,V,A) the groups $H_p(V,A)$ are zero for all p, because a deformation retraction of V onto A gives a homotopy equivalence of pairs $(V,A) \simeq (A,A)$, and $H_p(A,A) = 0$. The deformation retraction of V onto A induces a deformation retraction of V/A onto A/A, so the same argument shows that the lower left horizontal map is an isomorphism as well. The other two horizontal maps are isomorphisms directly from excision. The right–hand vertical map q is an isomorphism since q restricts to a homeomorphism on the complement of A. From the commutativity of the diagram it follows that the left–hand q_* is an isomorphism.

39.3 $D_p(A)$ is a free abelian group with a basis $\{(f_\alpha)_*(\gamma) : f_\alpha$ is a characteristic map for a p-cell of $A\}$, and inclusion simply maps each p-cell of A to itself in X.

(a) According to Five–lemma, the isomorphisms between the homology groups of CW complexes and simplicial homology groups induce isomorphisms between the relative ones.
$$\begin{array}{ccccccccc} H_p(\mathcal{D}(A)) & \longrightarrow & H_p(\mathcal{D}(X)) & \longrightarrow & H_p(\mathcal{D}(X,A)) & \longrightarrow & H_{p-1}(\mathcal{D}(A)) & \longrightarrow & H_{p-1}(\mathcal{D}(X)) \\ \downarrow\simeq & & \downarrow\simeq & & \downarrow & & \downarrow\simeq & & \downarrow\simeq \\ H_p(A) & \longrightarrow & H_p(X) & \longrightarrow & H_p(X,A) & \longrightarrow & H_{p-1}(A) & \longrightarrow & H_{p-1}(X) \end{array}$$
(b) Let $X_p = X^p \cup A$ for all integers p. Repeat the proof of Theorem 39.4, replacing $H_i(X_p)$ by $H_i(X_p,A)$ and $H_i(X)$ by $H_i(X,A)$, we get $H_i(X_p,X_{p-1}) \simeq H_i(X,A)$ (for a detailed proof, see [Jiang] Section 3.2, 3.3, pp.99–104).

Plus, the isomorphism between $\mathcal{D}(X,A) \simeq \bigoplus_\alpha H_p(B^p_\alpha, S^{p-1}_\alpha)$ and $H_p(X_p,X_{p-1})$ is given by
$$\bigoplus_\alpha H_p(B^p,S^{p-1}) \overset{\oplus(f_\alpha)_*}{\longrightarrow} H_p(X_p,X_{p-1}), \quad H_p(X_p,X_{p-1}) \overset{\oplus(q_\alpha)_*}{\longrightarrow} \bigoplus_\alpha H_p(B^p,S^{p-1}),$$
where the index α is taken over all p-cells in $X-A$, f_α is the characteristic map for e^p_α, and q_α is the quotient map collapsing $X^p - e^p_\alpha$ to a point.

39.4 (a) Let γ be the map induced by the following diagram
$$\begin{array}{ccc} (E_+^n, S^{n-1}) & \hookrightarrow & (S^n, E_-^n) \\ \simeq\downarrow g & & \downarrow h \\ (\sigma, \operatorname{Bd}\sigma) & \xrightarrow{\gamma} & (S^n, x_0) \end{array}$$
where E_-^n is the hemisphere with the south pole x_0, g is a homeomorphism, h is a homotopy equivalance stretching E_+^n to cover S^n, and the inclusion induces an isomorphism in homology since
$$H_n(E_+^n, S^{n-1}) \simeq H_n(S^n - U, E_-^n - U) \simeq H_n(S^n, E_-^n)$$
where U is a small disk neighborhood of x_0, and the first isomorphism is by homotopy equivalence and the second by excision.

Suppose that ϕ is multiplication by d, then it is the homomorphism induced by γ composite a map from S^n to S^n of degree d.

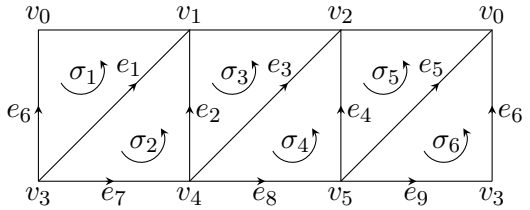
(b) This is because the inclusions induce an isomorphism $\oplus(i_\alpha)_*\colon \bigoplus_\alpha H_n(S_\alpha^n, p) \to H_n(X, p)$.

(c) For $p > 0$, B_{p-1}, a subgroup of the free group C_{p-1}, is also free and thus projective. It follows that there is a homomorphism $k_{p-1}\colon B_{p-1} \to C_p$ such that $\partial \circ k_{p-1} = \mathrm{id}_{B_{p-1}}$, and therefore $C_p = k_{p-1}B_{p-1} \oplus Z_p$.
$$\begin{array}{ccccccccc} & & & & & & B_{p-1} & & \\ & & & & & \overset{k_{p-1}}{\swarrow} & \| & & \\ 0 & \longrightarrow & Z_p & \longrightarrow & C_p & \xrightarrow{\partial} & B_{p-1} & \longrightarrow & 0 \end{array}$$
For $p = 0$, $H_0(C) = C_0/\partial C_1 = C_0/\partial U_1$, and $A \coloneqq k_{-1}H_0(C)$ is non–trivial.
$$\begin{array}{ccccccccc} & & & & & & H_0(C) & & \\ & & & & & \overset{k_{-1}}{\swarrow} & \| & & \\ 0 & \longrightarrow & U_1 & \xrightarrow{\partial} & C_0 & \xrightarrow{/\partial U_1} & H_0(C) & \longrightarrow & 0 \end{array}$$

(d) By (c), it suffices to construct a CW complex X whose cellular chain complex is isomorphic to $\partial\colon U_n \to Z_{n-1} \to 0 \to \cdots \to 0 \to \mathbb{Z}$ for each n (and ∂U_1 in place of Z_{-1} for $n = 0$) where the final $\mathbb{Z}$ is generated by a point in A, since $\partial = 0$ on Z_n and $\operatorname{im}\partial \subset \ker\partial$.

Both U_n and Z_{n-1} are free abelian groups, bases denoted $\{x_\alpha\}$ and $\{y_\beta\}$, and write $\partial x_\alpha = \sum_\beta d_{\beta\alpha} y_\beta$. Let $X^n = \bigvee_\alpha S_\alpha^n$. We construct X from X^n by attaching cells e_β^{n+1} via maps $f_\beta\colon S^n \to X^n$ such that the projection onto the summand S_α^n has degree $d_{\beta\alpha}$. Plus, generate A by other points if $|A| > 1$.

43.2 (a) Let us take
$$\sigma_2^*,\ \sigma_1^* - \sigma_2^*,\ \sigma_2^* - \sigma_3^*,\ \sigma_3^* - \sigma_4^*,\ \sigma_4^* - \sigma_5^*,\ \sigma_5^* - \sigma_6^*$$
as a basis for $Z^2(C, E)$, and
$$e_7^*,\ e_1^*,\ e_2^*,\ e_3^*,\ e_4^*,\ e_5^*,\ e_1^* + e_2^* + e_3^* + e_4^* + e_5^* + e_6^*,\ e_2^* + e_3^* - e_7^* + e_8^*,\ e_4^* + e_5^* - e_8^* + e_9^*$$
as a basis for $C^1(C, E)$. We see that $\delta e_i^* = \sigma_i^* - \sigma_{i+1}^*$ for $i = 1, \ldots, 5$; and
$$\delta e_6^* = \sigma_6^* - \sigma_1^*, \quad \delta e_7^* = \sigma_2^*, \quad \delta e_8^* = \sigma_4^*, \quad \delta e_9^* = \sigma_6^*.$$
Thus δ is surjective, i.e. $H^2(C, E) = 0$.



Note that

$\delta(e_1^* + \cdots + e_6^*) = 0$, and

$$\delta(e_2^* + e_3^* - e_7^* + e_8^*) = 0, \quad \delta(e_4^* + e_5^* - e_8^* + e_9^*) = 0.$$

That is to say, $e_1^* + \cdots + e_6^*, e_2^* + e_3^* - e_7^* + e_8^*, e_4^* + e_5^* - e_8^* + e_9^*$ constitute a basis for 1-cocycles; but they are 1-coboundaries as well, as

$$-\delta(v_3^* + v_4^* + v_5^*) = e_1^* + \cdots + e_6^*, \quad -\delta v_4^* = e_2^* + e_3^* - e_7^* + e_8^*, \quad -\delta v_5^* = e_4^* + e_5^* - e_8^* + e_9^*,$$

and therefore $H^1(C, E) = 0$. Finally, $H^0(C, E) = 0$.

$$0 \longrightarrow H^0(C, E) \longrightarrow H^0(C) \xrightarrow{\simeq} H^0(E)$$

(b) The exact sequence $\longrightarrow H^1(C, E) \longrightarrow H^1(C) \xrightarrow{\simeq} H^1(E)$ yields

$$\longrightarrow H^2(C, E) \longrightarrow H^2(C) \xrightarrow{\simeq} H^2(E) \longrightarrow 0$$

$$H^i(C) \simeq H^i(E) = \begin{cases} \mathbb{Z}, & i = 0, 1 \\ 0, & i = 2 \end{cases}.$$

43.3 (a), (b), (f).

44.3 This is inferred by a general fact that if a chain map between free chain complexes $\phi\colon C \to C'$ induces isomorphisms $\phi_*\colon H_p(C) \to H_p(C')$ on homology groups for all p, then it also induces isomorphisms $\phi^*\colon H^p(C'; G) \to H^p(C; G)$ on cohomology groups with any coefficient group G for all p. The proof of the fact is an application of the Five–lemma to the diagram below.

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathrm{Ext}(H_{p-1}(C), G) & \longrightarrow & H^p(C; G) & \longrightarrow & \mathrm{Hom}(H_p(C), G) & \longrightarrow & 0 \\ & & \uparrow\simeq & & \uparrow & & \uparrow\simeq & & \\ 0 & \longrightarrow & \mathrm{Ext}(H_{p-1}(C'), G) & \longrightarrow & H^p(C'; G) & \longrightarrow & \mathrm{Hom}(H_p(C'), G) & \longrightarrow & 0 \end{array}$$

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